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Tuning apicobasal polarity and junctional recycling in the hemogenic endothelium orchestrates the morphodynamic complexity of emerging pre-hematopoietic stem cells

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Abstract

Hematopoietic stem cells emerge in the embryo from an aortic-derived tissue called the hemogenic endothelium (HE). The HE appears to give birth to cells of different nature and fate but the molecular principles underlying this complexity are largely unknown. Here we show, in the zebrafish embryo, that two cell types emerge from the aortic floor with radically different morphodynamics. With the support of live imaging, we bring evidence suggesting that the mechanics underlying the two emergence types rely, or not, on apicobasal polarity establishment. While the first type is characterized by reinforcement of apicobasal polarity and maintenance of the apical/luminal membrane until release, the second type emerges via a dynamic process reminiscent of trans-endothelial migration. Interfering with Runx1 function suggests that the balance between the two emergence types depends on tuning apicobasal polarity at the level of the HE. In support of this and unexpectedly, we show that the zebrafish Pard3 isoform - Pard3ba - is sensitive to interference with Runx1 activity, in aortic endothelial cells. This supports the idea of a signaling cross talk controlling cell polarity and its associated features, between aortic and hemogenic cells. In addition, using new transgenic fish lines that express Junctional Adhesion Molecules and functional interference, we bring evidence for the essential role of ArhGEF11/PDZ-RhoGEF in controlling the HE-endothelial cell dynamic interface, including cell-cell intercalation, which is ultimately required for emergence completion. Overall, we highlight critical cellular and dynamic events of the endothelial-to-hematopoietic transition that support emergence complexity, with a potential impact cell fate.

eLife assessment

This **important** study presents a detailed characterization of two distinct cellular morphologies of haematopoietic stem cells undergoing endothelial to haematopoietic transition in zebrafish. It brings new information on how regulation of apico-basal polarity influences cellular behaviour, shape, and interaction with neighbouring cells. The evidence supporting the existence of these two distinct morphologies is **convincing**, using state-of-the-art confocal microscopy and image analysis of 2D-cartography.

Introduction

Hematopoietic stem cells (HSCs) endowed with full regenerative potential in adult vertebrates are generated during a narrow-time window of few days during embryonic development. These cells, at the root of all blood and immune cells in the body, emerge from intra-embryonic aortic vessels and, more specifically, from a specialized type of vascular tissue called the hemogenic endothelium (HE, (Wu and Hirschi 2021 [↗](#))). A series of seminal studies have evidenced the autonomous production of repopulating HSCs within an intraembryonic region called the Aorta-Gonad-Mesonephros (AGM), before their appearance in other hematopoietic organs (A. L. Medvinsky et al. 1993 [↗](#); Müller et al. 1994 [↗](#); Garcia-Porrero, Godin, and Dieterlen-Lièvre 1995 [↗](#); A. Medvinsky and Dzierzak 1996 [↗](#); Cumano, Dieterlen-Lievre, and Godin 1996 [↗](#)). Thereafter, the endothelial origin of HSCs was evidenced (Jaffredo et al. 1998 [↗](#); de Bruijn et al. 2002 [↗](#); T. E. North et al. 2002 [↗](#)). The direct visualization, in real-time, of the emergence of precursors of hematopoietic stem and progenitor cells (HSPCs) from the dorsal aorta, termed the Endothelial-to-Hematopoietic Transition (EHT), was finally achieved *in vitro* (Eilken, Nishikawa, and Schroeder 2009 [↗](#)), *ex-vivo* from mouse sections (Boisset et al. 2010 [↗](#)), and *in vivo* in the zebrafish embryo (Bertrand et al. 2010 [↗](#); Kissa and Herbomel 2010 [↗](#)), using live microscopy. This opened the way to a more detailed analysis of the characteristics of the HE at the transcriptional level, revealing its transient nature and its early hematopoietic commitment (Swiers et al. 2013 [↗](#)).

The HE is characterized by inherent heterogeneity and is not only contributing to the formation of precursors to long-term HSCs (pre-HSCs) but also to more restricted progenitors (Hadland and Yoshimoto 2018 [↗](#)). These progenitors can be born from HE sub-types that are found in extra-embryonic source such as the yolk-sac – in which case it gives rise to erythro-myeloid progenitors that will sustain erythro-myelopoiesis until birth (Frame et al. 2016 [↗](#)) -, or both from extra and intra-embryonic sources (the yolk sac and the dorsal aorta) in which cases progenitors are biased, such as for example in the mouse, toward T- and B-innate-type lymphoid progenitors (Yoshimoto et al. 2011 [↗](#); 2012 [↗](#)) or toward less restricted, multipotent progenitors (Hadland and Yoshimoto 2018 [↗](#); Dignum et al. 2021 [↗](#)). Importantly, while some of the non-HSC derived progenitors born during embryonic life only support the functions of the immune system during embryonic/ fetal life, others can persist in the adult to exert tissue-resident functions, as has been mainly described in the mouse (Ghosn et al. 2019 [↗](#)). However, studies in human embryos support the idea that developmental hematopoiesis is highly similar in mice and humans (Ivanovs et al. 2011 [↗](#); 2017 [↗](#)).

A key question regarding the issue of the capacity of the HE to give birth to cells endowed with different hematopoietic potential is whether this comes from HE cells of distinct origin (in which case the HE would be a mosaic of precursors with distinct potential) or if this results from extrinsic environmental cues that impose variability in the constantly evolving developmental context (for a discussion, see (Barone et al. 2022 [↗](#))).

To address this type of question, high-resolution experimental settings need to be developed, if possible at the single-cell resolution. Along this line, recent developments of transcriptomics, including single cell RNAseq, spatially resolved in situ transcriptomic approaches, and in situ barcoding, have been invaluable (Weijts, Yvernogeu, and Robin 2021 [↗](#)). Among those high-resolution approaches, cell fate mapping and lineage tracing approaches that include imaging technologies have been very instrumental, particularly when performed with the zebrafish model that reproduces many aspects of developmental hematopoiesis in higher vertebrates (Orkin and Zon 2008 [↗](#)). Recently, work performed in the zebrafish embryo has brought evidence for heterogeneity of hematopoietic stem cell precursors being born from the HE, in the ventral floor of the dorsal aorta and independently from HSCs, including a wave of transient T-lymphocytes (using temporally-spatially resolved fate-mapping, (Tian et al. 2017 [↗](#))) and lympho-myeloid biased progenitors born from a myeloid-lymphoid biased Spi2⁺ HE that appears to co-exist with an erythroid-biased HE (using single-cell RNA-sequencing, (Xia et al. 2023 [↗](#))).

While light starts to be shed on the molecular and signaling cues that appear to regulate HE sub-specification and a continuum from arterial endothelium, HE maturation, and subsequent heterogeneity in HSPCs (Zhu et al. 2020 [↗](#)), the essential molecular and cell biological properties that support HE functional plasticity remain to be determined.

Here, using the zebrafish embryo as a model, we complement our previous work describing essential molecular and mechanistic features of EHT cell emergence (Lancino et al. 2018 [↗](#)). With the support of high-resolution live imaging and the generation of new transgenic fish lines that express a functional marker of cell polarity (Podocalyxin, a sialomucin of the CD34 family (Nielsen and McNagny 2008 [↗](#))), we show that the HE is giving birth to two cell types with distinct cell polarity status; these cells also emerge with radically different morphodynamic characteristics, which raises the intriguing possibility of an incidence of EHT emergence complexity on downstream fate, after release from the aortic wall.

To substantiate the significance of apico-basal polarity control in the EHT, we investigate on the potential involvement of Pard3 proteins. These proteins, recruited by transmembrane receptors via their PSD-95/Dlg/ZO-1 (PDZ) domains (Buckley and St Johnston 2022), are at the root of apicobasal polarity initiation and are essential for the maintenance of apical membrane functional properties (Román-Fernández and Bryant 2016 [↗](#)). Among the four Pard3 isoforms expressed in the zebrafish (encoded by four different genes and non-including splicing variants), we describe the specific expression of the zebrafish Pard3ba isoform in the aorta, during the EHT time-window. We show that the expression of Pard3ba is sensitive to interference with the transcription factor Runx1 whose function is essential for EHT completion (Kissa and Herbomel 2010 [↗](#)) and for regulating HSC number (Adamo et al. 2009 [↗](#); T. E. North et al. 2009 [↗](#)). In link with apico-basal polarity and to address its incidence on the biomechanics of the emergence of the two EHT cell types that we describe, we generated new fish lines that express Junctional Adhesion Molecules (JAMs) fused with eGFP (eGFP-JAMs). The JAMs belong to tight junction complexes (Garrido-Urbani, Bradfield, and Imhof 2014 [↗](#)) and recruit Pard3 via direct interaction with their carboxy-terminus (Ebnet et al. 2003 [↗](#)), thereby allowing initiation of apico-basal polarity. With our eGFP-JAMs fish lines, we investigate on the function of ArhGEF11/PDZ-RhoGEF that was shown to be involved in disrupting junctional integrity in the context of the epithelial-to-mesenchymal transition as well as in the growth and migration of invasive cancer cells (more specifically one splicing variant of its c-terminus conserved in mammalian species, Itoh et al. 2017 [↗](#); Lee et al. 2018 [↗](#)). We reveal the function of ArhGEF11/PDZ-RhoGEF in the dynamic interplay between HE/EHT cells and their endothelial neighbors, particularly in cell-cell intercalation which is essential for EHT completion and the proper sealing of the aortic floor.

Results

Apicobasal polarity determines emergence types

Our previous work describing the morphodynamic evolution of cells emerging from the aortic floor through the EHT (hereafter designated as “EHT cells”), in the zebrafish embryo, revealed the unusual feature of a cell extruding from the plane of a tissue while maintaining its luminal/apical membrane until the very end of the release process, thus contributing to its peculiar crescent-shaped morphology (Kissa and Herbomel 2010 [↗](#); Lancino et al. 2018 [↗](#)), and see the cartoons in **Figure 1A** [↗](#)). However, to our knowledge, the polarity status of EHT cells has not been investigated so far and the maintenance of a *bona fide* apical domain has never been proven (with the luminal membrane enriched in apically targeted proteins and physically delimited by apical polarity factors and tight junction complexes (Rodriguez-Boulán, Müsch, and Le Bivic 2004 [↗](#); Buckley and St Johnston 2022)). Importantly, the fate of this apical-luminal membrane, after the release, may lead to cells potentially endowed with specific functional features. For example, this membrane surface may be directly and fully exposed to the extracellular space or released in the cytoplasm of EHT cells for recycling and/or degradation, after emergence completion (for examples of different scenarios, see **Figure 1A** [↗](#), and the legend for more details). Overall, this could lead to precursors of hematopoietic stem cells that, ultimately, may be differentially fated.

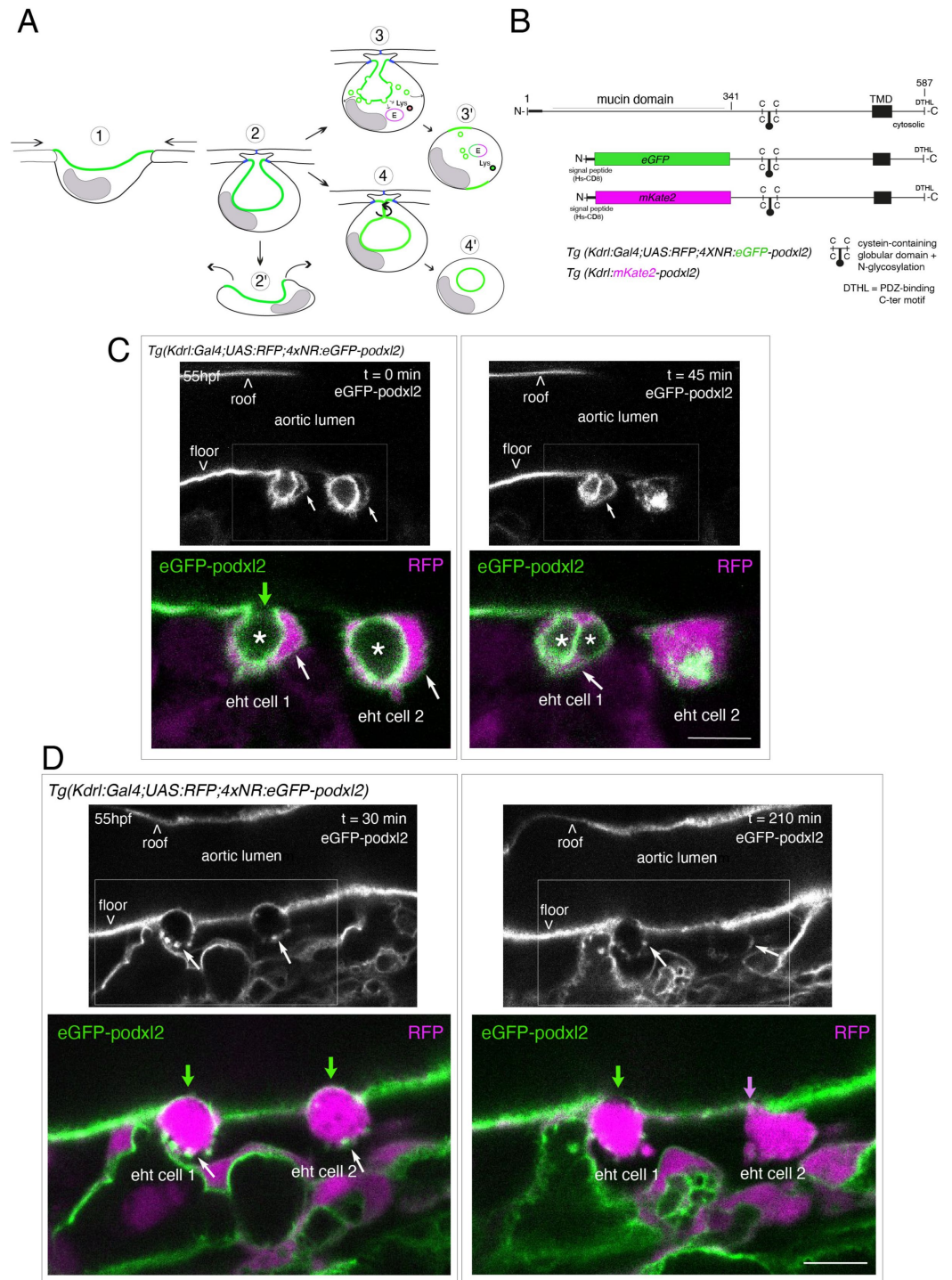


Figure 1

Tracing the evolution of the luminal membrane using the polarity marker Podocalyxin points at the biological significance of apicobasal polarity establishment in EHT cell emergence complexity.

A, cartoons depicting the early and late steps of EHT cells emerging from the aortic floor (steps 1 and 2, as previously described in the zebrafish embryo, see (Kissa and Herbomel 2010; Lancino et al. 2018)) and with hypothetical evolution of the luminal membrane (in green) before (3 and 4) and after the release (2', the cell detaches from the

endothelial layer via junction downregulation leading to exposure of the luminal membrane with the extracellular milieu; 3', the luminal membrane is consumed via endocytic recycling (E) and/or lysosomal degradation (Lys) prior to detachment; 4', the luminal membrane in 4 is released inside the cell (twisted arrow) before detachment. Grey area = nucleus. **B**, PodocalyxinL2 (Podxl2) construct designed to establish transgenic fish lines. Cartoons representing full length (top drawing) and deleted Podxl2 (amino-acid sequence 341-587) in which the mucin domain (serine/threonine-rich O-glycosylation domain) is replaced by either eGFP or mKate2. The tetracysteine-containing globular domain (subjected to N-glycosylation) was kept as favoring apical membrane retention. TMD, transmembrane domain; DTHL, C-terminal peptidic motif involved in partnership with PDZ domain containing proteins. **C-D**, EHT performing cells visualized using *Tg(Kdrl:Gal4;UAS:RFP;4xNR:eGFP-podxl2)* embryos and time-lapse sequences initiated at 55 hpf obtained with spinning disk confocal microscopy (imaging was performed at the boundary between the most downstream region of the AGM and the beginning of the caudal hematopoietic tissue). Top grey panels show the green, eGFP channels for eGFP-podxl2 only. Bottom panels show the merge between green and red (soluble RFP, in magenta) channels. Bars: 8 μ m. **C**, single plane images of 2 EHT pol+ cells extracted from a time-lapse sequence at t = 0 and t = 45 min, with the right cell (eht cell 2) more advanced in the emergence process than the left one (eht cell 1). Note the enrichment of eGFP-podxl2 at the luminal membrane (surrounding the cavity labelled with an asterisk) in comparison to the basal membrane (white arrow). Note also the evolution of the luminal membranes with time, with the aortic and eht cell 1 lumens still connecting at t = 0 (green arrow), the apparent fragmentation of the cytosolic vacuole (2 asterisks for eht cell 1 at t = 45 min) and the compaction of Podxl2-containing membranes for eht cell 2 at t = 45 min. More details on the evolution of the connection between the aortic/eht cell lumens are shown in **Figure 1 - figure supplement 1A**. **D**, single plane images of 2 EHT pol-cells extracted from a time-lapse sequence at t = 30 min and t = 210 min (see **Figure 1 - video 3** for the full-time lapse sequence), with the right cell (eht cell 2) slightly more advanced in the emergence than the left one (eht cell 1, with the latest attachment point between the emerging cell and the aortic floor (pink arrow)). Note, in comparison with the cells in panel C, the ovoid shapes of cells, the absence of enrichment of eGFP-podxl2 at luminal membranes (green arrows) and the accumulation of eGFP-podxl2 at basal membrane rounded protrusions (white arrows).

To address the polarity status of EHT cells, we raised transgenic fish lines that express endogenous Podocalyxin (Podxl2, (Herwig et al. 2011)). Podocalyxin was shown to take part in the formation of the preapical domain during polarization and in the regulation of its oriented organization, in tissue culture (Meder et al. 2005; Bryant et al. 2014). Its contribution to lumenization *in vivo*, in the mouse aorta, has been described and it involves negative charge repulsion induced by their glycosyl residues (Strilić et al. 2009; 2010).

We first attempted to express transiently, in the vascular system, the full-length Podxl2 zebrafish protein fused to eGFP at its extreme N-terminus. We failed to detect the fusion protein at any time point relevant for observing easily the EHT process (a time window ranging from 48 to 72 hpf (hours post-fertilization)); therefore, we designed a N-ter truncated form deleted of the mucin domain and that retains amino-acids 341-587 fused to eGFP at its N-terminus (**Figure 1B** and see **Methods**). Transient transgenesis revealed that this truncated version is detected and is targeted to the luminal membranes of EHT cells. We then raised 2 transgenic (*Tg*) fish lines that express the N-ter truncated form of Podxl2 fused to either eGFP (thereafter abbreviated eGFP-podxl2) and under the control of the *Kdrl:Gal4* driver *Tg(Kdrl:Gal4;UAS:RFP; 4xNR:eGFP-podxl2)*, or to mKate2 and under the control of the *Kdrl* promoter *Tg(Kdrl:mKate2-podxl2)*. We observed that eGFP-podxl2 is enriched at the luminal side of crescent-shaped EHT undergoing cells (see **Figure 1C**, **Figure 1 - figure supplement 1A** and **Figure 1 - video 1** (z-stack at t = 0) and their legends for the details of the luminal/apical membrane evolution through time). Thereafter, these cells will be referred to as EHT pol+ cells.

We also followed the cell after emergence and observed the evolution of the luminal/apical membrane appearing as internal pseudo-vacuoles. We illustrate the reduction of their volume via membrane retrieval and, ultimately, their remanence as an intracellular membrane compartment which we define as a post-EHT signature (**Figure 1 - figure supplement 1A** and **Figure 1 - video 2** (z-stack at t = 80 min)); we propose a model for the intracellular evolution of the luminal/apical

membrane (**Figure 1-figure supplement 1B**) which, unfortunately, cannot be traced after 2-3 hrs post-emergence because of the apparent short half-life of eGFP-podxl2 and of the drop in activity of the *Kdr1* promoter. Of notice, the pseudo-vacuoles are reminiscent of the cystic formations observed in EHT cells in the mouse and visualized by electron-microscopy (T. North et al. 1999 [\[1\]](#); Marshall and Thrasher 2001 [\[2\]](#)) and also of the vacuolar structures recently described in EHT cells in avian embryos (Sato et al. 2023 [\[3\]](#); for more details, see **Discussion**).

While imaging the EHT using the Podxl2 expressing lines that clearly delimitate the luminal membrane, we unambiguously identified a second type of emergence. This second cell type is primarily characterized by a round-to-ovoid morphology (cells never bend as crescent-shaped EHT pol+ cells, see **Figure 1D** [\[4\]](#) and **Figure 1** [\[5\]](#) – video 3 for a time-lapse sequence). Importantly, these cells do not show any enrichment of eGFP-podxl2 at the luminal membrane and will be referred to as EHT pol- cells. EHT pol- cells were observed in all other *Tg* fish lines that we are routinely imaging, including the *Tg(Kdr1:Gal4;UAS:RFP)* parental line that was used for transgenesis, thus excluding the possibility that these cells result from an artefact due to the expression of a deleted form of Podxl2 and/or to its overexpression (see also for example our new *Tg(kdr1:eGFP-Jam3b)* fish line **Figure 5C** [\[6\]](#) and **Figure 6B** [\[7\]](#)).

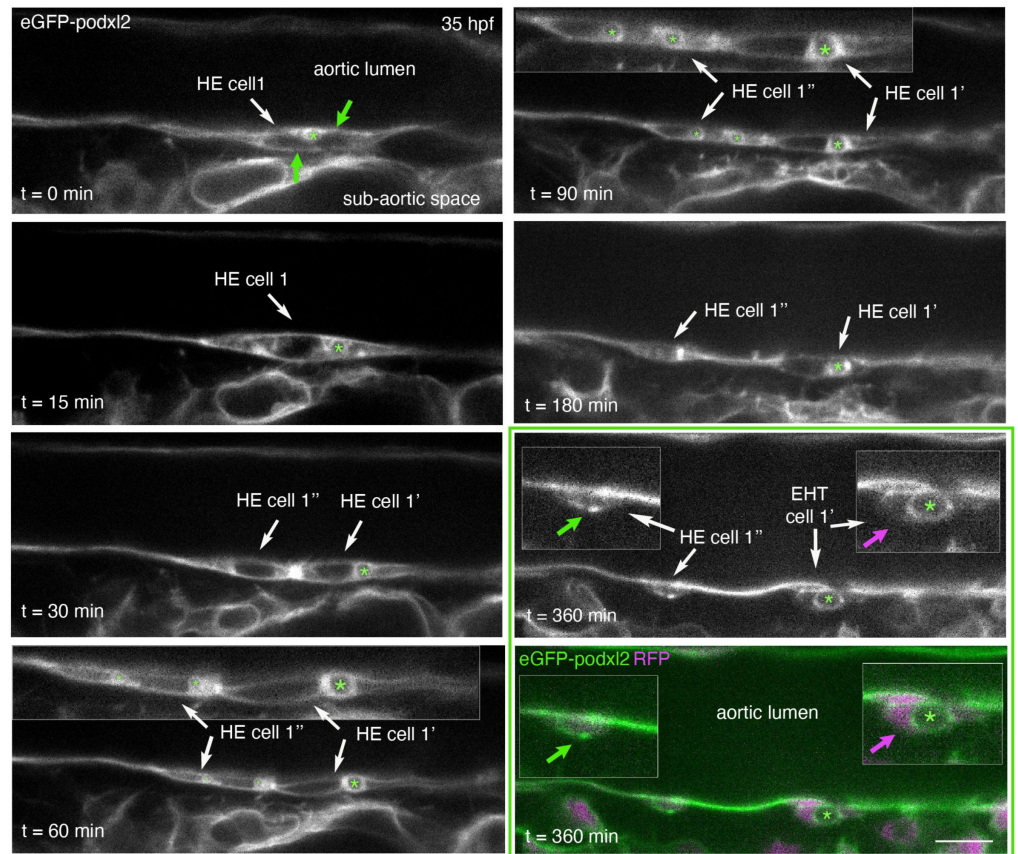
Finally, we have estimated that the ratio between EHT pol+ and EHT pol- cells is of approximately 2/1, irrespective of the imaging time window and of the localization of emergence along the aortic antero-posterior axis (starting from the most anterior part of the AGM (at the limit between the balled and the elongated yolk) down to the caudal part of the aorta facing the CHT). We observed that both EHT pol+ and EHT pol- cells divide during the emergence and remain with their respective morphological characteristics after completing abscission (**Figure 1 – figure supplement 2**); hence they appear as pairs of cells that exit the aortic wall sequentially (as shown **Figure 1C, D** [\[4\]](#)). We also observed that both EHT pol+ and EHT pol- cells express reporters driven by the hematopoietic marker CD41 (**Figure 1 – figure supplement 3**), which indicates that they are both endowed with hematopoietic potential. Altogether, our results show that hematopoietic precursor cells emerging from the aortic floor do so with heterogeneity in their morphodynamic characteristics. They also suggest that the control of apico-basal polarity may be at the root of these specific emergence types.

The immature HE is not polarized

EHT pol+ and EHT pol- cells appear to emerge from the hemogenic endothelium (HE), the latter constituting the virtually exclusive cell population of the aortic floor just prior to the initiation of the EHT time-window (around 28 hpf, see (Zhao et al. 2022 [\[8\]](#))). In this context, we addressed the polarity status of HE cells and its evolution throughout the EHT (**Figure 2** [\[9\]](#)). Surprisingly, confocal microscopy using our *eGFP-podxl2* and *mKate2-podxl2* expressing fish lines revealed that HE cells do not appear to be polarized, based on the absence of Podxl2 enrichment at luminal membranes, at the initiation of the EHT time-window and later (at approximately 28 - 30 hpf, **Figure 2 – figure supplement 1**, see also at 35 hpf **Figure 2A** [\[10\]](#), top left panel, green arrows and **Figure 2 – figure supplement 2** and at 48 – 55 hpf **Figure 3 – figure supplement 2A**). Interestingly, the cytoplasm of characteristic elongated HE cells located on the aortic floor is filled with more-or-less large membrane vesicles that carry eGFP-podxl2 (the largest vesicles reaching approximately 30µm in diameter). This suggests that HE cells contain a reservoir of eGFP-podxl2 membranes that may be subjected to exocytose; as such, HE cells may be comparable to endothelial cells organizing a vascular lumen and that have been proposed to exocytose large intracellular macropinocytic-like vacuoles when cultured in 3D extracellular matrices (Bayless and Davis 2002 [\[11\]](#); Davis, Bayless, and Mavila 2002 [\[12\]](#)) or, *in vivo*, in the zebrafish model (Kamei et al. 2006 [\[13\]](#); Lagendijk, Yap, and Hogan 2014 [\[14\]](#)). This finding is unexpected since HE cells are assumed to possess aortic cell characteristics (i.e exhibit an apicobasal polarity) as they are supposedly integrated in the aortic wall contemporarily to aortic precursors (Jin et al. 2005 [\[15\]](#)) and may have

been taking part in the lumenization of the ancestral vascular cord, a process that takes place around 18-20 hpf. Consequently, loss of apicobasal polarity features of HE cells at 28-30 hpf may be part of the programme that initiates the EHT process.

A *Tg(Kdrl:Gal4;UAS:RFP;4xNR:eGFP-podxl2)*



B

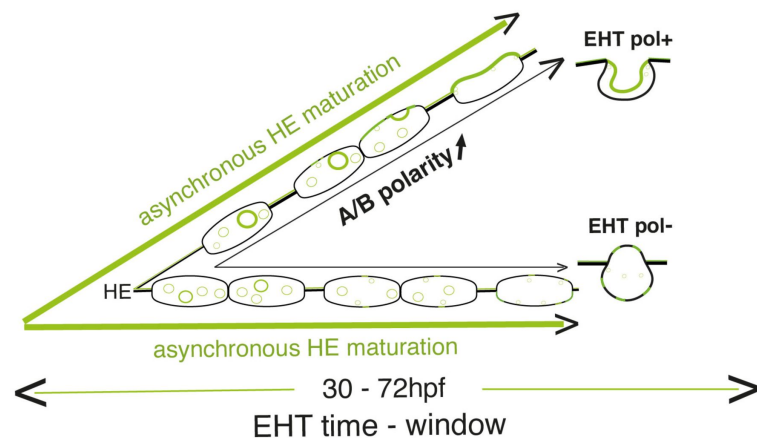


Figure 2

Immature HE is not polarized and controls membrane delivery of intra-cytosolic vesicular pools.

A, *Tg(Kdrl:Gal4;UAS:RFP;4xNR:eGFP-podxl2)* embryo imaged using spinning disk confocal microscopy. Black and white images show eGFP-podxl2 only. Images (single z-planes) were obtained from a time-lapse sequence (initiated at 35 hpf) lasting for 435 min (7.25 hrs), with intervals of 15 min between each z-stack. Example of an HE cell with equal partitioning of eGFP-podxl2 between luminal and abluminal membranes (at $t = 0$ min), with eGFP-podxl2 containing intra-cytosolic vesicles (one labelled with a green asterisk) and undergoing mitosis at $t = 30$ min (HE cell 1' and HE cell 1'' are daughter cells). Note the inheritance of the largest micropinocytic-like vacuole by HE cell 1' and its maintenance over time until EHT emergence initiation at $t = 180$ min (green asterisk in 1.5x magnified areas at $t = 60$ and 90 min). At $t = 360$ min (green box) EHT is proceeding and both fluorescence channels are shown; bottom panel: green (eGFP-podxl2), magenta (soluble RFP). The magenta arrow points at the basal side of the EHT pol + cell (EHT cell 1', on the right) that does not contain any detectable eGFP-podxl2; on the contrary, eGFP-podxl2 is enriched at the luminal/apical membrane (note that exocytosis of the large vacuolar structure may have contributed to increase the surface of the apical/luminal membrane (the green asterisk is surrounded by the apical/luminal membrane of the EHT pol + cell)). The green arrow points at the abluminal membrane of the EHT cell derived from HE cell 1'' (EHT cell 1'') and that contains eGFP-podxl2 (with no evidence of a significant expansion of a luminal/apical membrane); this indicates that this cell is more likely to be an EHT pol- cell that did not sort the vesicular cargo to the luminal/apical membrane. Bar = 10 μ m. **B**, hypothetical model summarizing the evolution of HE cells involving the tuning of apicobasal polarity to lead to cells competent for giving birth to either EHT pol+ or EHT pol- cells (including the release of large vesicular macropinocytic-like vacuoles preferentially toward the luminal membrane of future EHT pol + cells). The polarity status of HE cells is proposed to evolve asynchronously throughout the entire EHT time window, leading to place-to-place ability to give birth to EHT cells (emergence of EHT pol+ and EHT pol- cells are both observed until 72 hpf, see main text).

Although technically difficult for long hours (because of important variations in the volume of the balled yolk that trigger drifting of embryos), we have been able to follow over time non-polarized HE cells and to visualize the evolution of their vesicular content, starting at the initiation of the EHT time-window (around 35 hpf, see [Figure 2](#) and [Figure 2 – figure supplement 2](#)).

Interestingly, we could follow a dividing HE cell for which the vesicular content labelled with eGFP-podxl2 appeared to partition unequally between daughter cells ([Figure 2A](#), $t = 30$ min, HE cell 1' inherits the largest macropinocytic-like vacuole (green asterisk) and emerges unambiguously as an EHT pol+ cell ($t = 360$ min, EHT cell 1'), with eGFP-podxl2 enriched at the apical/luminal membrane (surrounding the green asterisk) and virtually undetected at the basal membrane (magenta arrow)). This suggests that asymmetric inheritance of cytosolic vesicles containing apical proteins may contribute, presumably after delivery to the luminal membrane, to specify the apical membrane of EHT pol+ cells.

Altogether, these results support the idea that the HE, at the initiation of the EHT time-window, is not polarized. Subsequently, HE cells establish – or not – apical and basal membrane domains, which characterizes EHT pol+ or EHT pol- cell types, respectively (see our hypothetical model [Figure 2B](#)). In the case of EHT pol+ cells and while emergence is proceeding, apicobasal polarity is maintained (if not reinforced) until the release.

Interfering with Runx1 function alters HE maturation, the EHT progression, and the balance between EHT cell types

To provide functional support to our findings and hypotheses on apico-basal polarity control in the gradual maturation of the HE throughout the EHT time-window, we thought of interfering with the activity of the transcription factor Runx1 whose expression is sensitive to fluid sheer stress (Adamo et al. 2009).

To interfere with Runx1 function, we generated a *Tg* fish line that expresses a truncated form of Runx1 (dt-runx1; fish line *Tg(Kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)* thereafter abbreviated *Tg(dt-runx1)*) deleted from its transactivation domain and carboxy-terminus but retaining its DNA-binding domain (the Runt domain (aa 55-183, (Kataoka et al. 2000 [↗](#); Burns et al. 2002 [↗](#); Kalev-Zylinska et al. 2002 [↗](#)), see **Figure 3C** [↗](#)). Importantly, in this *Tg* fish line, dt-runx1 expression is restricted to the vascular system, hence excluding expression in the brain region that express Runx1 endogenously (see the aforementioned articles) and preventing biases owing to potential interference with neuronal functions. In addition, owing to the expression of eGFP concomitantly to dt-runx1 (the C-terminal eGFP is cleaved from dt-runx1 via a T2a site for endopeptidase, see cartoons at the top of panel C), this fish line allows for the easy selection of embryos for imaging and for phenotypic analysis. In preliminary experiments aimed at addressing the localization of dt-runx1 as well as its stability, we expressed it transiently and measured its proper targeting to the nucleus (the construct also contains a double HA (2xHA) epitope for immuno-detection, see **Figure 3C** [↗](#)).

Figure 3

Interference with Runx1 function alters emergence efficiency and induces a bias toward EHT pol+ cells.

A, *Tg(kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)* mutant embryos imaged using spinning disk confocal microscopy and analyzed in the AGM/trunk region. Z-projections of the dorsal aorta obtained from 52 - 55 hpf embryos. **Top panel**: fluorescence from the red channel is shown for the *Tg(kdrl:Gal4;UAS:RFP)* control. **Bottom panels**: fluorescence in the green channel only is shown for the mutants (eGFP, released from the dt-runx1-eGFP cleavage). The black asterisks point at emerged cells that are in close contact with the aortic floor. Green arrows: EHT pol+ cells on aortic floor; magenta arrows: EHT pol+ cells in the lateral aortic wall; blue arrows: uncharacterized emerging cells; he: hemogenic cells. **B**, quantitative analysis of the dt-runx1 mutant phenotype. The analysis was carried out on *Tg(kdrl:Gal4;UAS:RFP)* control embryos (n = 3) and on *Tg(kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)* mutant embryos (n = 7). **Left top**: hemogenic cell count (comprising hemogenic cells, labelled by "he" on panel **A**, as well as uncharacterized emerging cells, labelled by blue arrows on panel **A**). **Left center**: EHT pol+ cell count (sum of EHT pol+ cells emerging ventrally and laterally, respectively labelled by green and magenta arrows on panel **A**). **Left bottom**: EHT pol- cell count (not shown on panel **A**). **Right**: percentage of cell types in control and dt-runx1 embryos (proportions relative to total number of hemogenic and EHT undergoing cells). Statistical comparisons have been performed using two-sided unpaired Wilcoxon tests, all p-values are displayed. **C, top**: cartoons representing the zebrafish full-length runx1a amino-acid sequence and the dt-runx1 mutant deleted from the trans-activation domain and of the C-terminus (note that the construct encodes for a C-terminal fusion with eGFP that is released upon expression via a cleavable T2A peptide (introduced between the 2xHA tag and the N-terminus of eGFP, to prevent from potential steric hindrance). nls, nuclear localization signal. **Bottom**: image of an anti-HA tag immunofluorescence obtained after z-projection of the dorsal aorta of a 50 hpf *Tg(dt-runx1)* embryo. Note the localization of the 2xHA-tagged dt-runx1 protein in nuclei (some of them are pointed by red arrowheads) and of eGFP in nuclei and the cytosol of aortic cells. Bar: 25µm.

To characterize further the *Tg(dt-runx1)* fish line, we addressed the incidence of dt-Runx1 expression on HE and EHT cell populations during the EHT time-window as well as later on, at 5 days post-fertilization (5dpf) in the thymus, as a readout of hematopoiesis efficiency. We made the following observations:

Firstly, using 48 - 55 hpf *Tg(dt-runx1)* embryos and live imaging we observed, in comparison to controls, a significant accumulation of HE cells (see **Figure 3A**, top and bottom panels (he, arrowheads), see also the quantifications **Figure 3B**, top graph, median value of 17 cells for controls and 25 cells for mutants). In the mutant condition, the bulk of HE cells also include uncharacterized EHT cells (**Figure 3A**, middle panel, blue arrows); hypothetically, these may be EHT cells at an early phase of their emergence, including cells that may have evolved as EHT pol+ cells in unperturbed conditions, owing to the absence of luminal membrane invagination. We also observed the unusual tendency for the accumulation of EHT pol+ cells (**Figure 3A**, green and pink arrows), with a maximum value of 3 cells per embryo for controls that reached the unusual - never observed in control conditions -, values of 11, 9 and 8 cells for 3 mutants out of 7 (**Figure 3B**, middle graph). In addition to the increase in the proportion of EHT pol+ cells in the ventral side of the aorta (**Figure 3B**, right panel, with a value of 4% for controls and reaching 7% for mutants, consistently with the increase in the absolute count per embryo), we also observe an increase in the proportion of EHT pol+ cells in lateral sides of the dorsal aorta (**Figure 3A**, pink arrows and **Figure 3B**, right graph, with a value of 4% for controls and reaching 9% for mutants). Finally, we did not observe any significant modification in the number of EHT pol- cells (same median values for both controls and mutants, the values oscillating between 0 and 1 cell per embryo, with only one example of an embryo exhibiting 2 apparent EHT pol- cells (**Figure 3B**, bottom graph)).

Secondly, using 48 - 55 hpf *Tg(dt-runx1)* embryos and controls *Tg(Kdrl:eGFP)* we confirm, by whole mount in situ hybridization performed with an RNAscope probe specific for *myb* (**Figure 3 – figure supplement 1**), that the cells that accumulate in the aortic floor of *dt-Runx1* mutants are indeed of hematopoietic nature (see the cells with RNAscope spots in magenta or marked by blue arrows, **Figure 3 – figure supplement 1A**, zoomed areas). Unfortunately, chemical fixation impaired the maintenance of EHT pol⁺ and EHT pol⁻ specific cell shape, preventing us from classifying emerging cell populations. However, cell counts after segmentation using the Imaris software confirmed, based on cell localization, the specific increase of hemogenic/EHT cells in aortic floors of mutants (median values of 34 and 16.5 for mutants and controls, respectively, **Figure 3 – figure supplement 1B** (right panel) while no difference was measured for endothelial roof cells between mutants and controls (left panel). Quantification of *myb* positive cells confirmed the increase of hemogenic cells in the aortic floors of mutants in comparison to controls, albeit with slightly smaller numbers (median values of 20.5 and 8.5 in mutants and controls, respectively (**Figure 3 – figure supplement 1C**, middle panel)), most probably because *myb* becomes detectable only when the EHT process has initiated. In comparison to floor cells, very few roof cells express *myb* (median values of 2.0 and 1.5 in mutants and controls, respectively (**Figure 3 – figure supplement 1C**, left panel)) consistently with the idea that the aortic roof is not hemogenic (Zhao et al. 2022 [\[1\]](#)). Importantly, the increase of pre-hematopoietic cells in the aortic floor in the mutants correlates with a decrease in hematopoietic cells lodging in the sub-aortic region, reinforcing the idea of impairment in EHT cell emergence (**Figure 3 – figure supplement 1C**, right panel). Finally, quantification of the number of *myb* RNAscope spots per segmented hemogenic cells highlights the important increase in expression by hemogenic cells in the floor of *dt-Runx1* mutants, in comparison to controls (with median values of 4 and 0, respectively (**Figure 3 – figure supplement 1D**, right panel). In comparison, virtually none of the roof cells contain more than 2 spots (**Figure 3 – figure supplement 1D**, with 2/162 and 1/220 cells for the mutants and controls, respectively, with median values of 0 for both conditions). Altogether, these last results confirm that pre-hematopoietic cells accumulate in the aortic floor of *dt-Runx1* mutants; the very substantial increase in the detection of *myb* mRNAs suggests an increase in the residential time in the aortic floor, which reinforces the idea that these cells are impaired in their release from the aortic floor.

Thirdly, between 30 and 55 hpf, using live confocal microscopy and contrarily to what we observed with our fish lines expressing either eGFP-podxl2 (**Figure 2** [\[2\]](#) and **Figure 2 – figure supplement 1 and 2**) or mKate2-podxl2 (**Figure 3 – figure supplement 2A**) in which HE cells appear to be non-polarized, embryos obtained from outcrossing *Tg(dt-runx1)* and *Tg(Kdrl:mKate2-podxl2)* fishes appear to contain polarized HE cells, based on enrichment of the polarity marker at luminal membranes (**Figure 3 - figure supplement 2B**).

Fourthly and occasionally, in live embryos, we observe - for EHT cells exhibiting invagination of the luminal membrane -, scattered cytosolic and sub-plasmalemmal pools of Podxl2-containing membranes and, consistently, the apparent decrease of Podxl2 enrichment at the apical/luminal membrane (see **Figure 3** [\[3\]](#) - **figure supplement 2C** and compare with **Figure 1C** [\[4\]](#)). In addition and occasionally as well, we observe the reversion of apparent EHT pol⁺ cells into apparent EHT pol⁻ cells (data not shown). These last two observations suggest that perturbing the control of apicobasal polarity in the HE, in the context of the *dt-Runx1* mutant, alters emergence morphodynamics, hypothetically and possibly more specifically in the case of EHT pol⁻ cells whose biomechanical features appear to require turning down the establishment of apical and basal membrane domains.

Fifthly, hematopoiesis is affected far downstream of emergence, as attested by the significant increase, at 5 dpf, in the number of hematopoietic cells in the thymus of *dt-Runx1* mutants, in comparison to control siblings (see **Figure 3 - figure supplement 3A** for images after cell

segmentation with the Imaris software accompanied by **Figure 3** [video 1](#) and **Figure 3 - figure supplement 3B** for quantifications showing significant increase in thymus volume (left panel) owed to the increase in cell number (middle panel) rather than in cell volume (right panel)).

Altogether, the results obtained upon expression of *dt-runx1* show impairment of hematopoiesis and suggest that, for both EHT cell types, the progression throughout EHT is perturbed, and so until the release. The accumulation of morphologically characterized EHT pol+ like cells that we observe may result from the sustained apicobasal polarity of the HE at early and later time points. Conversely, characterized EHT pol- cells did not accumulate, although we observed cells of uncharacterized morphology that could correspond to EHT pol- cells that did not evolve properly. In addition, our results suggest that precursors of HE cells are polarized, as is expected to be the case for non-hemogenic aortic cells, and that Runx1 is involved in controlling the molecular events that are tuning apico-basal polarity, starting at the initiation of the EHT time-window.

Interfering with Runx1 activity unravels its function in the control of *Pard3ba* expression and highlights heterogeneous spatial distribution of *Pard3ba* mRNAs along the aortic axis

We then explored the potential involvement of proteins of the *Pard3* family during the EHT. We anticipated that interfering directly with *Pard3* proteins would hamper the development of the aorta (in addition to other functions of polarized tissues that are essential for embryonic development) and rather searched for correlative evidence for the differential expression of *Pard3* isoforms in the hemogenic endothelium, EHT cells and the aortic endothelium, both in wild type and *dt-Runx1* expressing embryos. To achieve this task, we analyzed the expression of mRNAs encoding for the four *Pard3* isoforms expressed in the zebrafish (**Figure 4A** [video 1](#)) by combining qRT-PCR on FACS-sorted endothelial cells and whole-mount *in situ* hybridization at the cellular resolution, using RNAscope. All four *Pard3* isoforms are composed of 3 PDZ domains and 2 conserved regions upstream and downstream of these domains (CR regions, **Figure 4A** [video 1](#)).

Figure 4.

Pard3ba expression is highly sensitive to interference with Runx1 activity.

A, Pard3aa, ab, ba and bb zebrafish isoforms, with their accession numbers. The cartoon represents the common structures of the 4 isoforms, with the 3 sequential PDZ domains (PDZ1-3), and the CR1 and CR3 conserved regions involved in oligomerization and atypical protein kinase C (aPKC) binding, respectively. The magenta arrows show the positioning of qRT-PCR primers, located in the PDZ1 domain (see **Methods** for primer sequences). **B**, qRT-PCR analysis of genes expression levels of (**left**) the hematopoietic markers *myb* and *runx1*, (**right**) the 4 isoforms of Pard3 proteins (Pard3aa, ab, ba and bb) in cell populations isolated from FACS-sorted trunk vascular cells of 48 - 50 hpf control and mutant embryos (see **Figure 4** - **figure supplement 1**, **Figure 4** - **figure supplement 2** and **Methods** for cell isolation procedures). Graphs show the measured mean fold changes relative to the expression of *ef1a* and to the expression in RFP+ control cells. Statistical tests: two sided unpaired two samples Wilcoxon test, all p-values are displayed. Analysis was carried out on n = 3 for control and mutant conditions. **C**, representatives images (Imaris 3D-rendering) of RNAscope *in situ* hybridizations for Pard3ba in 48 - 50hpf *Tg(Kdrl:eGFP)* control embryos and *Tg(Kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)* mutant embryos. The aorta is outlined with the white dashed lines. The RNAscope signal was segmented into spots and classified based on its localization: in aortic endothelial and hemogenic cells (magenta spots) or in extra-aortic tissues (grey spots, essentially residing in the sub-aortic space). Magenta dashed boxes delineate examples of mRNA Enriched Regions (MERs) in control and mutant embryos. Scale bars: 10µm. **D**, Pard3ba spots count per aortic segment. **E**, Pard3ba spots count per hemogenic/EHT cells. **F**, cumulative Pard3ba spots count relative to hemogenic/EHT cell distance, averaged for each aortic segment, in control embryos (grey) and *dt-runx1* mutant embryos (orange). **G**, HE/EHT cell count relative to number of spots, all expressed per aortic segment (**left**), 50% Pard3ba spots attraction distance (in µm) (**middle**) or 75% Pard3ba spots attraction distance (in µm) (**right**). 50%/75% spots attraction distance corresponds to the distance from hemogenic/EHT cells within which 50%/75% of Pard3ba spots are located. Grey dash lines correspond to regression lines and light grey background correspond to the confidence interval (95%) of the regression model. Linear correlation coefficient (R, Pearson correlation) is displayed for each condition. For **D** and **E**, statistical tests: two sided unpaired two samples Wilcoxon test, all p-values are displayed. Analysis was carried out on n = 5 control embryos and n = 6 mutant embryos, 2 aortic segments per embryos. For **G**, p-values were calculated using a t-test.

We first sought to measure expression of these different isoforms in the vascular system of 48 hpf embryos and, more restrictively, in the trunk region. To do so, we dissociated and FACS- sorted cells of either whole embryos, or dissected trunks, using the *Tg(Kdrl:Gal4; UAS:RFP)* fish line in which soluble RFP is expressed in endothelial tissues (**Figure 4 – figure supplement 1A**). RFP+ vascular cells were found to be slightly enriched in the trunk region, in comparison to whole embryos (**Figure 4 – figure supplement 1A**, see the table with 2.10 versus 0.95 % of RFP+ cells in the trunk and the whole embryo, respectively). Using qRT-PCR on the FACS-sorted cells, we found that the hematopoietic markers *myb* and *runx1* were heavily detected in RFP+ cell extracts in comparison to RFP- cells (**Figure 4 – figure supplement 1B**); this suggests that part of the isolated RFP+ cells contain hemogenic and hematopoietic cells (at 48 hpf, it is expected that some newly born hematopoietic cells are retained in the sub-aortic space). When we measured the expression levels of the four Pard3 isoforms, we observed, in comparison to control RFP-cells, the enrichment of Pard3aa and Pard3ba in extracts from RFP+ cells isolated from whole embryos, and the opposite for Pard3ab and Pard3bb (**Figure 4 – figure supplement 1C**). These differences were conserved with extracts from RFP+ cells isolated from trunk regions, except for Pard3ba for which important variations in mRNA quantities were obtained, blurring conclusions (we shall come back to this variability beneath). These results suggest that, at least based on the vascular system of whole embryos, Pard3aa and Pard3ba exert functions in endothelial cells, with the possibility of having more specific functions in aortic and HE cells. To discriminate between cell types expressing Pard3aa and Pard3ba isoforms (and in particular vascular cell types), we used RNAscope on whole mount *Tg(Kdrl:eGFP)* embryos, at 48 hpf. Confocal images focused on trunk regions show that Pard3aa is strongly expressed in cells of the spinal cord (potentially neuronal cells and/or radial

glia), as well as in the notochord (**Figure 4 – figure supplement 2A**). Interestingly, Pard3ba mRNAs are expressed in more restricted areas, in majority in the spinal cord, albeit to a much lesser extent than for Pard3aa as well as in the region of the pronephric duct and the gut epithelium (**Figure 4 – figure supplement 2B**). By curiosity and because it appears to be the most highly expressed in the whole embryo (and hence may be ubiquitously expressed, including in the trunk region and in accordance to the Daniocell resource at <https://daniocell.nichd.nih.gov/> , Farrell et al. 2018 ; Sur et al. 2023 , we also investigated on the expression of Pard3ab mRNAs (**Figure 4 – figure supplement 2C**). We confirmed its relatively ubiquitous expression in comparison to Pard3aa and Pard3ba, with a massive expression in the spinal cord. To visualize more precisely expression levels in the vascular system in the trunk (more specifically in the aorta and the underlying vein), we zoomed in this region (**Figure 4 – figure supplement 2D**) and observed a homogeneous localization of Pard3aa and Pard3ab mRNAs as well as, strikingly in the case of Pard3ba, a quite heterogeneous localization of mRNAs unequally concentrated along the aortic axis (**Figure 4 – figure supplement 2D**, middle panel and see also **Figure 4C** , top panel; these will be referred thereon as Pard3ba mRNA enriched regions (MERs)). Of notice, in the whole aorta, we observed more RNAscope signals for Pard3ab in comparison to Pard3aa and Pard3ba, including in some HE/EHT cells (**Figure 4 – figure supplement 2D**, with positive HE/EHT cells indicated by blue arrows). This indicates that the Pard3ab isoform, consistently with the absence of variation upon dt-Runx1 expression (see beneath), is constitutively expressed in some HE/EHT cells (this may be also the case for the Pard3aa isoform but this should be taken with caution since its mRNAs are more in the limit of detection). We attempted to detect Pard3 isoforms expressed in the vascular system, in the trunk region of the embryo, with available antibodies raised against other species and failed, possibly because of inter- species recognition limitation.

We then analyzed Pard3 mRNA levels in the context of dt-Runx1 expression. As we did for *Tg(Kdrl:Gal4;UAS:RFP)* embryos, we isolated by FACS GFP⁺ and double positive GFP⁺/RFP⁺ cells dissociated from dissected trunks of *Tg(dt-Runx1)* 48 hpf embryos. For controls, we also isolated RFP⁺ cells from dissected *Tg(Kdrl:Gal4;UAS:RFP)* embryos (see **Methods** and **Figure 4 – figure supplement 3A**). qRT-PCR on isolated eGFP⁺ cells that express dt-Runx1 revealed an increase of approximately 50% of the *myb* and *runx1* mRNAs, in comparison to RFP⁺ control cells (**Figure 4B** , left panel), which is consistent with the accumulation of hemogenic/EHT cells that we observe in **Figure 3**  and in **Figure 3 – figure supplement 1**. Interestingly, we obtained very little to no variation in the expression levels of Pard3aa, ab and bb mRNAs in dt-Runx1 expressing eGFP⁺ cells in comparison to control RFP⁺ cells but, again, a high variability in the case of Pard3ba mRNAs (**Figure 4B** , right panel). Using RNAscope, the non-homogeneous localization of Pard3ba mRNAs observed in controls, along the aortic axis, was not only conserved but became even more obvious to the eye for dt-Runx1 mutants, owing to the increase in the density of RNAscope spots (**Figure 4C** , compare top and bottom panels). Confocal images on several controls and dt-Runx1 mutants highlight the heterogeneity in Pard3ba mRNA detection as attested by the variability in the localization and number of RNAscope spots (compare in particular the 5 mutants shown in **Figure 4 – figure supplement 3B**, right panels). Quantification of RNAscope spots per aortic segment and per hemogenic/EHT cell using Imaris revealed an increase of almost 300% for the former when comparing controls and mutants (median values of 65.00 and 167.50 for controls and dt-Runx1 mutants, respectively, **Figure 4D** ) and that virtually no Pard3ba mRNA was detected in the latter (median values of 0.00 spots per hemogenic/EHT cell for both controls and dt-Runx1 mutants, **Figure 4E** ). A precise representation showing the hemogenic/EHT cell count with RNAscope spots ranging from 0 to 4 reveals, for mutants in comparison to controls, a significant increase in the number of hemogenic/EHT cells not expressing Pard3ba (**Figure 4 – figure supplement 3C**, first panel from the left), as well as a slight tendency for an increase in the number of cells with 1 to 4 Pard3ba spots (**Figure 4 – figure supplement 3C**, second to fifth panels), with more significance for the population with 2 spots (middle panel). Overall, in either control or mutant conditions, we detected the presence of RNAscope spots in very few

hemogenic/EHT cells (15/115 for controls and 21/224 for dt-Runx1 mutants, with a maximum for 2 spots for controls and 4 spots for dt-Runx1 mutants, see [Figure 4E](#) and [Figure 4 – figure supplement 3C](#)).

Again, we obtained a large variability in the number of Pard3ba RNAscope spots per aortic segment (the number of spots per aortic segment ranged from 10 to 94 for controls and from 53 to 366 for the mutants, see [Figure 4D](#)); this mirrors what was obtained by qRT-PCR, particularly for dt-Runx1 mutants ([Figure 4B](#), right panel). This is not the case for Pard3aa and Pard3ab (see RNAscope images and quantifications [Figure 4 – figure supplement 4](#)), which is also consistent with qRT-PCR results ([Figure 4B](#), right panel). Importantly, expression of dt-Runx1 neither influenced the expression of Pard3aa, nor of Pard3ab ([Figure 4 – figure supplement 4](#)).

Finally and importantly, in wild-type embryos, the Pard3ba mRNAs are preferentially localized in the direct vicinity of hemogenic/EHT cells ([Figure 4F, G](#)). Indeed, more than 50% of Pard3ba spots are localized within 5 to 10 μm of hemogenic/EHT cells ([Figure 4G](#) middle panel), and more than 75% of spots are localized within a 5 to 20 μm distance ([Figure 4G](#) right panel). Interestingly, the number of hemogenic/EHT cells does not correlate with the absolute number of spots ([Figure 4F](#), left panel, $R = 0.34$), but rather it strongly correlates with the spatial densification of spots around hemogenic/EHT cells: the shorter the distance of the majority of spots are of hemogenic/EHT cells ([Figure 4F](#), middle and right panel, representing 50% and 75% of spots respectively), the higher the hemogenic/EHT cell count ($R = -0.84$ and $R = -0.83$ respectively). This shows that AECs expressing Pard3ba mRNAs are mostly in direct contact with cells committed to undergo EHT and suggests a signalling crosstalk occurring between them. This densification of spots in MERs is reinforced upon expression of dt-Runx1 ([Figure 4F](#), left panel): we found that virtually all Pard3ba spots are localized within a 10-15 μm distance from hemogenic/EHT cells.

Altogether, these results unveil the Runx1-dependent control of the Pard3ba isoform specifically, during the EHT. In the aorta, AECs appear to be the main cell type that expresses this Pard3 isoform and, unexpectedly, expression in these cells is sensitive to interference with our dt-Runx1 mutant. One plausible explanation is that AECs respond to Runx1-mediated signalling taking place in HE/EHT cells (see [Discussion](#) for other possibilities and further details). This suggests that a specific signalling axis is taking place between these different cell types that would control the expression of Pard3ba mRNAs and, consequently, downstream apico-basal polarity associated features carried out by the Pard3ba protein isoform (example: the recycling mode of junctional complexes between EHT pol⁺ cells and endothelial neighbours, in comparison to EHT pol⁻ cells, see beneath and [Discussion](#)). This potential intercellular signalling is supported by the proximity of Pard3ba mRNAs expressed in AECs to HE/EHT cells in MERs. These results also strengthen the importance of the interplay between HE/EHT cells and their direct endothelial neighbours (this interplay has been highlighted in [Lancino et al. 2018](#); and see also beneath).

Junctional recycling between EHT cell types and endothelial neighbours is differentially controlled

To investigate the functional links between apicobasal polarity and the peculiarities of EHT pol⁺ and EHT pol⁻ emergence processes, we sought to follow the dynamics of Junctional Adhesion Molecules (JAMs) that belong to tight junction complexes ([Garrido-Urbani, Bradfield, and Imhof 2014](#)). During apicobasal polarity establishment in epithelial and endothelial tissues, these molecules recruit the Pard3/aPKC complex, via a PDZ-binding peptide located in their extreme carboxy-terminus ([Figure 5A](#), left cartoons and [Itoh et al. 2001](#); [Ebneth et al. 2001](#); [2018](#)). In addition, JAMs are expressed in the vascular system and, in relation to this work, JAM3b is expressed in the aorta of the zebrafish embryo and promotes endothelial specification and hematopoietic development ([Kobayashi et al. 2020](#)).

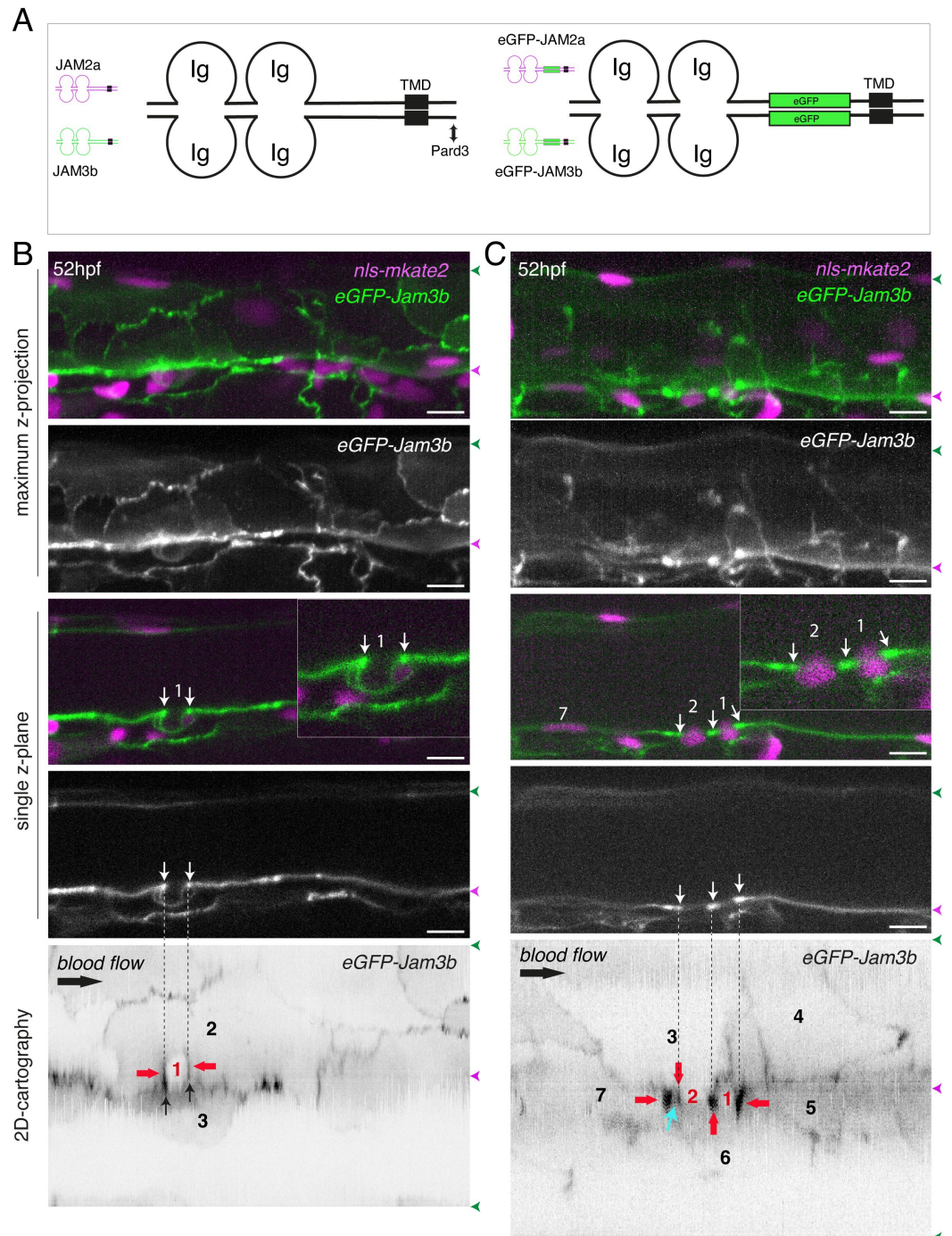


Figure 5

eGFP-Jam3b localization is reinforced at antero-posterior sites of the endothelial/EHT interface and at tri-cellular junctions.

A, cartoons representing homodimers of full-length JAMs with the C-terminal cytosolic part interacting with Pard3 (JAMs interact with the first PDZ domain of Pard3) as well as the constructs generated in this study and containing eGFP inserted between the Immunoglobulin-like (Ig) domains and the trans-membrane region (TMD). The constructs were obtained for zebrafish JAM2a and JAM3b. **B**, **C**, 52 hpf *Tg(kdrl:eGFP-Jam3b; kdrl:nls-mKate2)* embryos were imaged in the trunk region (AGM) using spinning disc confocal microscopy. The panels are either maximum z-

projections (top two) or single plane z-sections (bottom two, focusing on the aortic floor) of aortic segments, with either the merged nls-mKate2 and eGFP-Jam3b fluorescence signals (magenta and green) or the eGFP-Jam3b signal only (black and white images). Bottom of the Figure: 2D-cartographies obtained after deploying aortic cylinders and showing the eGFP-Jam3b signals only. The white/black vertical dashed lines show the correspondence of the antero-posterior junctional reinforcements in the single z-sections (white arrows) on the respective 2D-cartographies resulting from the deployment of the aortic walls (obtained from the z-stacks). **B**, example of an EHT pol+ cell (cell 1, white arrows point at reinforcement of signal at antero-posterior junctions). On the 2D cartography, cell 1 (red) is contacting endothelial cells 2 and 3; note the reinforcement of eGFP-Jam3b signals along antero-posterior membrane interfaces perpendicular to blood flow (red arrows) as well as at the two tri-cellular junctions visible between cells 1, 2 and 3 (black arrows). **C**, example of two EHT pol- cells (cells 1 and 2, white arrows point at reinforcement of signal at antero-posterior junctions). On the 2D cartography, cells 1 and 2 (red) are contacting endothelial cells 3, 4, 6 and 3, 6 respectively; note the reinforcement of eGFP-Jam3b signals along antero-posterior membrane interfaces perpendicular to blood flow (red arrows) and endothelial cell 6 that has intercalated between endothelial cell 7 and EHT pol- cell 2 (blue arrow). In right margins, magenta and green arrowheads designate the aortic floor and roof, respectively. Bars = 10 μ m.

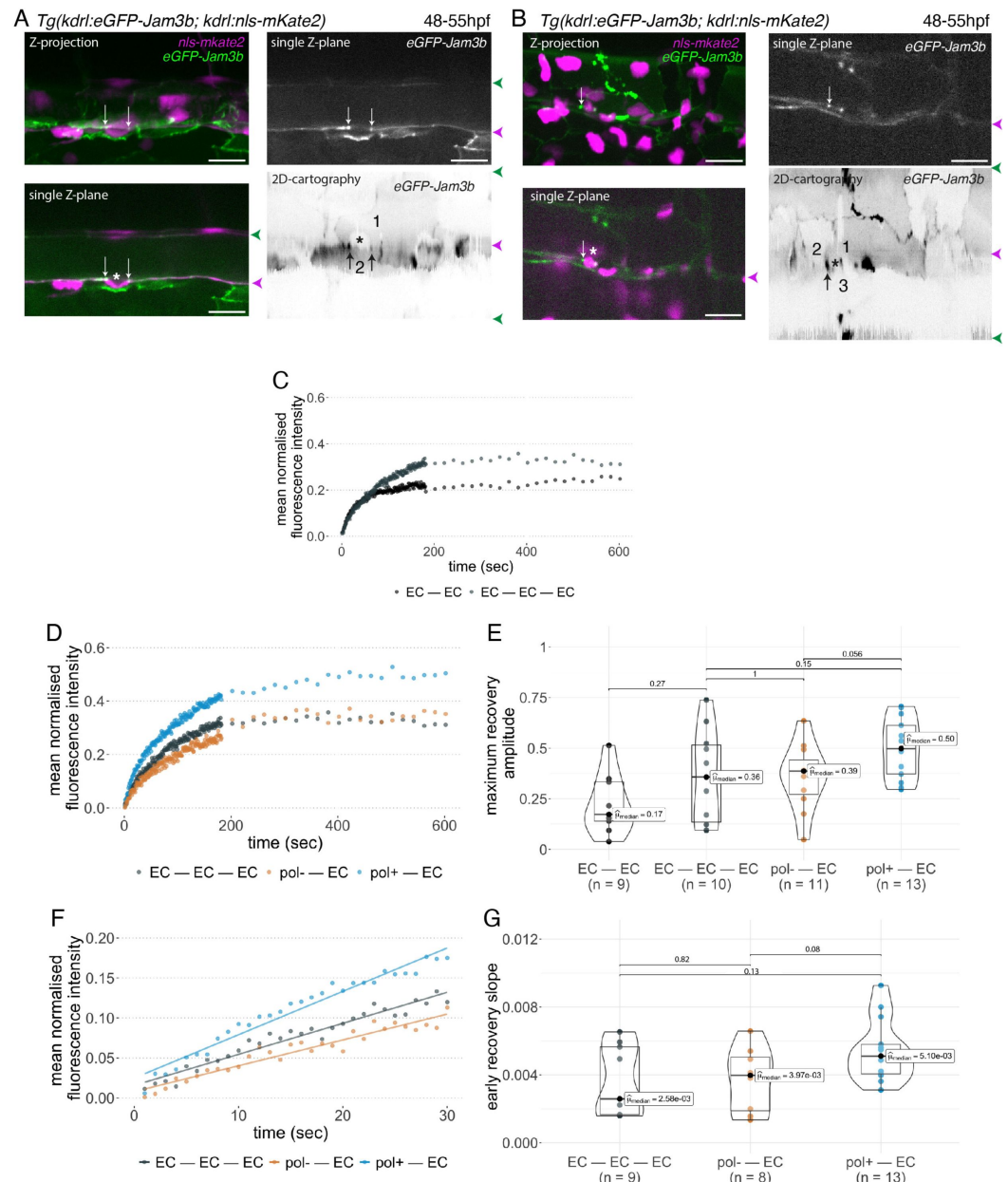


Figure 6

Junctional recycling at tri-cellular contacts is differentially controlled between the two EHT types.

48–55 hpf *Tg(kdrl:eGFP-Jam3b; kdrl:nls-mKate2)* embryos were imaged using spinning disc confocal microscopy and illuminated for Fluorescence Recovery After Photobleaching (FRAP) in the trunk region (AGM, Aorta Gonad Mesonephros). **A, B**, panels are either maximum z-projections (top left) or single plane z-sections (bottom left and top right, focusing on the aortic floor) of aortic segments, with either the merged nls-mKate2 and eGFP-Jam3b fluorescence signals (magenta and green) or the eGFP-Jam3b signal only (black and white images). White arrows point at reinforcement of signal at antero-posterior junctional pools of an EHT pol+ cell (**A**) or of an EHT pol- cell (**B**), both marked by asterisks. Bottom right: 2D-cartographies obtained after deploying aortic cylinders and showing the eGFP-Jam3b signals only. Black arrows point at antero-posterior junctional pools, in particular at tri-junctional regions that exhibit increase in signal density (well visible in **A**, black arrows). 2 and 3 endothelial cells are contacting the EHT pol+ cell (**A**) and the EHT pol- cell (**B**), respectively. In right margins, magenta and green arrowheads designate the aortic floor and roof, respectively. Bars = 20µm. **C – G**, FRAP analyses. EGFP-Jam3b junctional pools corresponding to the brightest spots inside junctional regions of interest were bleached for FRAP measurements (these high intensity pools were localized at the level of bi- and tri- junctions for endothelial cells (EC) and in tri-junctional regions for EHT pol+ and EHT pol- cells; all these junctional pools were systematically visualized by deploying each aortic segment before bleaching as shown in the 2D-cartographies in **A** and **B** as well as in **Figure 6 – figure supplement 1**, see also **Methods**). FRAP analysis concerned 3 types of junctional interfaces: between endothelial cells (EC – EC, black and grey), EHT pol- and endothelial cells (pol- – EC, brown), EHT pol+ and endothelial cells (pol+ – EC, blue). **C, D**, evolution of mean fluorescence intensity for each type of junctional interface over time (10 min), after photobleaching (t = 0s). **E**, median maximum amplitude of recovery of all determinations and for each type of junctional interface (maximum of simple exponential fitted curves). **F, G**, early fluorescence recovery. Early evolution (over the first 30s) of the mean fluorescence intensity for each type of junctional interface over time after photobleaching (t = 0s). **F**, the fitted lines correspond to linear regressions of mean fluorescence intensities. **G**, median values of fluorescence recovery slopes (linear regressions) of all determinations and for each type of junctional interface. **E, G**, the number of biological replicate (n) is stated on the plots. Statistical tests: two sided unpaired two samples Wilcoxon test.

We envisaged a scenario whereby EHT pol+ cells, whose longitudinal interface with endothelial neighbours shrinks along the X axis of the X, Y 2D-plane, may have less mobile junctional pools than EHT pol- cells whose entire junctional interface moves along the X, Y, Z 3D-axes (**Figure 5 – figure supplement 1**). In the case of EHT pol- cells, the consumption of the junctional interface with adjoining endothelial cells appears to result from the converging migration of endothelial neighbours crawling over the luminal membrane, based on interpretation of our time-lapse sequences (**Figure 1- video 3**). In this context, we favoured the analysis of junctional pools localized at antero-posterior sites of emerging cells as we have shown that they are enriched with tight junction components (Lancino et al., 2018). In addition and in the case of EHT pol+ cells, it is conceivable that these adhesion pools – spatially restricted owing to apicobasal polarity –, contribute to anchoring the emerging cell in the 2D- plane (**Figure 5 – figure supplement 1**, top panel).

To achieve our goal, we designed 2 constructs in which eGFP is introduced in the extracellular domains of the two JAM2a and JAM3b molecules (**Figure 5A**, right cartoons). To investigate their localization and proper targeting at junctional interfaces, these constructs were expressed transiently and ubiquitously, using the *Hsp70* heat shock promoter. We observed that the 2 fusion proteins are efficiently targeted at cellular contacts (**Figure 5 – figure supplement 2B, C and F**, orange and white arrows) as well as at the apical side of polarized epithelia such as for example cells of the ependymal canal (**Figure 5 – figure supplement 2A, E**, white arrows). They were also observed at protrusions of epithelial cells and at T-tubules of muscle fibres (for JAM2a and JAM3b,

respectively, see **Figure 5 - figure supplement 2D, G**). We then established *Tg* fish lines expressing eGFP-Jam2a and eGFP-Jam3b under the control of the vascular *kdr1* promoter (*Tg(kdr1:eGFP-Jam2a)* and *Tg(kdr1:eGFP-Jam3b)*). Using these fish lines and spinning disk confocal microscopy, we observed a remarkable efficiency of targeting to intercellular junctions, for both proteins (data not shown for eGFP-Jam2a; for eGFP-Jam3b, see **Figure 5B** and **C**; in the maximum z-projections, the green signal is enriched at intercellular junctions; in single z-plane images, an EHT pol⁺ cell and 2 EHT pol⁻ cells are visible in the left and right panels, respectively). Deployment of the aortic wall into 2D cartographies using an algorithm described previously (Lancino et al. 2018) allows to point precisely at junctional pools established at the interface between endothelial and EHT cells. This emphasizes on the enrichment of eGFP-Jam3b at antero-posterior poles of EHT pol⁺ and EHT pol⁻ cells (**Figure 5B, C** bottom panels, EHT and aortic endothelial cells (red and black numbers, respectively) are visualized as if the eye is located inside the aortic lumen; red arrows point at junctional reinforcements located at antero-posterior poles of EHT cells, black arrows in the left panel point at tri-cellular junctions and the blue arrow in the right panel points at intercalation of endothelial cells 3 and 6 between the EHT pol⁻ cell (cell 2) and the endothelial neighbour 7).

Using double transgenic *Tg(kdr1:eGFP-Jam3b; kdr1:nls-mKate2)* embryos at 48 - 55 hpf, we addressed the recycling capacity of junctional pools using Fluorescence Recovery After Photobleaching (FRAP; for more details, see **Methods**). The labelling of nuclei with nls- mKate2 allowed to point at EHT pol⁺ and EHT pol⁻ cells unambiguously because eGFP-Jam3b is essentially targeted to junctional interfaces and not labelling luminal/abluminal membranes (**Figure 6A, B**), except in few cases (**Figure 5B**, for an EHT pol⁺ cell); additionally, nuclei of EHT pol⁺ cells have a crescent shape (see **Figure 6A**).

First and to set up our protocol, we spotted bi- and tri-junctional contacts between endothelial cells (**Figure 6 - figure supplement 1**). Recycling parameters (fluorescence intensity recovery with time and maximum recovery amplitude (that addresses the mobile pool), **Figure 6C**) showed that bi-junctional contacts are less mobile than tri-junctions, with a higher dispersion of maximum recovery amplitude values for the latter (**Figure 6E**); this introduced a clear limitation for statistical significance of the results, although clear tendencies were observed for mean fluorescence intensity recovery (**Figure 6C**) and median values for maximum recovery amplitude, when recycling reached its equilibrium (**Figure 6E**). We then focused at antero-posterior sites of EHT pol⁺ and EHT pol⁻ cells and more specifically in the region of tricellular junctions (shared by 1 EHT cell and 2 endothelial neighbours (**Figure 6A, B**, 2D-cartographies, black arrows) and for more examples in 2D-cartographies, see **Figure 6 - figure supplement 1**, black arrows) that are clearly the most enriched with eGFP-Jam3b. In each experiment, eGFP-Jam3b pools at tricellular junctions between endothelial cells (EC-EC-EC) were also spotted for a comparative analysis. Measurements of fluorescence recovery intensities revealed a tendency of increase in the mobile fraction of eGFP-Jam3b at EHT pol⁺ - EC versus EHT pol⁻ - EC junctional interfaces (see **Figure 6D** for fluorescence recovery curves and **Figure 6E** for median values (50% and 39% maximum recovery amplitudes after 10 min, respectively)). Although smaller, differences in recovery were also measured between EHT pol⁺ - EC and EHT pol⁻ - EC junctional interfaces when focusing on the earliest time points (the first 30 seconds, **Figure 6F**), with the median value of early recovery slopes for the EHT pol⁺ - EC versus EHT pol⁻ - EC junctional interfaces increased by 128% (**Figure 6G**). Altogether and unexpectedly regarding our initial scenario, these results indicate that tri- junctional pools localized at antero-posterior poles of EHT cells and enriched with eGFP-Jam3b molecules are significantly more dynamic for EHT pol⁺ cells in comparison to EHT pol⁻ cells. Since EHT pol⁺ cells - by virtue of apicobasal polarity establishment possibly assemble an apical endosome -, this should favour a faster recycling of eGFP-Jam molecules (see **Discussion** for more details).

ArhGEF11/PDZ-Rho GEF plays essential functions during EHT progression

Junctional maintenance and recycling are dependent on intracellular membrane trafficking, supported by sub-cortical actin remodelling and actomyosin contractility, which are controlled mainly by GTPases of the Rho family (Ridley 2006 [↗](#); Olayioye, Noll, and Hausser 2019 [↗](#)). Owing to the significance of apicobasal polarity control on EHT features, as suggested by our work, we investigated which proteins may be essential for actin/actomyosin regulation and focused on regulators of Rho GTPases, in particular Rho GEFs that catalyse exchange of GDP for GTP to positively regulate their activity (Rossman, Der, and Sondek 2005 [↗](#)). As for Pard3 proteins, several of these Rho GEFs contain one or several PDZ domain(s) that target most proteins to complexes acting at the apical side therefore interlinking actin/actomyosin regulation with cell polarity (Mack and Georgiou 2014 [↗](#); Ebnet and Gerke 2022 [↗](#)). We focused on 9 PDZ-domain containing Rho GEFs, all encoded by different genes in the zebrafish (**Figure 7-figure supplement 1**): ArhGEF11/PDZ-RhoGEF (thereafter shortened as ArhGEF11), ArhGEF12a, ArhGEF12b, PRex1, PRex2, Tiam1a, Tiam1b, Tiam2a, Tiam2b. We first investigated their expression by Whole mount In Situ Hybridization (WISH) and found that all 9 mRNAs are detected in the aorta, and for the vast majority at 30-32 and 48-50 hpf (**Figure 7-figure supplement 2**). Then, using qRT-PCR (**Figure 7-figure supplement 1B**), we measured and compared their expression levels in the trunk region (at 35 and 48 hpf), for dt-runx1 expressing embryos and controls. We found that, in comparison to controls, ArhGEF11, ArhGEF12b, Tiam1b and Tiam2a are significantly reduced upon dt-runx1 expression at 48 hpf, hence being positively controlled in the wild type condition, when the emergence of EHT cells is peaking. This is consistent with a functional link between these Rho-GEFs, actomyosin and the control of junctional dynamics during EHT progression. We finally decided to focus on ArhGEF11 for the following reasons: (i) in comparison to Tiams that act on Rac1, ArhGEF11 and ArhGEF12 (which are close relatives and can form heterodimers (Chikumi et al. 2004 [↗](#))), are mostly acting on RhoA which is controlling apical constriction via the RhoA-Myosin II signalling axis; as we have shown previously (Lancino et al. 2018 [↗](#)), EHT progression requires the constriction of circumferential actomyosin; (ii) ArhGEF11 was shown to regulate the integrity of epithelial junctions by interacting directly with the scaffolding protein ZO-1, hence connecting inter-cellular adhesion with RhoA-Myosin II (Itoh et al. 2012 [↗](#)); (iii) ArhGEF11 mediated apical constriction is essential during tissue morphogenesis such as, for example, the neural tube formation in which epithelial cells, submitted to mediolateral contractile forces, constrict at their apical side thus triggering inward bending of the neural plate which leads to the formation of the neural tube (Nishimura, Honda, and Takeichi 2012 [↗](#)). The EHT may share features with this process, i.e the anisotropic distribution of contractile forces controlling the plane of emergence; (iv) mammalian ArhGEF11 exhibits alternative splicing in its C-terminal region that controls tight junction stability via the regulation of RhoA activity (Lee et al. 2018 [↗](#)) as well as cell migration and invasion (Itoh et al. 2017 [↗](#)).

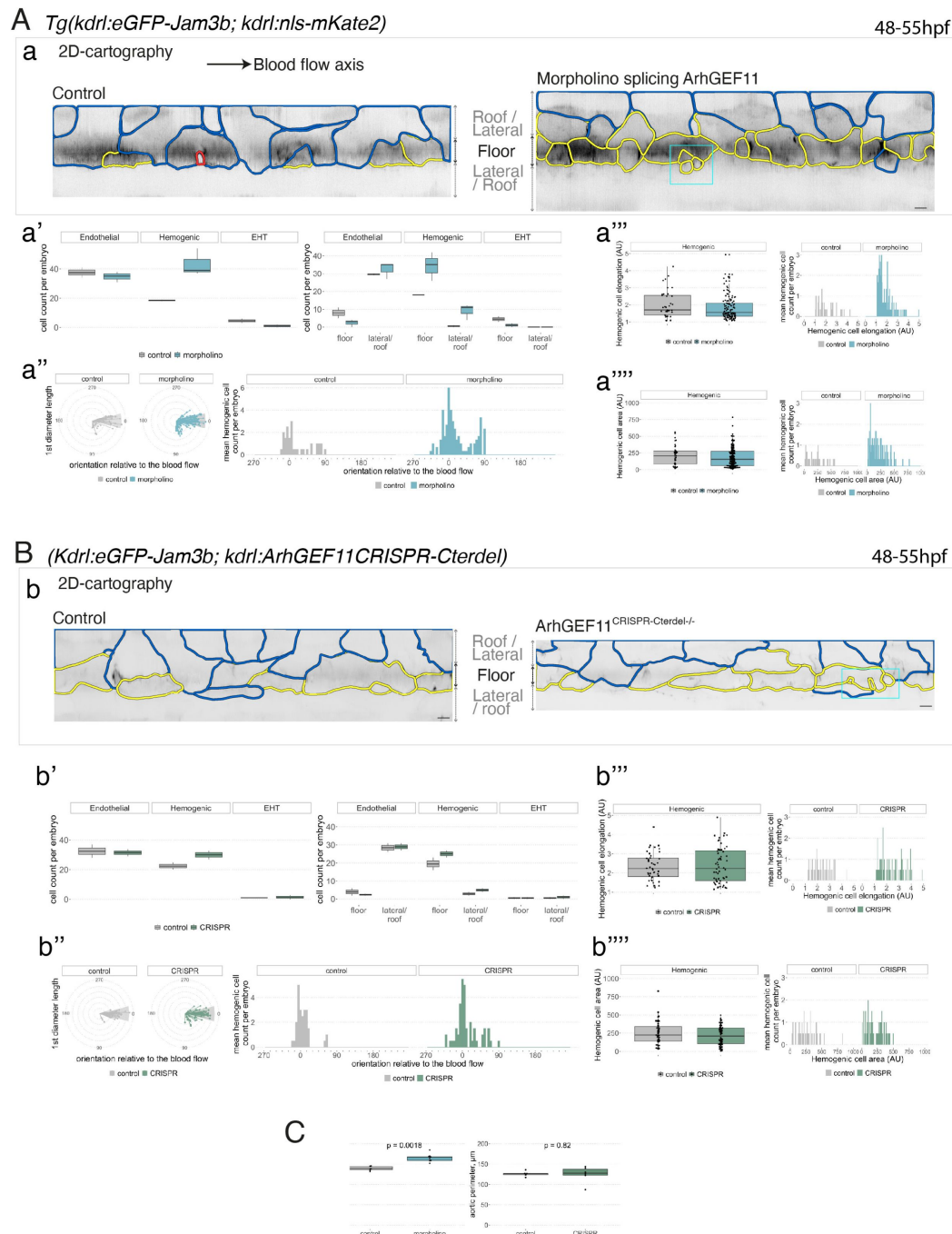


Figure 7

Interfering with ArhGEF11/PDZ-RhoGEF function leads to the accumulation of hemogenic cells and impairs EHT progression.

A-C, numeration and morphometric analyses of aorta and cell types for *Tg(kdrl:eGFP-Jam3b; kdrl:nls-mKate2)* ArhGEF11 exon 38 splicing morpholino-injected and control embryos (**A**), or for (*Kdrl:eGFP-Jam3b; kdrl:ArhGEF11CRISPR-Cterdel/+*) homozygous ArhGEF11 C-ter deletion mutants and control siblings (**B**). 48-55 hpf embryos were imaged using spinning disk confocal microscopy. **Aa, Bb**, 2D-cartographies obtained after deploying aortic cylinders and showing the eGFP-Jam3b signals only with cell contours delineated either in blue (endothelial cells), yellow (hemogenic cells, see **Methods** for their morphological definition), red (morphologically characterized EHT cells, for controls), and small

cells delineated by cyan boxes (morphologically uncharacterized EHT cells and putative post-mitotic cells remaining as pairs, included in the numeration as hemogenic cells). Cellular contours have been semi-automatically segmented along the cellular interfaces labelled with eGFP-Jam3b (see **Methods**). Scale bars: 10µm. **Aa', Bb', left:** numeration of endothelial, hemogenic and EHT-undergoing cells according to the position of their geometrical center (either on the aortic floor, or on the roof, or on the lateral side), for each condition; **right:** number of endothelial, hemogenic and EHT-undergoing cells in each condition calculated from the segmentation of 3 x 2D-projections per embryo and covering the entire aortic regions in the trunk. **Aa'', Bb'', left:** length of hemogenic cells (in the longest axis) in function of their orientation (°) relative to the blood flow axis (0 – 180°); **right:** distribution of the orientation of hemogenic cells relative to the blood flow axis, displayed as a mean distribution of cells per embryo. **Aa''', Bb'''**, hemogenic cell elongation factors in arbitrary Units (scale factor given by the ratio between the first- and the second-best fitting ellipse diameters, the minimum value being 1 for a non-elongated object) represented as boxplot distribution of all segmented cells (**left**) or as the distribution of cell elongation factor per embryo (**right**), for controls and for interfering conditions as indicated. **Aa''', Bb''''**, hemogenic cell area represented as boxplot distribution of all segmented cells (**left**) or as the distribution of cell area per embryo (**right**), for controls and for interfering conditions as indicated. **C**, aortic perimeter (in µm) for controls and mutant conditions as indicated. Statistical tests: two sided unpaired two samples Wilcoxon test. For the ArhGEF11 exon 38 splicing morpholino condition, analysis was performed on 2 x control (non-injected embryos) and 3 x embryos injected at the one-cell stage; for the CRISPR mutant condition, analysis was performed on 2 x wild-type siblings for control and 2 x homozygous mutant embryos whose DNA mutation was confirmed by sequencing. 3 consecutive aortic segments per embryo were analyzed to cover the whole length of the dorsal aorta, in the trunk region (covering a distance of 990µm per embryo).

To confirm the potential function of ArhGEF11 at the junctional interface between HE/EHT and endothelial cells, we first investigated its intracellular localization. We attempted to detect the full-length form upon expression of a GFP fusion protein, in the vascular system, and failed to do so. We then generated a truncated form that retains the N-terminal fragment encompassing the PDZ and RGS domains (see **Figure 7 – figure supplement 3A**) fused with eGFP in its C-terminus. Upon transient expression in the vascular system, we visualized its localization at the interface between endothelial and hemogenic cells progressing throughout the EHT, with an apparent increase in density at antero-posterior regions between adjoining EHT cells (**Figure 7 - figure supplement 3B**).

In line with the formerly described function of a splicing variant of ArhGEF11 in controlling tight junction integrity, particularly during egression of cells from the skin epithelium in the mouse (Lee et al. 2018 [\[4\]](#)), we questioned the potential role of such variant in EHT. This variant, referred to as the mesenchymal form, results from the inclusion of a peptide encoded by exon 37 or exon 38 in mouse and human, respectively (Shapiro et al. 2011 [\[5\]](#); Itoh et al. 2017 [\[6\]](#); Lee et al. 2018 [\[4\]](#)). This insertion locates in the degenerated C-terminal region of the protein, which is predicted to be relatively poorly organized in its 3D structure. Upon amplifying the full-length sequence of zebrafish ArhGEF11 for investigating its localization, we cloned fragments encoding – or not – for the insertion of a peptide of 25 amino-acid residues (75 base pairs corresponding to exon 38). Although variable when compared with other species (see **Figure 7 – figure supplement 4C, bottom panel**), this peptide is inserted in the same region as it is for mammals and may correspond to an ArhGEF11 variant functionally equivalent and involved in the regulating of junctional stability (see **Discussion**).

To investigate the function of ArhGEF11 on the junctional interface between HE/EHT and endothelial cells, and more specifically of the isoform containing the exon 38 encoded peptide, we used both morpholino (MO) and CRISPR-based strategies. We designed a splicing MO at the 3-prime exon/intron boundary of exon 38 that interferes with its inclusion in the encoding mRNA (**Figure 7 - figure supplement 4A**). This MO did not trigger any deleterious effect on the gross morphology of zebrafish embryos (**Figure 7 – figure supplement 4B**) and blood circulation was normal in comparison to control embryos. We attempted to generate CRISPR-based genomic

deletion mutants of exon 38, both using Cpf1 and Cas9 enzymes and failed (see **Methods**). However, using CRISPR/Cas9, we obtained a deletion mutant triggering a frame shift downstream of exon 38 and introducing a premature stop codon few amino-acid residues downstream thus leading to a sequence encoding for an ArhGEF11 C-terminal deleted form (see **Methods** and **Figure 7-figure supplement 4C**). Unlike the variant skipping exon 38 induced by the splicing MO, expression of the CRISPR/Cas9 C-ter deletion mutant triggered, around 24 hpf, a severe retardation of blood circulation initiation in approximately 80% of the embryos obtained from incrossing ArhGEF11^{CRISPR-Cterdel/+} heterozygous mutant fishes. From 24 to 48 hpf, approximately 50% of these embryos recovered a normal blood circulation (suggesting that these embryos are probably heterozygous for the mutation), approximately 35% remained with a severe phenotype (characterized by a large pericardiac oedema) and approximately 15% died, see **Figure 7-figure supplement 4D** and **Methods**). This indicates essential functions of the C-terminal region of ArhGEF11, in agreement with previously published data on the mammalian protein (Chikumi et al. 2004 [\[1\]](#)).

We then characterized more in depth the MO and CRISPR phenotypes (**Figure 7** [\[1\]](#)) and performed a quantitative analysis of the number and morphology of HE, EHT and adjoining endothelial cells, based on confocal images and subsequent segmentation of cell contours (using 2D deployment of the aortic cylinder).

For the morphants, we found approximately a double amount of hemogenic cells in the aortic floor in comparison to controls (altogether localized on the floor and on the lateral sides of the aorta, see **Figure 7A** [\[1\]](#), panels in **(a)** for 2D-cartographies to visualize cell shapes, with the accumulation of hemogenic cells delineated in yellow for one morphant on the right and panels in **(a')** for cell counts of endothelial, hemogenic and EHT cells either in the entire aorta in trunks (left) or, more precisely, the aortic floor and lateral aortic side and the roof (right)); see also **Figure 7-figure supplement 5** for additional examples). Increase in the number of hemogenic cells in morphants in comparison to controls also concerned cells oriented more perpendicularly to the blood flow axis (toward the Y-axis, **Figure 7A** [\[1\]](#), see panels in **(a'')**), consistently with variation in emergence angle, as we have shown previously (Lancino et al. 2018 [\[2\]](#)). This is accompanied by a significant reduction in the number of morphologically characterized EHT cells (**Figure 7A** [\[1\]](#) and **Figure 7-figure supplement 5**). Similar results were obtained for the CRISPR homozygous deletion mutants, particularly regarding the accumulation of hemogenic cells, although with a smaller amplitude (**Figure 7B** [\[1\]](#), with the panels **(b - b')**) mirroring the presentations of results obtained for morphants and internal controls in **Figure 7A** [\[1\]](#); see also **Figure 7-figure supplement 6** for additional examples). Consistently, a significant increase in the aortic perimeter was found, at least for the morphants (**Figure 7C** [\[1\]](#)). Of notice also, hemogenic cells in morphants are on average less elongated in comparison to control and to the CRISPR homozygous mutants (as well as less dispersed according to this parameter, compare **Figure 7A** [\[1\]](#), panel **(a''')** with **Figure 7B** [\[1\]](#), panel **(b''')**), with visualization of the dispersion on right panels). Importantly, we observed for the morphants an accumulation of a less elongated cell population standing below the median value (**Figure 7A** [\[1\]](#), panel **(a''')**); to some extent, this is reproduced as well when measuring hemogenic cell area, compare **Figure 7A** [\[1\]](#), panels in **(a''')** with **Figure 7B** [\[1\]](#), panels in **(b''')** for morphants and CRISPR mutants, respectively). Finally, the population of small cells in the morphants are of uncharacterized morphology, which means that they appear neither as EHT pol+ nor as EHT pol- cells and may represent abortive emergences (**Figure 7A** [\[1\]](#) and **Figure 7-figure supplement 5**, cells in cyan boxes).

Additionally, we assessed the impact of MO interference on ArhGEF11 exon38 splicing on hematopoiesis using the *Tg(CD41:eGFP)* fish line. By performing live imaging on *Tg(CD41:eGFP) X Tg(Kdrl:nls-mKate2)* outcrosses (see **Figure 7-figure supplement 7A** and **B** for control embryos and morphants, respectively), we quantified the number of hemogenic/EHT cells in the aortic floor, based on fluorescence intensities as well as the number of newly generated hematopoietic stem and progenitor cells (HSPCs, that continue to express the hematopoietic eGFP reporter but

stopped to express the vascular mKate2 reporter, **Figure 7 – figure supplement 7C**). We show that, in agreement with our morphometric analysis, an increase in the number of hemogenic/EHT cells in the aortic floor in morphants in comparison to control embryos (**Figure 7 – figure supplement 7D**, with median values of 14.0 and 22.5 hemogenic/EHT cells for controls and morphants, respectively). Additionally, we observe, despite the high variability in both controls and morphants, a relative diminution in the number of newly generated HSPC precursors (**Figure 7 – figure supplement 7E**, with median values of 22.0 and 16.5 HSPCs for controls and morphants, respectively). This is consistent with our previous results showing an accumulation of HE/EHT cells in aortic floors of morphants and highlights the negative impact of this accumulation on the generation of HSPCs.

Altogether, the morphants and CRISPR mutant phenotypes show that the progression throughout EHT is significantly impaired supporting the idea that ArhGEF11 exerts important functions in the process. Increase in the frequency of elongated cells for the CRISPR mutants in comparison to morphants suggests that interference triggered by the C-terminal deletion of ArhGEF11 is more effective at early time points. Interference with inclusion of exon 38 in morphants triggers an increase in the number of hemogenic cells of smaller area and reduced elongation indicating a more efficient progression throughout emergence in comparison to the CRISPR mutants, albeit with impairment at later stages. This may be due to alteration in the dynamics of endothelial cell intercalation which is required for emergence completion (which is also compatible with the accumulation of pairs of cells that may be post-mitotic and that do not spread apart because this also requires aortic cell intercalation).

ArhGEF11/PDZ-RhoGEF and its variant containing the exon 38 encoded sequence control junctional dynamics to support reduction of the hemogenic/endothelial interface and cell-cell intercalation

To investigate how ArhGEF11 controls the dynamic interplay between HE/EHT and endothelial cells, we characterized further the MO and CRISPR phenotypes, at the junctional level. To do so, we performed FRAP experiments using the *Tg(Kdrl:eGFP-Jam3b)* fish line that was injected at the one cell stage with the exon 38 splicing MO, or incrossing (*Kdrl:eGFP-Jam3b* x ArhGEF11^{CRISPR-Cterdel/+}) heterozygous fishes to obtaining homozygous mutants and wild type alleles for control siblings.

FRAP experiments were focused on the hemogenic (HE)/endothelial (EC) membrane interface, with the support of images obtained after 2D deployments of aortic segments (**Figure 8A** [↗](#)). Pools of *eGFP-Jam3b* molecules localized at bi-junctional (HE-EC, green arrows in panel **A**) and at tri-junctional interfaces (HE-EC-EC (magenta arrows in panel **A**) and HE-HE-EC (black arrows in panel **A**)) were bleached and subsequently imaged for FRAP analysis. In the case of the MO treatment (in which case HE cells accumulate on the aortic floor, see the 2D cartography **Figure 8A** [↗](#))) we measured, in comparison to controls, an increase in FRAP recovery parameters for *eGFP-Jam3b* pools at the tri-junctional HE-HE-EC interface, with a prominent tendency for an increase in the recovery speed at early time points (**Figure 8B** [↗](#), panel **(b''')**), values obtained from fluorescence intensity recoveries whose mean values are represented by curves in panel **(b'')**; compare with maximum recovery amplitudes **(b')**, with values obtained from the plateaus of fluorescence recovery intensities whose mean values are represented in panel **(b)**). Results are different for the ArhGEF11^{CRISPR-Cterdel/+} homozygous mutants for which we did not observe any obvious effect at HE-HE-EC tri-junctions (**Figure 8C** [↗](#), with the panels **(c - c''')**) mirroring the presentations of results obtained for morphants and internal controls in **Figure 8B** [↗](#)) but, rather, a small tendency for an increase in the mobile pool at HE-EC-EC tri-junctions (that has the particularity to concern 2 endothelial cells that contact the longitudinal membrane of one HE cell, see **Figure 8A, D** [↗](#)).

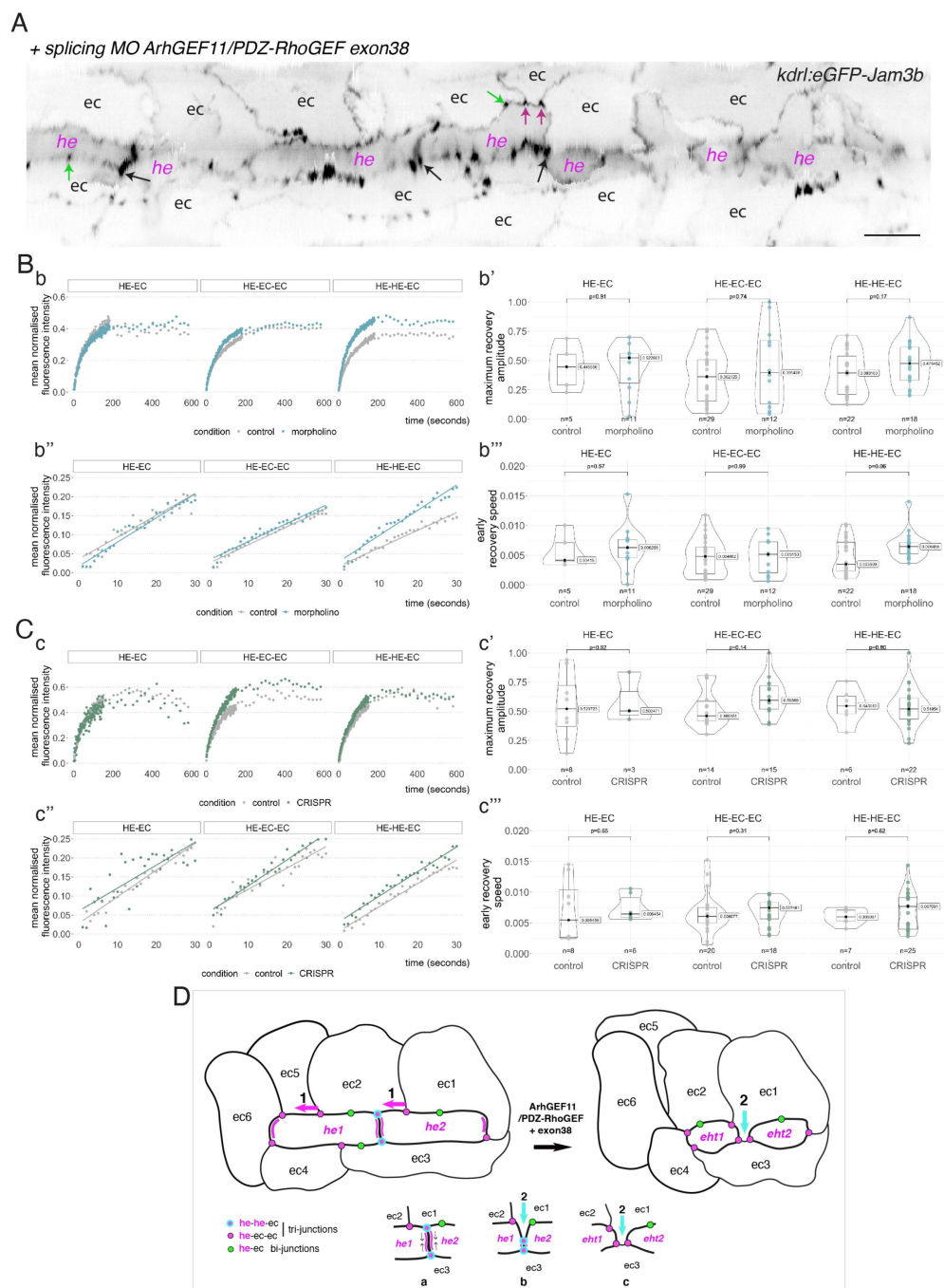


Figure 8

Interfering with the function of ArhGEF11/PDZ-RhoGEF suggests an activity at the interface between endothelial and hemogenic cells that relies on restricting the mobility of junctional pools at tri-cellular junctions, with a prominent role of the +exon 38 variant during endothelial cell intercalation.

A, 2D-cartography of an aortic segment of a *Tg(kdrl:eGFP-Jam3b; kdrl:nls-mKate2)* embryo injected at the one cell stage with the ArhGEF11 exon 38 splicing morpholino. All the arrows pointing at reinforced junctional contacts between cells were bleached and imaged for FRAP analysis (the bleached areas correspond to the spots of highest intensities in the regions of interest, as visualized at the fluorescent confocal microscope, see *Methods*). Black and Magenta arrows point at he-he-ec and he-ec-ec tri-cellular junctions, respectively (he: hemogenic cell; ec: endothelial cell).

Green arrows point at he-ec bi-cellular junctions. ec: endothelial cell, he: hemogenic cell. Bar = 20µm. **B, C**, FRAP analysis of bleached eGFP-Jam3b localized in regions of interest in controls (grey) and ArhGEF11 exon 38 splicing morpholino injected embryos (blue) or homozygous ArhGEF11^{CRISPR-Cterdel/+} mutants (green)). The number of biological replicates (n) is stated on the plots. Statistical tests: two sided unpaired two samples Wilcoxon test. Bb, Cc, evolution, after photobleaching (at t = 0s), of the mean fluorescence intensity per condition at HE-EC, HE-EC-EC and HE-HE-EC bi- and tri-junctions over time (10 min). Bb', Cc', median values for maximum amplitude of recovery (maximum of simple exponential fitted curves, see *Methods*). Bb'', Cc'', early evolution, after photobleaching (at t = 0s), of the mean fluorescence intensity per condition over time (for the first 30 seconds). The fitted lines correspond to the linear regression of the mean fluorescence intensities. Bb''', Cc''', median fluorescence recovery slopes (linear regression). **D**, Model (2D deployment of the aortic wall) representing the endothelial/hemogenic dynamic interplay and the proposed function of ArhGEF11 and of its +exon 38 peptide encoding variant at junctional and membrane interfaces. This interplay involves 2 essential dynamic events requiring junctional remodeling: **1** (left cartoon, magenta arrows), the mobility of he-ec-ec tri-junctional contacts accompanying the movement of endothelial cells (ex: for ec1 and ec2) along lateral sides of hemogenic cells which is required to decrease the number of adjoining cells (see Lancino et al. 2018 [\[1\]](#)). This takes place contemporarily to - or in alternance with -, the contraction of HE and EHT cells as they are progressing throughout the emergence and reducing contacting membrane surfaces along the longitudinal axis (the reduction in contacting membrane surfaces also involves membrane retrieval, hypothetically via endocytosis). Our data obtained with the CRISPR deletion mutants (a slight tendency to increase, on average, the turnover and mobile pool of these he-ec-ec junctions) suggest that ArhGEF11, in the wild type condition, should be slowing down the recycling of junctional components at tri-junctions, which hypothetically should contribute to increase adhesion strength. This may be required also to stabilize the junction-cytoskeleton interface involved in controlling the contraction/shrinkage of HE cells along the longitudinal axis, a hypothesis that is compatible with the mutant phenotype observed in this study, i.e the increase in frequency of more elongated HE cells and the decrease in HE cell progression throughout EHT (see **Figure 7** [\[2\]](#)); **2** (right cartoon, cyan arrow and bottom cartoons a-c), the intercalation of an endothelial cell to isolate 2 adjacent hemogenic cells or 2 daughter cells after mitosis (not depicted). This is mandatory for EHT progression and completion which requires adjoining endothelial cells to protrude membrane extensions that will anastomose to seal the aortic floor (see Lancino et al. 2018 [\[1\]](#)). The accumulation of adjoining cells of rather small length and apparently impaired in EHT progression that we describe in **Figure 7** [\[2\]](#) upon MO interference (that may also indicate impairment in abscission completion) suggests that the ArhGEF11 +exon 38 peptide encoding variant is more specifically involved in controlling the remodeling of the he/he interface that leads to endothelial cell intercalation (bottom cartoons, the remodeling of he-he-ec junctions is leading to he-ec-ec junctions and takes place between b and c). The increase in the recycling parameters that we measure in this interfering condition (mobile pool and early speed of recovery) indicates that the activity of ArhGEF11, and in particular of its +exon 38 peptide encoding variant, negatively controls junctional recycling. Hypothetically, the junctional adhesion strengthening triggered by reducing the recycling of junctional components at the he/he interface may be required during the early phase and progression of intercalation to support increase in membrane tension and environmental constraints (intercalation contributes to reducing the length/surface of the he/he membrane interface preceding junctional remodeling (cartoons b and c); this reducing membrane interface is in addition submitted to the shear stress imposed by blood flow).

Altogether and including our results presented **Figure 7** [\[2\]](#), these results reinforce the idea that ArhGEF11 and its variant encoded by exon 38 are involved in critical steps of EHT progression, partly by controlling the dynamics of junctional pools at the level of tri-junctions established between adjoining hemogenic and endothelial cells (see the model **Figure 8D** [\[3\]](#) and its legend for a detailed analysis of mechanistic issues). The results also suggest that the +exon38 peptide encoding variant is prominently involved in controlling the HE-HE-EC interface and more specifically during intercalation, consistently with the accumulation, upon MO interference, of EHT cells having progressed throughout the emergence but appearing to be slowed down when reaching completion (cells of smaller area and less elongated, see **Figure 7** [\[2\]](#)). Finally, ArhGEF11 is controlling RhoA and, most probably, its function in EHT progression also involves actin and actomyosin activities that are involved in the contraction of HE and EHT cells (see Lancino et al. 2018 [\[1\]](#); this is coupled with the consumption of longitudinal and transversal membrane interfaces, see also the model **Figure 8D** [\[3\]](#) and its legend).

Discussion

Here, we bring new insight into the cell biology and mechanics of the EHT, using live confocal microscopy, RNAscope (to reveal the expression of mRNAs at the cellular resolution), and genetic interference. We show that the emergence of EHT cells is more complex than previously shown, with two types of EHT cells of highly different morphodynamics and apico-basal polarity status. We bring evidence suggesting that these differences take root in HE maturation that is primarily characterized by loss of apico-basal polarity that is then tuned by the activity of the transcription factor Runx1. This leads to two types of EHT cells whose respective emergence characteristics appear to rely, partly at least, on controlling the recycling of junctional complexes directly linked to apico-basal polarity features (the JAMs). Also, we reveal essential aspects of the interplay between HE/EHT cells and adjoining aortic endothelial cells. The first is based on a potential signalling cross talk, whose molecular components remain to be identified, controlling apico-basal polarity and its associated features, in space (and possibly also in time, because of EHT asynchrony over the EHT time-window). The second involves the control of the dynamics of junctional interfaces to reduce contacting surfaces between aortic and HE/EHT cells and to promote cell-cell intercalation for the ultimate release of EHT cells and proper sealing of the aortic floor.

EHT morphodynamics in the zebrafish embryo in comparison to amniotes

EHT cell emergence in the zebrafish embryo is relatively unique; cells extrude from the aortic floor toward the sub-aortic space and not in the aortic lumen, probably because of mechanical constraints on the aortic wall (discussed in [Lancino et al. 2018](#)). However, as a common feature in amniotes and the zebrafish, EHT cells appear to contain large vacuolar structures (which is also the case for HE cells in the zebrafish embryo, as we show); recently, this has been precisely described in chick and quail embryo models ([Sato et al. 2023](#)). In amniotes, in which EHT cells are round, these vacuoles appear to emanate from the abluminal membrane (facing the sub-aortic space) and not from the luminal membrane. In addition, these structures disappear before EHT completion, based on water chase by aquaporin activity ([Sato et al. 2023](#)); it has been suggested that they contribute to exert mechanical force to initiate the detachment of the cell from the endothelial layer. We do not know the membranes from which the pseudo-vacuolar structures that we observe originate, particularly in the HE (they may be formed by transcytosis from the basal membrane, by endocytosis from the luminal membrane, or coming directly from the biosynthetic pathway). However, we can exclude that they contribute to the rounding of EHT pol-cells, as is the case for avians, since we do not see them during EHT pol-emergence. For EHT pol+ cells, according to our results (see [Figure 2A](#) and our hypothetical model [Figure 2B](#)), the most plausible scenario is that the apical/luminal membrane is established by fusion of cytosolic vacuolar structures or redistribution of internal pseudo-vacuoles connected to the luminal membrane. Subsequently, the dynamic properties of the EHT pol+ cell luminal/apical membrane allows its rapid evolution after EHT completion (see [Figure 2 – figure supplement 1](#)). We propose that this is compatible with the activity of aquaporins in late EHT pol+ cells and/or shortly after their release, for water chase and vacuolar regression as proposed in our hypothetical model (see [Figure 1 – figure supplement 1B](#) and the legend for details). Water chase activity would then be a common theme controlling the morphology and dynamics of EHT cells throughout vertebrates, although not for the same purposes.

Characteristics of the HE and complexity of pre-hematopoietic stem cell emergence

Heterogeneity in the identity and hematopoietic potential of cells emerging from the HE in the AGM has been recently shown in co-culture systems and *in vivo* in the mouse (Ganuza et al. 2017 [↗](#); Dignum et al. 2021 [↗](#)) and in the zebrafish model (Tian et al. 2017 [↗](#); Xia et al. 2023 [↗](#)); however, knowledge on the fundamental cell biological and molecular events leading to this complexity is still lacking. Here, we highlight functional aspects that may significantly contribute to this complexity and that take root in the cellular plasticity of the HE. We bring evidence suggesting that HE cells that become competent to initiate the EHT time-window (around 28- 30 hpf) tune their apicobasal polarity status which endows them with the ability to support two emergence processes with radically different morphodynamic characteristics; eventually, this would lead to differentially fated cells, which remains to be determined.

While the ability of the HE to regulate its polarity features may be inherited from molecular cues involved in its upstream intra-aortic specification (such as for example NOTCH (Robert- Moreno et al. 2008; Gama-Norton et al. 2015 [↗](#); Bonkhofer et al. 2019 [↗](#)), TGFβ (Monteiro et al. 2016 [↗](#)), Gata2 (Butko et al. 2015 [↗](#); Daniel et al. 2019 [↗](#))), or any other of the more upstream factors involved in hemogenic specification (Zhao et al. 2022 [↗](#))), our results suggest that Runx1 is taking part in this regulation, consistently with its hematopoietic and EHT-inducing activity (T. North et al. 1999 [↗](#); Kalev-Zylinska et al. 2002 [↗](#); Kissa and Herbomel 2010 [↗](#); Lancrin et al. 2012 [↗](#)). Our experiments using an interfering form of Runx1 (dt-Runx1) that produces a relatively weak and subtle phenotype in comparison to morpholino treatment that prevents aortic cells to convert into HE cells (Bonkhofer et al. 2019 [↗](#)) or genetic mutation and gene knockout abrogating hematopoiesis (T. E. North et al. 2009 [↗](#); Sood et al. 2010 [↗](#); Gao et al. 2018 [↗](#)), provided the opportunity to observe that, at 30 hpf (the timing point at which a majority of HE cells are still paving the aortic floor), elongated HE cells appear to be maintained and to exhibit apicobasal polarity, with the absence of apparent cytosolic vacuolar structures (as has been described in the mouse in Runx1 interfering condition (T. North et al. 1999 [↗](#))); this correlates with the subsequent accumulation of EHT pol⁺ cells. This is in favour of the idea that the polarity status of HE cells has a direct incidence on the cell type that will arise, with a stronger chance to evolve as an EHT pol⁺ cell if apical and basal domains are already determined (which is the case for HE cells upon dt-Runx1 expression) and a lesser chance if these domains are not established and the pool of intracellular vacuoles/pseudo-vacuoles is reduced (see the hypothetical model **Figure 2B** [↗](#)).

A key question arising from our observations is what would control the balance orienting toward an EHT pol⁺ or an EHT pol⁻ type of emergence. Cell division may be orienting toward EHT pol⁻ emergence type (which would also be the reason why these cells have a round-shape morphology in comparison to EHT pol⁺ cells). The frequency of EHT pol⁺ versus EHT pol⁻ cells that we have estimated and that is of approximately 2/1 is not incompatible with this possibility since divisions take place in the HE at a relatively high frequency. Indeed, it has been estimated that 50% of HE cells that constitute the aortic floor at 28 hpf will divide and will undergo emergence thereafter during the 28 - 72 hpf time window (Zhao et al. 2022 [↗](#)). However, we observe that both EHT pol⁺ and EHT pol⁻ cells can divide during the emergence process (see also our former work Lancino et al. 2018 [↗](#) for EHT pol⁺ cells), which is followed by the full- recovery of their respective morphodynamic characteristics. Hence, this excludes the possibility of a short term and purely mechanical effect of cell division on emergence morphodynamics. In link with cell division however, there exists the possibility that the pool of large vesicular structures potentially awaiting for exocytosis in HE cells having lost their apicobasal polarity may become asymmetrically inherited after division. This would endow the cell remaining with the largest pool to undergo EHT pol⁺ emergence after exocytosis and expansion of the apical/luminal membrane. Our results **Figure 2** [↗](#) would support this possibility but what controls this asymmetry would remain to be established.

Still, regarding the control of the balance orienting toward an EHT pol⁺ or an EHT pol⁻ type of emergence, the intrinsic nature of the HE, i.e. its acquaintance with the aortic differentiation programme (Bonkhofer et al. 2019 [\[1\]](#)), raises the possibility of a partial recovery of an aortic cell phenotype in the case of EHT pol⁺ emergence, thus supporting its very unique mechanobiology. Therefore, the regulation of local signalling between aortic and HE cells may tune the balance toward the one or the other fate; this could be the case, for example, for adjoining HE cells in comparison to isolated ones that would then receive unequal signalling cues owing to the difference in homotypic versus heterotypic contacting surfaces. In this scenario, NOTCH signalling which is determinant for aortic and HE specification (Gama-Norton et al. 2015 [\[2\]](#)) may play a significant role, particularly via the ratio of the Jag1/Dll4 ligands that is fundamentally involved in HE versus aortic specification (Robert-Moreno et al. 2008 [\[3\]](#); Bonkhofer et al. 2019 [\[1\]](#)).

Finally, we cannot totally exclude that EHT pol⁺ and EHT pol⁻ cells would originate from different precursors. On this line, it has been shown that the aortic wall can home a transient veinous-fated progenitor cell population that will lead to the underlying vein via selective sprouting from 21 to 23 hpf (Herbert et al. 2009 [\[4\]](#)). This option would require a massive expansion of a minor fraction of the cell population remaining in the aortic floor since it is expected that most of it has been moving out toward the sub-aortic region, which is not in favour of this possibility. In addition, recent scRNAseq analysis of floor and roof aortic cells at 21 and 28 hpf has led, for both timing points, to two well defined clusters defined only by aortic and hematopoietic signatures (Zhao et al. 2022 [\[5\]](#)).

Spatially restricted control of Pard3ba mRNAs by Runx1

Recent work has identified Runx1 targets in the zebrafish embryo at 29 hpf (Bonkhofer et al. 2019 [\[1\]](#)); however, the results do not point at any of the conventional apicobasal polarity organizers as being direct targets although Pard3 and Pard6 mRNAs are clearly downregulated in the HE in comparison to aortic roof cells. Our work now strengthens the idea that understanding the mechanisms underlying the EHT may require more than simply deducing from expression of mRNAs in the HE (including the corresponding proteins), as exemplified by members of the Pard3 family. Also, annotation of protein isoforms in reference databases needs to be improved so as to gain precise and accurate insight on mechanistic processes.

Our results reveal the unique sensitivity of Pard3ba mRNAs expression to Runx1 interference, in comparison to the Pard3aa and ab isoforms. Also, the dt-Runx1 mutant phenotype with enhanced expression of Pard3ba mRNAs in MERs highlighted and strengthened the existence of these specific aortic regions in wild type animals, which could not have been so obviously pointed at because of its relatively low and variable expression. Importantly, the expression of Pard3ba mRNAs is mainly detected in AECs and we cannot for sure say that hemogenic/EHT cells also express this mRNA because very few spots are detected and those may be at the limit of the technical background (although we measured a tendency to increase their detection upon dt-Runx1 expression, indicating that they may indeed be expressed in HE/EHT cells but at very low levels, in wild type embryos).

The sensitivity of AECs on Runx1 interference is somehow unexpected since Runx1 is rather expected to be involved in HE maturation as well as during EHT. This raises the possibility of a direct signaling cross talk between HE/EHT cells and aortic adjoining cells. In this scenario, HE/EHT cells exposed to Runx1 signaling would instruct AECs that subsequently regulate their apico-basal polarity features (via Pard3ba expression); this would then feedback on HE/EHT cells. Another plausible explanation is that AECs have been exposed to interference on Runx1 activity upstream of the EHT time-window, at a timing point at which aortic and HE cell types have been sharing a common ancestor, a possibility that has been debated and reinforced in the zebrafish model (Bonkhofer et al. 2019 [\[1\]](#); Zhao et al. 2022 [\[5\]](#)). Of notice, our dt-Runx1 mutant is expressed under the vascular kdrl promoter that may be active sufficiently early (Liao et al. 1997 [\[6\]](#)) to allow

efficient interference with endogenous Runx1 activity on a relatively long time scale; this may take place in a more or less restricted population of precursor cells, according to the spatially restricted localization of Pard3ba MERs.

The increase of Pard3ba mRNAs detection, upon dt-Runx1 mutant expression, suggests that endogenous Runx1 negatively controls this specific mRNA in wild type embryos. This is consistent with our results since there are two cell types/stages during the EHT process for which we observe a negative regulation of apico-basal polarity, i.e the maturing HE and EHT pol- cells, as visualized upon expression of eGFP-podxl2 and mKate2-podxl2 (the fluorescent fusion proteins are partitioned equally between the luminal/apical and the basal membranes). Altogether, this reinforces the idea that Runx1 negatively controls apico-basal polarity in HE and EHT pol- cells; at least in part, it does so via the Pard3ba isoform that we hypothesize acts as a tuner, consistently with the high sensibility of its mRNA expression to Runx1 activity and the recurrent variability in expression that we are measuring. Owing to the apparent stochasticity of the EHT process in time and space (emergence takes place asynchronously during the time window of the EHT that ranges from approximately 28 to 72 hpf and proceeds from place to place, all along the floor of the dorsal aorta), this may be what contributes to the experimental variability that we are measuring at a given time using qRT-PCR and RNAscope (for all determinations of Pard3ba mRNAs in comparison to other isoforms; of notice, variations are even larger for the dt-Runx1 mutants, which is consistent with the high sensitivity to dt-Runx1 expression levels that are expected to be relatively different from one transgenic embryo to the other).

Altogether, our results are consistent with the hypothesis that the variability in expression of Pard3ba mRNAs reflects the response to intrinsic fluctuations of the system over time and space and reinforce the idea that these conditions, mediated by Runx1 and environmental constraints imposed by the blood flow, tune apico-basal polarity. This tuning scenario would also explain the specific case of EHT pol+ cells that would then proceed as such upon exposition to positive apico-basal signaling when Runx1 activity is decreased, consistently with the tendency to accumulate EHT pol+ cells upon expression of our dt-Runx1 mutant. Unfortunately and presently, we cannot show a spatial proximity between EHT pol+ cells and Pard3ba MERs *in situ* (or the opposite for EHT pol- cells) owing to technical limitations with fixed embryos that do not allow the maintenance of EHT cell shapes. Resolving this issue in the future will be very informative, also to place cellular morphodynamics and gene expression regulation throughout the EHT, with spatial and temporal resolution.

The complexity of pre-HSC emergence and heterogeneity of hematopoietic stem cell and progenitor populations

Currently, we do not know what would be the features that would endow the two cell types to behave differently after their release. Since post-EHT cells have to migrate in transient developmental niches (i.e the CHT, the thymus, the pronephric region), these features may provide them with more or less ability to migrate in the sub-aortic space and pass throughout the wall of the underlying vein to conquest more distant niches (Murayama et al. 2006 [\[1\]](#)), or to remain in a local AGM niche and be exposed to specific signalling there, in the proximity to the aorta as it is the case for hematopoietic clusters in mammals and avians (Ciau-Uitz et al. 2014 [\[2\]](#); Jaffredo and Yvernogeu 2014 [\[3\]](#)). Beside their potential differences in colonizing different niches which would impact on their subsequent fate, these cells may also contribute to more or less transient waves of hematopoietic stem cells and progenitors that would be specific to the developmental period such as, for example, a restricted sub-set of T-lymphocytes that was described before (Tian et al. 2017 [\[4\]](#)). In the specific case of the Tian et al. work, the AGM was shown to produces transient T-lymphocytes proposed to be independent of long-lived HSCs according to a gradient that increases from the anterior to the posterior part of the trunk region, as well as from the aorta in the tail (the posterior blood island in the tail region that was proposed to give rise to hematopoietic precursors related to mammalian erythro-myeloid progenitors, (Bertrand et al. 2007 [\[5\]](#))). Since the EHT pol+

and EHT pol⁻ cells that we describe in our work are equally produced from the aorta in the trunk and in the tail regions (thus irrespective of the antero-posterior axis of the aorta), the one or the other of the two EHT types most probably does not directly relate to the first wave of T-lymphopoiesis arising from the aortic endothelium described in [Tian et al. 2017](#).

The cell biology behind emergence control and morphodynamics

The intrinsic capacity of the HE, that we show here to be able to support two highly distinct emergence processes, in the zebrafish embryo, is very unique. Indeed, not only are they taking place contemporarily, but also can the two cell types proceed throughout emergence at a distance shorter than 50 µm, which accommodates the intercalation of a single adjoining endothelial cell. This may reflect the remarkable properties of the vascular system that needs to allow cellular extrusion and transmigration. The latter is particularly relevant in the context of developmental hematopoiesis since hematopoietic precursors that travel via blood circulation need to pass throughout the aortic wall to conquest their distant niches.

In the present work, we provide some of the mechanistic insights into the specificities of the two emergence types and that rely on two fundamental aspects of cellular biology, i.e apicobasal polarity establishment and junctional mobility. The establishment and maintenance of apicobasal polarity in a cell extruding from its environment, as is the case for EHT pol⁺ cells, is at odds with the fundamental mechanisms of cell extrusion irrespective of the context into which the process is taking place ([Nieto et al. 2016](#); [Gudipaty and Rosenblatt 2017](#); [Pei et al. 2019](#); [Staneva and Levayer 2023](#)); this places the emergence of EHT pol⁺ cells as an extreme case of extrusion and it is somehow unexpected to observe this type of emergence when the aortic wall can extrude EHT pol⁻ cells that appear to emerge contemporarily, apparently according to a more conventional mode. During the EHT time-window, Runx1 is directly controlling the emergence programme by regulating Rho-dependent cytoskeletal functions in HE cells derived from mouse ES lines ([Lie-A-Ling et al. 2014](#)) and inducing the expression of Gfi1ab in the zebrafish ([Bonkhofer et al. 2019](#)) that belongs to the family of transcriptional repressors of the arterial programme expressed in the HE (Gfi1 and Gfi1b in the mouse, ([Lancrin et al. 2012](#))). This programme downregulates key molecules involved in the maintenance of the aortic endothelium among which Cdh5 (VE-Cadherin) and the downregulation of cadherins is a hallmark of extrusion. Thus, for EHT cells, and particularly EHT pol⁺ cells, downregulation of VE-Cadherin should be compensated by other mechanisms that allow for the maintenance of aortic integrity. In this context of adhesion downregulation, the JAMs (that are expressed endogenously in the vascular system and that were used in this study) appear to be a well-suited substitute. As part of tight junction complexes, they can strengthen adhesion and, in addition, since they are also involved in cell migration (in particular leukocyte trans-endothelial migration, see ([Ebnet 2017](#))), they could in theory support the extrusion of EHT pol⁻ cells that do so according to a seemingly migration-type emergence mode (inferred from our time-lapse sequences) more or less related to trans-endothelial migration. In the case of EHT pol⁺ cells, we propose that JAMs may significantly contribute to the re-establishment of apicobasal polarity and to the strengthening of adhesion while EHT pol⁺ cells should undergo strong mechanical tension (owing to cellular bending). Interestingly, Jam3b appears to be a direct and positively regulated target of Runx1 ([Bonkhofer et al. 2019](#)), which is supporting its aforementioned potential functions during EHT.

In regard to the different modes of junctional recycling between aortic and either EHT pol⁺ or EHT pol⁻ cells for which we bring evidence using FRAP, we hypothesize that differences take root in the regulation of cell polarity features, such as for example the organization of an apical endosome in the case of EHT pol⁺ cells (which may be why recycling of eGFP-Jam molecules appears to be faster in EHT pol⁺ cells in comparison to EHT pol⁻ cells, in relation to the recycling pathways undertaken by JAM molecules). In addition, since JAMs interact with Pard3 ([Ebnet et al. 2003](#)) this recycling, that would provide renewed JAM/Pard3 complexes at membrane junctions and sustained activation of the Pard3/Pard6/aPKC complex ([Ebnet 2017](#)), may contribute to the maintenance/reinforcement of EHT pol⁺ cell polarity throughout the entire emergence process.

Also, the regime of recycling may be regulated by an adjacent aortic endothelial cell that expresses *Pard3ba* and that controls trans-cellular JAMs pairing via signalling involving apico-basal polarity features. Finally, this turnover may be adapted to the mechanical tensions EHT pol+ cells are exposed to, in comparison to EHT pol- cells, owing to their specific morphodynamic features (ex: cellular bending); these tensions may be sensed by tri-cellular junctions which would be preferential sites for the renewal of junctional components whose half-life is reduced owing to submission to mechanical forces.

The cell biology behind emergence control and mechanics: the function of ArhGEF11/PDZ-RhoGEF in the dynamic interface between aortic and HE cells

Finally, regarding inter-cellular junctions and the control of their dynamics all along the EHT process, we also addressed the dynamic interface between aortic and HE cells. As for EHT cells, we focused more specifically on tri-junctional interfaces that have been proposed to sense tension and transmit the information to other types of junctional complexes (Bosveld, Wang, and Bellaïche 2018); in the EHT context, the contracting hemogenic endothelium is additionally exposed to high mechanical tension owing to wall shear stress imposed by the blood flow and that was proposed to regulate emergence efficiency (Lundin et al. 2020; Campinho et al. 2020; Chalin et al. 2021). In our work, we highlight one essential aspect of controlling junctional recycling which is during cell-cell intercalation which concerns either contacting HE cells lying on the aortic floor or HE/EHT cells after cytokinesis (as mentioned before, cell division is rather frequent during the EHT time window (Zhao et al. 2022)). In this context, we addressed the function of ArhGEF11/PDZ-RhoGEF which provides a functional link between cell polarity (with its PDZ-binding motif and its RGS domain that couples to G- protein coupled receptor signalling to regulate planar cell polarity (Nishimura, Honda, and Takeichi 2012)), cell adhesion (it binds to the tight junction associated regulatory protein ZO-1, (Itoh et al. 2012)), and cytoskeletal regulation/contraction (Itoh et al. 2012; Itoh 2013), all three essential mechanistic components of morphogenetic movements and tissue remodelling. Since morphodynamic events leading to cell extrusion and to the control of cell migration do not only rely on the regulation of mRNAs levels but also on alternative splicing (Pradella et al. 2017) our interest for investigating the function of ArhGEF11 in EHT was raised by the fact that the protein undergoes changes in its biological properties via the alternative splicing of a small exon which leads to a modification of its C-terminus. We identified a splicing event in this region (concerning exon 38) for a potential zebrafish isoform ortholog. However, the precise function of this variant for the zebrafish protein, that was shown in the case of the mammalian isoform to induce cell migration and motility (Itoh et al. 2017) and to bind to the Pak4 kinase thus leading to the subsequent destabilization of tight junction complexes via loss of RhoA activation (Lee et al. 2018), remains to be established. On this line, our results consistently indicate that preventing exon 38 insertion interferes with EHT progression (as well as deleting the C-terminus of the protein as shown with our CRISPR mutant) and appears to impair intercalation of aortic cells between adjoining HE cells. Hence, the +exon 38 isoform, by regulating the dynamic interplay between HE and endothelial cells, would favour one of the essential steps of the EHT process leading to its completion and requiring the complete isolation of EHT cells to ensure the ultimate sealing of the aortic floor (see our former work (Lancino et al. 2018)). It remains to be established, as our measurements indicate, why the ArhGEF11 +exon 38 isoform appears to slow down junctional recycling which is counterintuitive with increasing the HE/endothelial dynamic interface. However, upon interference, we have been following Jam3b recycling which is only a sub-population of tight junction components that should be at play in the system. Finally, the EHT process on its whole may be submitted to a complex regime of alternating contraction and stabilization phases as we have shown previously in the context of EHT pol+ emergence and that may reflect the necessity to adapt to mechanical constraints imposed by the environment as well as the rearrangements of the HE/EHT/aortic cellular interface (Lancino et al. 2018). In this context, back and forth regulatory mechanisms, particularly involved in the control of the RhoA-Myosin II signalling axis partially regulated by

ArhGEF11, may locally and at specific timing points change the turnover of junctional molecules, thus blurring the correlation between gross phenotype (here the organization of the HE/aortic interface) and very local and dynamic molecular events (for a complement to the discussion, see also the legend of the model presented **Figure 8C** [↗](#)).

Overall, our work highlights the complexity of pre-hematopoietic stem cell emergence as well as some of the essential molecular and mechanistic aspects involved, at the interface with aortic endothelial neighbours. We show that the aorta, in the zebrafish embryo, produces two fundamentally different types of EHT cells and propose that this results from specific functional features of the HE regulated by the transcription factor Runx1. Among these features, apico-basal polarity and its regulation appear to be essential for the spatial and temporal control of EHT cell emergence which is an asynchronous process. This would support the production of cell types potentially endowed with different cell fate potential, subsequently reinforced by the type of niche into which these cells will establish homing. It now remains to be established if the different cell types that we describe are indeed leading to cells that will eventually hold specific hematopoietic fate and properties.

Materials and Methods

Key Resources Table

For the exhaustive list of all primers, see Materials and Methods - Supplementary Table.

Reagent (species) resource	type or	Designation	Source reference	or	Identifiers	Additional information
recombinant DNA reagent		pG1-flk1-MCS-tol2	(Jin et al. 2005)		N/A	
recombinant DNA reagent		pEGFP-C1	Clontech		N/A	
recombinant DNA reagent		Transposase pCS-zT2TP	(Suster et al. 2011)		N/A	
recombinant DNA reagent		pmKate2-f-mem	Evrogen		Cat# FP186	
antibody		goat anti-rabbit Alexa Fluor 488	Invitrogen		Cat# A11070	1/400
antibody		rabbit anti-GFP	MBL		Cat# 598	1/300
antibody		mouse anti-HA	Sigma		Cat# 12ca5	1/50
antibody		goat anti-mouse HRP-conjugated	Thermo Fisher Scientific		Cat# G-21040	1/300
sequence-based reagent		Morpholino sih	(Sehnert et al. 2002)		ZDB-MRPHLNO-060317-4	CATGTTTGCTCTG ATCTGACACGCA
sequence-based reagent		Morpholino ArhGEF11 exon 38	This paper		N/A	GAAATAAATGAAG CCCCACCTCCGT
sequence-based reagent		Oligo for CRISPR: Alt-R® CRISPR-Cas12a (Cpf1) crRNAs -1	This paper		N/A	TATCACACACACA TCACCTTCTA
sequence-based reagent		Oligo for CRISPR: Alt-R® CRISPR-Cas12a (Cpf1) crRNAs -2	This paper		N/A	TTTCTCAGCGCTC CTGACAGATG
sequence-based reagent		Oligo for CRISPR: Alt-R® CRISPR-Cas9 crRNAs	This paper		N/A	AGCCAATCGTCTG AGGACGG
sequence-based reagent		Oligo for CRISPR: Alt-R® tracrRNA	IDT		Cat# 1072532	
sequence-based reagent		Oligo for WISH: RNAscope™ Probe-Dr-myb-C3	ACD-Biotechnie		Cat No. 558291-C3	Cat No. 558291-C2
sequence-based reagent		Oligo for WISH: RNAscope™ Probe-Dr-pard3ab-C3	ACD-Biotechnie		Cat No. 1282521-C3	
sequence-based reagent		Oligo for WISH: RNAscope™ Probe-Dr-pard3ba-C2	ACD-Biotechnie		Cat No. 1309581-C2	
sequence-based reagent		Oligo for WISH: RNAscope™ Probe-Dr-pard3aa-C2	ACD-Biotechnie		Cat No. 1305321-C2	
software, algorithm		Imaris	Oxford Instruments		https://imaris.oxinst.com/	v10.1.0
software, algorithm		CRISPRscan	(Moreno-Mateos et al. 2015)			N/A

Reagent (species) resource	type or	Designation	Source reference	or	Identifiers	Additional information
software, algorithm		Zen Pro2 software	Zeiss		https://www.zeiss.com/microscopy/en/products/software/zeiss-zen.html	N/A
software, algorithm		QUANTSTUDIO™ DESIGN ANALYSIS 2	& Thermofisher		https://apps.thermofisher.com/apps/da2/	N/A
software, algorithm		MetaMorph software	Molecular Devices		https://www.moleculardevices.com/products/cellular-imaging-systems/acquisition-and-analysis-software/metamorph-microscopy	N/A
software, algorithm		Acquisition-Imaging analysis-LAS X	Leica		http://www.leica-microsystems.com	N/A
software, algorithm		Image Analysis-Fiji	NIH		https://imagej.net/Fiji	N/A
software, algorithm		Image Analysis Icy	(De Chaumont et al. 2012)		http://icy.bioimageanalysis.org/	N/A
software, algorithm		Image Analysis TubeSkinner plugin for Icy	(Lancino et al. 2018)		https://icy.bioimageanalysis.org/plugin/tubeskinner/	N/A
software, algorithm		Figures-Illustrator CC 2017.1.1	Adobe		http://www.adobe.com/cn/	N/A
software, algorithm		Active contour plugin	(Dufour et al. 2011)		https://icy.bioimageanalysis.org/plugin/active-contours/	N/A
software, algorithm		NIS software	Nikon		https://www.microscope.healthcare.nikon.com/en_EU/products/software/nis-elements	N/A
software, algorithm		Rstudio	RStudio Team (2020)		http://www.rstudio.com/	N/A
software, algorithm		ggplot2	(Wickham 2016)		https://ggplot2.tidyverse.org	N/A
software, algorithm		readr	(Wickham, Hester, and Bryan 2022)		https://github.com/tidyverse/readr	N/A
software, algorithm		cowplot	(Wilke 2020)		https://wilkelab.org/cowplot/	N/A
software, algorithm		dplyr	(Wickham et al. 2023)		https://github.com/tidyverse/dplyr	N/A
software, algorithm		wesanderson	(Ram and Wickham 2018)		https://github.com/karthik/wesanderson	N/A

Reagent (species) resource	type or	Designation	Source reference or	Identifiers	Additional information
software, algorithm		matrixStats	(Bengtsson et al. 2023) 09/02/2024 14:57:00	https://github.com/HenrikBengtsson/matrixStats	N/A
software, algorithm		stringr	(Wickham 2022)	https://github.com/tidyverse/stringr	N/A
software, algorithm		ggsci	(Xiao 2018)	https://github.com/nanxstats/ggsci	N/A
software, algorithm		ggpubr	(Kassambara 2020)	https://rpkgs.datanovia.com/ggpubr/	N/A
software, algorithm		ggstatsplot	(Patil 2022)	https://indrajeetpatil.github.io/ggstatsplot/index.html	N/A
software, algorithm		ggbeeswarm	(Clarke and Sherrill-Mix 2017)	https://github.com/eclarke/ggbeeswarm	N/A
software, algorithm		DT	(Xie, Cheng, and Tan 2022)	https://github.com/rstudio/DT	N/A
software, algorithm		SciViews	(Grosjean 2019)	https://www.sciviews.org/SciViews/	N/A
software, algorithm		nls.multstart	(Padfield and Matheson 2020)	https://github.com/padpadpadpad/nls.multstart	N/A

Contact for reagent and resource sharing

Further information and requests for resources and reagents should be directed to and will be fulfilled by the corresponding Author, Anne A. Schmidt (anne.schmidt@pasteur.fr).

Zebrafish husbandry

Zebrafish (*Danio rerio*) of the AB background and transgenic fish carrying the following transgenes *Tg(kdrl:ras-mCherry)* (Chi et al. 2008 [\[1\]](#)); *Tg(kdrl:nls-mKate2)*; *Tg(kdrl:Gal4;UAS:RFP)*; *Tg(kdrl:eGFP)* (Jin et al. 2005 [\[2\]](#)); *Tg(CD41:eGFP)* (Lin et al. 2005 [\[3\]](#)); and the fish lines generated in this study: *Tg(kdrl:Gal4;UAS:RFP;4xNR:eGFP-podxl2)*; *Tg(kdrl:mKate2-Podxl2)*; *Tg(kdrl:eGFP-Jam2a)*; *Tg(kdrl:eGFP-Jam3b)*; *Tg(kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)*; *Tg(ArhGEF11^{CRISPR-Cterdel/+})* were raised and staged as previously described (Kimmel et al. 1995 [\[4\]](#)). Adult fish lines were maintained on a 14 hr light/10-hr dark cycle. Embryos were collected and raised at either 28.5 or 24°C in N-Phenylthiourea (PTU, Sigma Aldrich, Cat# P7629)/Volvic source water (0.003% final) to prevent pigmentation complemented with 280 µg/L methylene blue (Sigma Aldrich, Cat# M4159). Embryos used for imaging, extracting mRNA, or for WISH ranged from developmental stages 28-to-60 hpf and larvae used for imaging were of 3-5 dpf, precluding sex determination of the animals. The fish maintenance at the Pasteur Institute follows the regulations of the 2010/63 UE European directives and is supervised by the veterinarian office of Myriam Mattei.

mRNA extraction and cDNA synthesis

Total RNA was extracted from whole 48 hpf embryos for cDNA cloning or from pooled trunks at the desired developmental stages (30-32 hpf, 48-50 hpf time windows for qRT-PCR experiments) (~30 individuals per tube). Briefly, embryos were anesthetized using balneation in embryo medium supplemented with tricaine methanesulfonate (MS-222, Sigma-Aldrich Cat# A5040), at a final concentration of 160 µg/ml. RNA was extracted via organic extraction using TRIzol reagent (Invitrogen Cat# 15596026) according to the manufacturer's guideline. gDNA contaminant was

removed using TURBO™ DNase (Invitrogen Cat# AM2238) treatment according to the manufacturer's guideline. Total RNA was stored at -80°C. Reverse transcription of mRNA was performed using SuperScriptIV (Invitrogen Cat# 18090010) with OligodT primer. cDNA samples were stored at -20°C until further processing (for quantitative real-time PCR, PCR, cloning and sequencing).

Transient and stable transgenesis

The *kdr1* promoter (*flk*) (Jin et al. 2005 [\[1\]](#)) was used to drive endothelial expression of mKate2-podxl2, eGFP-Jam2a and eGFP-Jam3b, with eGFP (Clontech) and mKate2 (Evrogen) cDNAs amplified using overlapping primers for Gibson cloning. eGFP-podxl2, ArhGEF11(PDZ-PRD-RGS-eGFP) and dt-runx1-eGFP were amplified for cloning into a pG1-4XNR vector (built from the 4XNR (non-repetitive) 4X UAS sequence less susceptible to methylation than the 14X UAS described in (Akitake et al. 2011 [\[2\]](#))). For the sequence of all the designed cloning primers, see the Key Resources Table.

Podocalyxin-like 2 (Podxl2), ArhGEF11/PDZ-RhoGEF, JAM2a and JAM3b full length sequences were amplified from pools of 48 hpf whole embryo cDNA (see before for mRNA extraction and cDNA synthesis methodologies). All constructs generated in this study were obtained using the Gibson assembly assay (NEB, Cat # E2611S).

For eGFP-Jam2a and eGFP-Jam3b cloning, we used the JAM2a signaling peptide sequence (included in the 5-prime amplification primers and encoding for the signal peptide:

mlvcvslililhsvpvsvptvssr) and, for both constructs, the eGFP sequence was inserted in frame upstream of the sequence encoding for the transmembrane domain; for the eGFP-Jam2a construct, the eGFP was inserted upstream of amino-acid D221 (DLNVAA) of the NCBI Reference Sequence: NP_001091734.1; for the eGFP-Jam3b construct, the eGFP was inserted upstream of amino-acid D243 (DINIAG) of the NCBI Reference Sequence: NP_001076332.2. For transient ubiquitous and inducible expression, the eGFP-Jam2a and eGFP-Jam3b fusion constructs were inserted into a KpnI pre-digested pG1 plasmid containing tol2 sites and the HSP70-4 promoter. For stable transgenesis using the *kdr1* (*flk1*) promoter, constructs cloned into the pG1-HSP70 plasmids were extracted using XhoI/NotI restriction enzymes and inserted into the XhoI/NotI pre-digested pG1-flk1 vector using DNA T4 ligation (NEB, Cat # M0202S).

For the Podxl2 N-ter deletion mutant that has been used for establishing stable *Tg* fish lines, the cDNA encoding for the peptide sequence starting with amino-acid G341 (GGTEYL; fragment 341-587) of the NCBI Reference Sequence: XP_692207.6 was fused with either eGFP or mKate2 cDNAs. The sequence encoding for the signal peptide of the human CD8 (malpvtalllplallhaarpsqfrvs) was introduced upstream (5-prime) of the eGFP or mKate2 sequences (included in the 5-prime primers designed for the Gibson cloning strategy). For the construct designed to obtain the *Tg(kdr1:Gal4;UAS:RFP;4xNR:eGFP-podxl2)* fish line, 2 overlapping fragments were amplified for cloning into a pG1-tol2_4xNR vector after digestion with NcoI, with the following couples of primers: 4XNR_CD8-eGFP_fw and delPodxl2_eGFP_rev; delPodxl2_fw and Podxl2_pG1-4XNR_rev. For the construct designed to obtain the *Tg(kdr1:mKate2-Podxl2)* fish line 2 overlapping fragments were amplified for cloning into a pG1-flk1 vector after digestion with EcoR1, with the following couples of primers: pG1-flk1_CD8_mKate2_fw and delPodxL2-mKate2_rev; mKate2-delPodxL2_fw and Podxl2_PG1-flk1_rev.

For the ArhGEF11(PDZ-PRD-RGS-eGFP) construct, the cDNA encoding for the peptide ending upstream of the DH domain at Valine 709 of the amino-acid sequence (GenBank: AY295347.1; PALDEDV; fragment 1-709) was fused upstream of the sequence encoding for eGFP, for expression using the pG1-4xNR vector. The cDNA was amplified using the GB_4xUAS-gef11_fw; GB-eGFP-RGS_rev couple of primers.

For the construct designed to obtain the *Tg(kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)* fish line, the Runx1 sequence amplified omits the ATG (replaced by an nls) and ends downstream of the Runt domain (ending at amino-acid sequence PAHSQIP). The construct allows for the expression of the dt-runx1 protein under the control of the 4xNR driver using the same strategy as for the above PodxL2 constructs. The dt-runx1 cDNA construct was amplified from the zebrafish Runx1 cDNA sequence (Kalev-Zylinska et al. 2002 [\[1\]](#)) using the couple of primers GB- 4xnr-nlsRunx1_fw; GB-eGFP-T2A-2xHA-delRunx1_rev. The sequence of these primers each allowed for the introduction, in the protein sequence, of a nls (encoded by the fw primer) and of a 2xHA-T2A peptide (encoded by the rev primer that overlaps with the 5-prime sequence of eGFP).

For all constructs, cloning reactions were used to transform NEB 5-alpha Competent E. coli (NEB, Cat # C2987) and resistant clones then used for miniprep plasmid preparations (Nucleospin plasmid, Macherey-Nagel, Cat # 740588.50), followed by sequencing. Plasmid DNA from clones containing correct insertions were then used to transform E. coli TOP10 competent cells (Invitrogen, Cat # C404010) and purified using the endotoxin free NucleoBond Xtra Midi kit (Macherey Nagel, Cat# 740420.10). Transgenesis was then performed by co- injecting 1 nl of plasmid (at 25 ng/μl) mixed with tol2 transposase mRNA (at 25 ng/μl) transcribed from linearized pCS-zT2TP plasmid using the mMESSAGE mMACHINE SP6 kit (Ambion, Cat# AM1340). For stable transgenesis, embryos were screened for fluorescence between 24 hpf and 48 hpf and raised to adulthood. Founders with germline integration were then isolated by screening for fluorescence after outcrossing with AB adults and for establishment of the F1 generation.

CRISPR methodology and GEF11 mutant transgenic line screening

CRISPR/Cas9 mutagenesis and isolation of founders

A sgRNA was designed against the splice donor site at the end of exon 38 with the aim to interfere with the production of the splicing variant including its encoding sequence, using the CRISPRscan web tool (Moreno-Mateos et al. 2015 [\[2\]](#)). The sgRNA was obtained by annealing for 5 min at 95°C equal volumes of 100μM specific Alt-R® crRNAs (IDT, Cat# sequence: AGCCAATCGTCTGAGGACGG) with 100μM Alt-R® generic tracrRNA (IDT, Cat# 1072532). Cas9 Nuclease Reaction Buffer (NEB Cat# B0386A) was added to obtain a final 45μM sgRNA stock solution. For the generation of the sgRNAs-Cas9 complex, a mix containing 18μM sgRNA and 12μM Cas9 protein (EnGen™ Spy Cas9-NLS S. pyogenes, NEB Cat# M0646T) was incubated for 10 min at room temperature. For mutagenesis, 1 nl of the sgRNA-Cas9 complex was injected into one-cell stage embryos, that were subsequently raised until 48 hpf. Bulk genomic DNA (gDNA) was extracted from 60 injected and 60 non-injected control embryos. gDNA extraction proceeded as follows: pooled samples were incubated for 3 hrs at 55°C in lysis buffer (100mM NaCl, 20mM Tris-HCl pH 8, 25mM EDTA, 0.5% SDS and 2 μg/μl proteinase K), purified using phenol:chloroform. Precipitation was done in 0.1 volume of 3M NaAc and 2.5 volume of 100% ethanol for 20 min at -20°C. Samples were centrifuged for 30 min at 4°C before washing with 70% ethanol and resuspension in water. gDNA was used for PCR, using DNA polymerase Platinum™ SuperFi™ (Invitrogen Cat# 12351010) and the SM31-SM32bis couple of primers (TTTCACTTTCTCTCTGCCTCTTACA; AACTCTGTCCAGATGATTGAGGAGC) flanking the targeted region. PCR products were run on gel electrophoresis before gel extraction of bands of interest (with sizes corresponding to either a wild-type allele or a deleted allele) (Nucleospin gel and PCR clean-up, Macherey-Nagel Cat# 740609.50). Fragments were cloned into blunt-end TOPO vector (Invitrogen Cat# 45024), transformed into E. coli TOP10 competent cells and grown on selective medium. 12 colonies for each PCR product were sent for sequencing. Mutant sequences contained either a 7 nucleotides deletion, or a mix of 1,4 or 7 nucleotides deletions located at the same site (at the end of exon 38, see **Figure 7 - figure supplement 4A** for the 7 nucleotides deletion), attesting of the proper functioning of our guide.

A stable homogeneous mutant line was generated by injecting the sgRNA:Cas9 complex into one-cell stage *Tg(kdrl:Gal4;UAS:RFP)* embryos similarly to what was described above. 100 embryos were raised until adulthood. Adult founders were identified by crossing adult F0 with AB fishes, and single embryo genotyping was performed on their progeny (10 embryos per adult fish). Briefly, single embryos were incubated in 40µl 25mM NaOH 0.2mM EDTA and heated for 10 min at 95°C, before cooling for 10 min at 4°C and addition of 40µl 40mM Tris- HCl pH5 to stop the reaction. PCR was performed using 5µl of template and specific primers (couple SM31-SM34 – TTTCACCTTTCTCTCTGCCTCTTACA; ATAAATGAAGCCCCACCTCCGTCC – for wild-type (WT) allele and couple SM31-SM40 – TTTCACCTTTCTCTCTGCCTCTTACA; ATGAAGCCCCACCTCAGACGATTGGC– for mutant allele). Presence of WT or mutant allele for each single embryo was assessed by the presence/absence of amplification band on gel electrophoresis. Four F0 founders were isolated, that had a transmission rate in the germline ranging from 20% to 80%. Progeny of the F0 founders and subsequent generations were raised separately until adulthood, and genotyped. After complete anesthesia (in fish water supplemented with 160µg/ml tricaine methanesulfonate), a small tissue sample was collected from the caudal fin, placed into DNA extraction buffer (10mM Tris pH8.2, 10mM EDTA, 200mMNaCl, 0.5% SDS, 200µg/ml proteinase K) and incubated 3 hrs at 50°C. DNA precipitation and resuspension was performed as described above. 7µL gDNA was used as template for PCR amplification using the primer couple SM31-SM32bis. Sequenced PCR products showed that all founders harbored the same 7 nucleotides deletion (see **Figure 7 - figure supplement 4A**). All experiments relative to **Figure 7** and **Figure 8** were performed using the progenies of one heterozygous mutant F3 couple. For these experiments, genotyping on single embryos was performed on gDNA as mentioned above. Only WT homozygous and mutant homozygous embryos were kept for analysis.

Alternative Cpf1 approach

As an alternative approach to the CRISPR/Cas9 deletion, we attempted to generate a second mutant line, using to CRISPR/Cpf1(Cas12a) system to delete a 315 nucleotides region encompassing exon 38 (with specific guides located in the introns before and after exon 38). The sequence of guides that have been tested are: TATCACACACACATCACCTTCTA and TTTCTCAGCGCTCCTGACAGATG. However, due to the structure of the intronic regions (repeat rich) and the necessary presence of specific PAM sequence, we only generated off- target deletions of intronic regions, without deletions of exon 38, leading us to focus on the CRISPR/Cas9 approach.

Morpholinos and injections

Morpholinos were obtained from GeneTools (see Key Resources Table for sequences). The *sih* Tnnt2 translation start codon and flanking 5-prime sequence MO (Sehnert et al. 2002) as well as the ArhGEF11 exon 38 splice blocking MO (overlapping the exon 38/intron 38-39 boundary) were resuspended in ddH2O to obtain stock solutions at 2mM and 1mM, respectively. The *sih* MO (1.5ng) and ArhGEF11 exon 38 MO (3ng) were injected into 1-cell stage zebrafish embryos, after dilution in ddH2O. For *sih* morphants, embryos that were used in the experiments were checked for absence of heart beating 24 hrs after injection as well as before being used for dissection and RNA extractions at the 30-32 and 48-50 hpf time-windows. For measuring the efficiency of the ArhGEF11 exon 38 MO, total RNA was extracted from pools of injected and control embryos followed by mRNA reverse transcription. PCR on cDNA flanking the exon 38 was performed using DNA polymerase Platinum™ SuperFi™ (Invitrogen Cat# 12351010) and analyzed on gel electrophoresis before extraction, cloning and sequencing to verify the absence of the DNA sequence encoded by exon 38 (see **Figure 7 - figure supplement 4A**).

Whole mount single molecule fluorescent in situ hybridization (RNAscope)

48 hpf *Tg(kdrl:eGFP)* and *Tg(kdrl:Gal4;UAS:RFP;4xNR:dt-Runx1-eGFP)* embryos were fixed in 4% formaldehyde (Electron Microscopy Sciences, Cat#15712) diluted in PBS/0.1% tween20 (PBST) for 2.5 hrs at room temperature, rinsed with PBST and kept in 100% methanol at -20°C until use. On the day of the experiment, embryos were rehydrated upon sequential incubations in 75%/50%/25% MeOH/ddH₂O (each for 10 min at room temperature (RT)). Embryos were then incubated for 10 min in PBST. In subsequent steps of the RNAscope procedure, the Multiplex Fluorescent Reagent Kit v2 was used (Cat# 323100), that included H₂O₂; probe diluent (PD); wash buffer (WB); AMP1, AMP2, AMP3 buffers; HRP-C1, HRP-C2, HRP-C3 reagents; TSA buffer; HRP blocker. All incubations were performed in Eppendorf tubes, with a maximum of 20 embryos/tube, in a dry heating block. Embryos were then incubated with H₂O₂ for 10 min at RT, washed for 15 min in PBST at RT, incubated for 7 min at 37°C with proteinase K in PBST (1/2000 from a glycerol stock at 20mg/ml, Ambion, Cat#10259184), washed in PBST for 2×10 min at RT, and incubated for at least 2 hrs at 40°C in PD. Embryos were then incubated overnight at 40°C with the Pard3aa-C2, Pard3ab-C3, Pard2ba-C2 RNAscope probes.

Next morning, embryos were washed for 2×10 min in WB. From that step on, extreme caution should be taken because embryos will become totally transparent, owing to the harsh conditions of the WB (we recommend that all washing steps, upon buffer removal, are performed under a binocular loupe). Embryos were then incubated sequentially with AMP buffers during 30 min for AMP1 and AMP2 and 15 min for AMP3, at 40°C, with 10 min washes at RT between each AMP buffer incubation. Embryos were then incubated for 15 min at 40°C with RNAscope Multiplex FL v2 HRP-C2 for Pard3aa and Pard2ba probes, HRP-C3 for the Pard3ab probe, washed for 10 min at RT with TSA buffer, incubated for 30 min at 40°C with OPAL-570 (PNFP1488001KT) in TSA buffer (dilution 1/500), washed 2×10 min at RT in WB, incubated for 15 min at 40°C in HRP blocker and finally washed 2×10 min in WB and 2×10 min in PBST at RT. Embryos were then embedded in 1% agarose and imaged by confocal microscopy, ideally immediately and no more than 2 days after mounting. Since eGFP can maintain its fluorescence property after fixation, Pard3 RNAscope spots could be relatively easily superposed to the fluorescence of eGFP expressed in endothelial cells.

Images need to be analyzed with 3D-rendering (for example, with Imaris) to ensure the localization of RNAscope spots in a given cells.

Whole mount chromogenic in situ hybridization

For the sequence of all the primers used to amplify probes (relative to PDZ-RhoGEFs, **Figure 7 – figure supplement 2**), see the Key Resources Table. Whole-mount chromogenic in situ hybridization was performed as described in [Lancino et al. 2018](#) [\[1\]](#). Probes were synthesized using the manufacturer recommendation (T7 RNA polymerase, Promega, Cat# P2075, DIG-nucleotides, Jena Bioscience, Cat# NU-803-DIGX).

Images were captured with the Zeiss Axio ZOOM V16 microscope with the Zen Pro2 software, with a brightfield transmitted light optics. Post-processing steps were performed using the Extended-Depth Focus method to combine in focus regions from multiple z-planes and convert into in a transmitted light z-stack to generate a unique in-focus image.

Immunofluorescence detection of the HA-tagged dt-runx1 deletion mutant

Whole mount immunostaining was performed on hand-dechorionated 48 hpf *Tg(dt-runx1)* embryos to assess the localization of the 2XHA-tagged dt-runx1 protein in aortic cells. Briefly, embryos were fixed in 4% methanol free PFA (Polysciences Cat# 040181) for 3 hrs at room

temperature (RT) and washed in PBS/0.1% tween20 (PBT). Embryos were treated in successive baths of milliQ water supplemented with 0.1% tween20, then in cold acetone for 10 min at -20°C and again in milliQ water plus 0.1% tween20. They were then washed in 1X HBSS (Invitrogen Cat# 14025), and permeabilized for 45 min at RT in 1X HBSS, 0.1% tween20, 5mM CaCl₂ and 0.1mg/ml collagenase (Sigma-Aldrich Cat# C9891). Embryos were then rinsed in PBSDT and incubated in blocking solution 1× sheep serum (SS, Sigma Cat# S2263) for at least 4 hrs at RT. They were then incubated overnight at 4°C with primary antibodies in blocking solution: rabbit anti-GFP (MBL Cat# 598; 1/300) and mouse anti-HA (Sigma, Cat# 12ca5; 1/50). On the next day, embryos were washed several times in PBSDT and endogenous peroxidase activity was inactivated by treatment with 6% H₂O₂ (Sigma-Aldrich, Cat# H1009) for 1 hr at 4°C. Embryos were then incubated in the NGS for at least 4 hrs and incubated overnight at 4°C with secondary antibodies: goat anti-rabbit Alexa Fluor 488 (Invitrogen Cat# A11070; 1/400) and goat anti-mouse HRP-conjugated (Thermo Fisher Scientific Cat# G-21040; 1/300). Finally, embryos were rinsed several times first in PBSDT and after in PBT before HRP fluorescent revelation. Embryos were incubated for 45 min at RT in the dark in imidazole buffer (0.1 M imidazole (Sigma-Aldrich, Cat# I5513) in PBT supplemented with 1% H₂O₂) with Cy3 Tyramide Reagent (Cy3 NHS sigma Cat# PA13101, tyramide sigma Cat# T-2879, dimethyl formamide sigma Cat# T-8654 and triethylamine sigma Cat# T-0886). Final washes in PBT and in 6% H₂O₂ for POD inactivation were performed before embryos were mounted in low-melting agarose in 1X PBT for fluorescence confocal imaging.

Quantitative real-time PCR gene expression analysis

Preparation of samples

In a first line of experiments, expression of *pard3* family genes by qRT-PCR as well as of some hematopoietic markers *myb* and *runx1* was evaluated in our tissue of interest at 48 - 50hpf. Total RNA was obtained from FACS sorted vascular cells of control embryo using an incross of *Tg(Kdrl:gal4;UAS:RFP)* as well as non-vascular cells (as described below). Cells were collected from either whole embryos or from dissected trunks. Subsequently, to determine the impact of *runx1* interference on *pard3* family genes, total RNA was obtained from dissected trunk regions of 48-50hpf embryos from either incross of control *Tg(Kdrl:gal4;UAS:RFP)* or mutant *Tg(Kdrl:dt-runx1-eGFP)* fish lines. Analyses were carried on 3 biological replicates.

Cell dissociation and Fluorescent Activated Cell Sorting

Single cell suspension for FACS sorting was prepared using an optimized protocol adapted from multiple sources (Manoli and Driever 2012 [↗](#); Samsa et al. 2016 [↗](#); Bresciani, Broadbridge, and Liu 2018 [↗](#)). All steps were performed with cooled solutions (4°C), and samples were kept on ice throughout the processing to preserve the viability of the cells. Briefly, embryos were anesthetized using balneation in embryo medium supplemented with tricaine methanesulfonate (MS-222, Sigma-Aldrich ref #A5040), at a final concentration of 640 µg/ml. Embryos were placed into a petri dish, rinsed in PBS supplemented with tricaine, and a majority of the medium was removed to prevent movement. Embryos were cut using a needle, and were then pooled in tubes and placed on ice until cell dissociation. Embryos trunks or whole embryos (around 120 or 60 respectively) were washed twice in PBS before centrifugation (300 x g for 1 min @4°C) and the supernatant was discarded. Embryos were incubated in 1ml TrypLE medium (Life Technologies ref #12605-010) for 10 min on ice, with gentle pipetting every 3 mins, first with a P1000 pipette and subsequently with a P200 pipette. Samples were then centrifugated for 7 min at 300g to remove the TrypLE and the cells were resuspended in 500µl FACSmax medium (Genlantis ref#T200100). Ultimately, samples were passed through a 40µm cell strainer moistened with FACSmax buffer onto a 35 mm cell culture dish using a syringe plunger. The cell strainer and the dish were washed with 300µl of FACS max and the flow-through was transferred to a FACS tube and kept on ice until sorting.

Cell sorting was performed on a BD FACS AriaIII cell sorter. For all samples, gating was done on SSC-A vs. FSC-A to collect cells, and then on FSC-A vs. FSC-H to keep only single cells. For control embryos, RFP positive cells as well as RFP negative cells were collected. For dt-runx1 expressing mutant, all GFP positive cells (both RFP positive and RFP negatives) were collected, as well as RFP single positives cells (not used for qPCR) and double negative cells as internal controls. FACS sorted RFP single positives cells from dt-runx1 expressing mutants were not used as control for the study of gene expression for 2 reasons. First, due to the mosaicism of the line, very few cells were RFP single positive compared to the GFP positive cells. Second, RFP single-positive cells from dt-runx1 mutant embryos would have been indirectly exposed to runx1 interference (they would have been potentially in direct contact with their neighbouring endothelial cells that express the dt-runx1 mutant) and thus would not be a true no interference condition.

RNA extraction from FACS-sorted cells

Total RNA was extracted from sorted cell fractions using the RNeasy Qiagen kit (Qiagen, Cat # 74104). Briefly, cells were sorted directly into 100µl lysis buffer, and RNA extraction was done immediately according to the manufacturer's instructions. RNA was eluted twice with 15µl RNA free H₂O. Reverse transcription of mRNA was performed using SuperScriptIV (Invitrogen Cat# 18090010) with OligodT primer. cDNA samples were stored at -20°C until further processing (for quantitative real-time PCR).

Real-Time qPCR

For the sequence of all the designed qRT-PCR primers, see the Key Resources Table. Primer couples were individually tested to assess their specificity as well as their efficiency. qRT-PCR was performed using Takyon Rox SYBR 2x Master mix blue dTTP kit (Eurogentec Cat# UF-RSMT-B0701), with starting concentrations of template of around 10ng/µl and primer concentrations of 0.5µM, on a QuantStudio3 system (Applied Biosystems). Each reaction was performed in technical triplicates and in 3 biological replicates. Ct was determined automatically by the QUANTSTUDIO™ DESIGN & ANALYSIS 2 software on the Thermofisher cloud and exported to be analyzed manually. Standard deviation was calculated within the technical triplicates, and when superior to 0.3, obvious outliers were removed (Dixon's Q test). The delta-delta-Ct method (double normalization to endogenous control — zebrafish elongation factor ef1α — and to control sample) was used to compare gene expression in control and altered conditions. The delta-Ct method (normalization to endogenous control ef1 α) was used to investigate expression of genes of interest in whole embryo and trunk vascular cells versus non-vascular cells.

In vivo confocal imaging

Embryos, dechorionated manually, were anesthetized using tricaine (Sigma Aldrich, Cat# A5040). They were then embedded on the side position in 1% low melting agarose (Promega, Cat# V2111) in a glass bottom 60µ-Dish (35mm high; Ibidi, Cat# 81156). To avoid movements and pigmentation during image acquisitions, 1x tricaine /1x PTU Volvic water was incorporated to the low melting agarose and, after solidification, 1ml of 1x tricaine /1x PTU Volvic water was added before sealing the dish.

Embryos were imaged with 2 confocal microscope systems. For the results presented **Figure 1** [↗](#), z-stacks and time-lapse sequences were obtained using a Leica TCS SP8 inverted confocal microscope as described in [Lancino et al 2018](#) [↗](#).

For all the other confocal microscopy results, embryos were imaged using an Andor (Oxford Instruments) spinning disk confocal system (CSU-W1 Dual camera with 50µm Disk pattern and single laser input (445/488/561/642nm), LD Quad 405/488/561/640 and Tripl 445/561/640 dichroic mirrors), equipped with a Leica DMi8 fluorescence inverted microscope and CMOS cameras (Orca Flash 4.0 V2+ (Hamamatsu)). Imaging was performed using a 40x water immersion objective (HC

PL APO 40x/1.10 Water CORR CS2), a LEDs light source (CoolLED pE-4000 16 wavelength LED fluorescence system), and the acquisitions were piloted with the support of the MetaMorph software.

RNAscope *in situ* hybridization were imaged similarly to live sample, using a spinning disk confocal microscope (see *in vivo* confocal imaging paragraph above) after embedding the embryos in 1% low-melting agarose (Promega, Cat# V2111) in PBS 1X in a glass bottom 60μ- Dish (35mm high; Ibidi, Cat# 81156).

Morphological and morphometric analysis of aortic and hemogenic cells

For the morphometric analysis of the aorta and aortic cells, large z-stack of 48-55 hpf Tg(kdrl:eGFP-JAM3b; kdrl:nls-mKate2) or Tg(kdrl:eGFP-JAM2a; kdrl:nls-mKate2) encompassing the whole aortic depth were acquired with optimal z-resolution (0.3μm z-step). For each embryo, 3 contiguous z-stacks of 330μm width were acquired, allowing us to image the entirety of the AGM region, from the anterior to the posterior end of the elongated yolk.

Samples

For ArhGEF11 splicing morpholino experiment, 3 splicing morpholino injected embryos and 2 control non injected siblings were kept for the analysis.

For ArhGEF11 CRISPR Cter deletion and when performing the experiments, imaging was done, for putative mutants, on embryos that had shown a delay in initiation of circulation at 24 hpf, regained circulation at 48 hpf albeit with a clearly visible oedema in the cardiac region (non-altered embryos were kept as sibling controls). All imaged embryos whose signal allowed us to generate segmentable 2D-cartography from the whole length of the aorta were kept and processed further for genotyping. Genotyping confirmed that the circulation problems at 24 hpf and the remaining oedema at 48 hpf corresponded to embryos bearing the mutation.

For CRISPR mutant phenotype analyses, 2 mutant embryos and 2 wild type siblings were kept, after genotyping.

Image analysis

The following image analysis was performed on Icy software.

For each segment, we generated a 2D-cartography of the eGFP-Jam signals at intercellular contacts of aortic cells, using the Tubeskinner plugin, according to the protocol described in [Lancino et al 2018](#) [\[4\]](#). Briefly, the aorta was rotated to face the lumen, a circular ROI fitting the interior of the aorta was initialized by the experimenter on the first plane, and for each plane sequentially, the algorithm will more accurately fit the ROI to the aortic perimeter and collect the fluorescence intensity all along its longitudinal axis. The collected intensity was then projected on a 2D-plane, where the Y-axis of the 2D-cartography corresponds to the perimeter of the aorta, and the X-axis corresponds to the X-axis of the original z-stack.

Before further processing, a metadata table recapitulating the dates and conditions of experiments for each embryo was generated, and a random label was assigned to rename each acquisition. This allowed for the later part of the analysis, that involves semi-automated segmentation and classification by the user, to be blind and avoid potential bias.

Each 2D-cartography was then manually pre-segmented using the ROI Polygon tool to draw the contour of each aortic cell, using the original 3D z-stack as reference. Due to technical limitation (decreased fluorescence signal collection on the farthest lateral side of the aorta), only about 2/3 of

the 2D-cartographies could be segmented (comprising the floor, the right lateral side of the aorta and the roof). The manual segmentation was then improved using the “Active Contour” plugin, that improved the fit of the cellular contour to the fluorescence signal and slightly dilated the cell contours until the ROIs formed a joint paving. All morphometric descriptors were extracted for each cell. The number of neighbours was automatically generated for each cell by counting the number of touching adjacent ROIs.

Additionally, cells were manually classified into cell types according to morphometric parameters using as reference the original 3D z-stack and in particular the nls-mKate2 nuclear signal. EHT-undergoing cells were positioned on the floor of the aorta, have reduce antero-posterior axis length, luminal membrane inward bending for EHT pol+ cells and luminal membrane outward bending for EHT pol- cells, and a thickened nucleus. Hemogenic cells whose criteria were established in [Lancino et al 2018](#) (elongated in the antero-posterior axis, with lateral membranes not trespassing the equatorial plane of the aortic wall), are positioned on the floor of the aorta, systematically have a reduced width perpendicular to the blood flow and, importantly, a thickened nucleus that protrudes out of the aortic plane. Finally, endothelial cells have a flattened nucleus positioned high on the lateral side or on the roof of the aorta.

Measurement of the aortic perimeter

An accurate measurement of the aortic perimeter was obtained using the z-stack generated for the aortic cells’ morphometric analysis. Briefly, the z-stack was rotated to face the lumen of the aorta and a ROI fitting the aortic perimeter was drawn by the user and optimized by the “Cell Contour” plugin. The length of the perimeter was then extracted from the ROI statistics.

Fluorescence Recovery After Photobleaching (FRAP) measurements and analysis

Measurement of junctional turn-over was performed on 48-55 hpf *Tg(kdrl:eGFP-JAM3b; kdrl:nls-mKate2)* embryos using a Nikon Ti2e spinning disk microscope equipped with a sCMOS camera (Photometrics, Prime 95B, pixel size 11µm, 1200×1200 pixels, QE 95%) and a 40x water objective (Numerical Aperture 1.15, Working Distance 0.6mm, xy pixel size of 0.27µm). Embryos were mounted as previously described.

Samples

For ArhGEF11 splicing morpholino phenotypic analysis, 10 different embryos (injected with the splicing morpholino) obtained from 3 separate experiments and 30 control embryos (non-injected siblings or control morpholino injected siblings) obtained from 6 separate experiments were analyzed.

For EHT analysis, 16 embryos over 4 separate experiments were analyzed.

For ArhGEF11 CRISPR/Cas9 Cter deletion phenotypic analysis, 8 mutant embryos and 12 wild type siblings over 3 separate experiments were analyzed.

Overall, the exact number of interfaces analyzed is written on each figure panel.

Acquisitions

For each acquisition, a wide two channels z-stack of the aorta was generated to capture the whole aortic volume and generate a 2D-cartography of the aortic environment (Icy, plugin Tubeskinner). This 2D-cartography of the cellular contours and the z-stack with the cellular contours and nuclei

were used jointly to identify un-ambiguously cellular and junctional types and define one or multiple single point regions of interest (ROIs) for subsequent bleaching and fluorescence recovery measurement.

The sequence of acquisitions before/after bleach focused only on the eGFP-Jam3b signal to speed up the acquisition and improve the temporal resolution. The 5 steps of the acquisition can be recapitulated as follows and we obtained/performed: (1) a single z-stack of 20μm centered on the plane of focus, (2) a 30 sec time-lapse acquisition on the plane of focus to measure average pre-bleach intensity with an image every second, (3) a high intensity stimulation of the ROIs (25% 300ms 488 FRAP laser exposure, leading to the bleaching of the Jam3b-eGFP signal), (4) a 3 min time-lapse acquisition on the plane of focus to measure fast post-bleach recovery with an image every second, and (5) a 7 min time lapse acquisition of 8μm z-stack centered around focus plane with a 20 sec time interval to measure slow post-bleach recovery. Each step was save separately as .nd2 files.

Signal quantification and recovery analysis

File were converted from .nd2 to .tiff using FIJI batch converter, scale factor 1.00, bilinear interpolation, no average when downsizing, images read using bioformat.

To collect fluorescence intensity, ROIs for each stimulation point was manually drawn using Fiji and named with an individual label using Fiji ROI manager. A metadata table was manually curated with all information for each ROI (junction type, condition, date of experiment, embryo number). An additional ROI was created to collect background intensity for subsequent corrections. Acquisitions with clear embryo movement or drifting were either completely removed from the dataset (~5% of the acquisitions) or only kept for early recovery analysis (see below) if the drift only occurred in the last step (fifth) of the acquisition.

Fluorescence intensity for each ROI was collected and saved automatically as a .csv file for each acquisition using a homemade Fiji macro.

All further analysis were performed using R for automatization.

Briefly, for each ROI: the fluorescent signal was normalized to the background signal ($I_{\text{background}}$ normalized = $I_{\text{roi}} - I_{\text{background}}$), then normalized to mean intensity pre-bleach ($I_{\text{normalized}} = I_{\text{background normalized ROI post-bleach}} / I_{\text{background normalized ROI pre-bleach}}$), and finally scaled ($I_{\text{full scaled normalized}} = (I_{\text{normalized}} - \min(I_{\text{normalized}})) / (1 - \min(I_{\text{normalized}}))$).

A single exponential equation was used to fit the data using the nls:multistart package for parameter optimization:

$$F(t) = A \cdot (1 - e^{-t/\tau})$$

With A the maximum amplitude of the curve, t the time, and τ a constant defining the growth rate of the curve.

The fitted curve allowed to extract quantitative information describing the recovery of each type of junction in different conditions: the mobile fraction of the bleached pool (corresponding to A, the amplitude of recovery) and the time of half recovery, related to the speed of recovery, that can be extracted from τ :

$$T_{\text{half-recovery}} = \ln(0.5) / (-\tau)$$

A more precise description of the recovery speed was extracted from the observation of the early time period (until 30s post bleach), where the fluorescence intensity increases uniformly. A first-degree polynomial equation was fitted to the data (using `lm()` function from the stats package), and the coefficient corresponding to the slope of the fitted curve was used as a metric for speed of fluorescence recovery.

RNAscope image analysis

Myb

Phenotypic analysis of the impact of the *dt-runx1* mutant on hematopoiesis was evaluated by investigating the *in situ* expression of *myb*, using the single-molecule *in situ* hybridization RNA-scope method, performed in control *Tg(Kdrl:eGFP)* and *dt-runx1* mutant embryos *Tg(Kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)* (analysis was performed on 4 and 3 biological replicates (embryos) for control and mutants respectively). For each embryo, 2 consecutive aortic segments were analyzed (each spanning over 5-6 intersegmentary vessels). It is roughly equivalent to the length to 3 consecutive aortic segments that we use for our live analysis (each spanning over 3-4 intersegmentary vessel). Discrepancies between live and fixed samples are probably due to the morphological alterations caused by the fixation and permeabilization steps of the RNAscope protocol. 3D-rendering of the aorta and the sub-aortic space were generated using the Imaris Software (Oxford Instruments, version 10.1.0). Both cell volume (GFP signal) and RNAscope spots were simultaneously and automatically segmented using the “Cells” tool from Imaris, that segment individual cells (generates an outer surface) as well as RNA-scope signal within cells. Cells were subsequently manually classified, based on their localization and the shape of the cell. Roof endothelial cells are located dorsally to the aortic floor, and are flat and elongated. Due to morphological alteration of the tissues and the impossibility to discriminate EHT pol⁻, EHT pol⁺ cells and the uncharacterized hemogenic cells we observed *in vivo*, our hemogenic / EHT cells category comprises both characterized hemogenic cells (elongated cells with thick nuclei, located on the aortic floor), uncharacterized hemogenic cells and putative EHT-undergoing cells. Sub-aortic cells were classified based on their localization, and they comprise a mix of endothelial cells from the underlying posterior cardinal vein (visible only in the control embryos and not in the *dt-runx1* mutant embryos, in which arterial structures are more specifically labelled compared to venous structures) as well as newly generated hematopoietic cells. Finally, the “Statistics” tool from Imaris allowed us to extract cell and RNA spots count information. The data plotting and statistical analysis was performed using R software.

Pard3

Pard3 isoforms (*Pard3ba*, *Pard3ab*, *Pard3aa*) *in situ* expression analysis was done similarly to that of *myb*, except that a second RNAscope signal segmentation was performed independently of the “Cells” segmentation, using the “spots” tool. This allowed to capture the majority of the *Pard3* signal, that is expressed in aortic cells, but with a discrepancy in the intra-cellular localization of RNA transcript, that could not be detected when restricting the segmentation of RNAscope spots to spots located inside the GFP signal. After RNAscope segmentation using the “Spots” tool, manual curation allowed us to select specifically RNAscope spots located in aortic tissues for subsequent analysis. The “Statistics” and “object-object Statistics” tools were used to extract relevant spatial and count information for subsequent analysis. A key information that we used for the spatial analysis was the average distance of *Pard3ba* spots to Hemogenic / EHT cells, as well as the cumulative number of spots within a distance from Hemogenic / EHT cells, to investigate spatial correlation between Hemogenic / EHT cells and *Pard3ba* expression. Analysis was performed in control *Tg(Kdrl:eGFP)* and *dt-runx1* mutant embryos *Tg(Kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)* (5 and 6 biological replicates (embryos) for control and mutants respectively, with 2 consecutive aortic segments analyzed per embryo).

Dt-runx1 phenotype analysis – cell count

For the live dt-runx1 phenotype analysis, large z-stack of 48-55 hpf dt-runx1 expressing mutant (eGFP positive) and siblings control (eGFP negative) *Tg(Kdrl:Gal4;UAS:RFP;4xNR:dt-runx1-eGFP)* embryos, encompassing the whole aortic depth were acquired with optimal z-resolution (0.3µm z-step). 3 and 7 biological replicates (embryos) for control and mutants respectively were analyzed. For each embryo, 3 contiguous z-stacks of 330µm width were acquired, allowing us to image the entirety of the AGM region. The resulting phenotype was assessed by manually counting EHT pol+ and pol- cells (taking into account their position, ventral or lateral) as well as hemogenic cells. Classification was done manually, based on morphological criterias. EHT pol+ cells presented the typical EHT pol+ morphology, with clear luminal membrane invagination, that could be observed even in laterally emerging cells. EHT pol- cells displayed a clear rounded ovoid, with a clear shortening of the antero-posterior axis. Hemogenic cells correspond to endothelial cells located on the aortic floor, that are elongated along the antero-posterior axis, have a narrowed left-right axis and a thickened nucleus. Non characterized hemogenic cells were also included in this category: they correspond to cells that had potentially already advanced in the emerging process but whose morphology did not fit any of the typical EHT pol- and pol+ characteristics (no clear invagination nor clear concave protrusion in the aortic lumen).

Thymus phenotype analysis

To question one aspect of the hematopoietic phenotype of our dt-runx1 mutant, dt-runx1 expressing mutants (GFP+ RFP-) as well as control siblings (RFP+ GFP-) were imaged at 5dpf to analyze their developing thymus *in vivo* (analysis was performed on 5 mutant and 5 control larvae). Subsequent image analysis was performed using the Imaris software (Oxford Instruments). Individual thymic cells were automatically segmented using the “Surface” tool. Cell count per thymus as well as morphometric information for each cell (volume) was automatically retrieve using the Imaris “Statistics” tool, allowing us to compare the distribution of cell volume in mutant versus control embryos, as well as the total cellular volume of the thymus (sum of individual cell volumes) for each larva.

Impact of ArhGEF11 exon38 splicing morpholino on hematopoiesis

After morpholino injection at the one cell stage, double transgenic *Tg(CD41:eGFP; kdrl:nls-mKate2)* embryos were raised until 48hpf and imaged using spinning disk confocal microscopy. The integrality of the dorsal aorta was imaged (corresponding to 3 contiguous segments), on 8 and 6 biological replicates (embryos) for controls and morphants respectively. Quantification of the number of hemogenic cells (double CD41:eGFP and kdrl:nls-mKate2 positive cells) as well as of hematopoietic cells (simple positive CD41:eGFP and kdrl:nls-mKate2 negative cells) was performed using a in house Fiji/R script. For the segmentation, only the eGFP channel is handled. An issue with the *Tg(CD41:eGFP)* background is the presence of circulating thrombocytes, that move faster than the speed of acquisition and thus generate multiple artefactual 2D objects (only presents in single Z-planes, compared to non-moving cells whose depth spans over 15-20 z-sections). These artefactual objects are removed during the pre-processing steps, by using a Faster Temporal Median algorithm (Jabermoradi et al. 2022 [\[1\]](#)). This algorithm, by calculating median value over rolling windows of 20 frames approximate cell depth) and subtracting it to the pixel values, removes the 2D outliers. The rest of the pre-processing steps consists of fluorescent signal filters to ease the subsequent segmentation (contrast enhancement, background subtraction, 3D gaussian blur). The cells are segmented using a TopHat algorithm and touching objects are separated using a 3D distance Watershed (Legland, Arganda-Carreras, and Andrey 2016 [\[2\]](#)). Once the objects are segmented, the total fluorescence intensity in both channels as well as the background fluorescence intensities are collected using the raw images (without any signal filters). The data are subsequently processed using R for plotting, threshold selection for classification as well as statistical comparisons.

Statistical analysis

Statistical tests used are described for each figure panel in the legend. Since our samples are usually small, the test used is an unpaired two samples non-parametrical Wilcoxon mean comparison test, without a priori knowledge on sample distribution or homoscedasticity. In the context of FRAP experiments (relative to [Figure 6](#) and [Figure 8](#)), for which we had larger sample size, we still decided not to assume that our data were normally distributed, because of the inherent variability of the biological events that are observed. Indeed, the EHT process is heterogeneous in time, with successive phases of contraction and stabilization, during which we can imagine that the dynamics of junctional recycling could vary. The distribution of speed and amplitude of recovery could thus correspond to a normal distribution, a bi- or multi-modal distribution, or a uniform distribution. In this case, the use of non-parametrical tests and the large variability we observed in the data reduces the power of statistical tests and our ability to confidently detect significant differences. In some cases, when our p-values were larger than the arbitrary threshold of 0.05, we refer to tendencies whose significance is legitimated by the overall consistency of the data (see for example between the results presented [Figure 7](#) and [Figure 8](#)).

Correlation analysis (relative to [Figure 4F](#)) was performed by calculating Pearson correlation coefficient and associated p-value (Student T-test) using R.

Data and software availability

All graphics and plots were generated using RStudio and packages including: [ggplot2](#) ([Wickham 2016](#)), [readr](#) ([Wickham, Hester, and Bryan 2022](#)), [dplyr](#) ([Wickham et al. 2023](#)), [stringr](#) ([Wickham 2022](#)), [ggstatsplot](#) ([Patil 2022](#)), [ggpubr](#) ([Kassambara 2020](#)), [wesanderson](#) ([Ram and Wickham 2018](#)), [cowplot](#) ([Wilke 2020](#)), [ggsci](#) ([Xiao 2018](#)), [ggbeeswarm](#) ([Clarke and Sherrill-Mix 2017](#)), [viridis](#) ([Garnier 2021](#)), [DT](#) ([Xie, Cheng, and Tan 2022](#)), [SciViews](#) ([Grosjean 2019](#)), [nls.multstart](#) ([Padfield and Matheson 2020](#)), [matrixStats](#) ([Bengtsson 2022](#)).

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Authors contributions

LT designed, performed and analyzed experiments, and wrote the Method section and the Figure legends. SM designed and conducted experiments. CV conducted experiments. AS conceived the project, designed, performed and analyzed experiments, wrote the manuscript with input from LT, and obtained funding.

Declaration of interests

The authors declare no competing interests.

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Reviewer #2 (Public Review):

In this study, Torcq and colleagues make careful observations of the cellular morphology of haemogenic endothelium undergoing endothelial to haematopoietic transition (EHT) to become stem cells, using the zebrafish model. To achieve this, they used an extensive array of transgenic lines driving fluorescent markers, markers of apico-basal polarity (podocalyxin-FP fusions) or tight junction markers (jamb-FP fusions). The use of the runx truncation to block native Runx1 only in endothelial cells is an elegant tool to achieve something akin to tissue-specific deletion of Runx1. Overall, the imaging data is of excellent quality. They demonstrate that differences in apico-basal polarity are strongly associated with different cellular morphologies of cells undergoing EHT from HE (EHT pol- and EHT pol+) which raises the exciting possibility that these morphological differences reflect heterogeneity of HE (and potentially HSCs, but this is not addressed in this manuscript) at a very early stage. They then overexpress a truncated form of Runx1 (just the runt domain) to block Runx1 function and show that more HE cells abort EHT and remain associated with the embryonic dorsal aorta. The revised version identifies pard3ab as differentially distributed in dtRunx mutants and correlates that distribution with a potential regulatory role on cell polarity. No direct evidence for their role in EHT is presented.

The manuscript has now been streamlined and reference to figures made much clearer. It provides for a clearer reading, and clearly a well thought out discussion of HE, polarity and the regulation of the EHT process. The evidence for the different cellular morphologies of cells undergoing EHT is strong, and the main claim that tuning apico-basal polarity and junctional recycling underlie morphological complexity of EHT (rather than of HSCs) is well supported by the data.

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Author response:

The following is the authors' response to the original reviews.

We wish to thank the Reviewers for their critical analysis of the article and for their suggestions and comments.

In addition and beside the point-by-point answer to the Reviewers, we wish here to emphasize on three essential points that have been raised: First, we never intended (nor pretended) to address the incidence of the two EHT cell emergence processes on downstream fate, after release from the aortic floor (see for example the last paragraph of our initially submitted manuscript). We only wished to bring evidence on cell biological heterogeneity of the HE, particularly relying on cell polarity control and polarity

Public Reviews:

We wish to thank the Reviewers for their critical analysis of the article and for their suggestions and comments.

In addition and beside the point-by-point answer to the Reviewers, we wish here to emphasize on three essential points that have been raised: First, we never intended (nor pretended) to address the incidence of the two EHT cell emergence processes on downstream fate, after release from the aortic floor (see for example the last paragraph of our initially submitted manuscript). We only wished to bring evidence on cell biological heterogeneity of the HE, particularly relying on cell polarity control and polarity reestablishment/reinforcement in the case of EHT pol⁺ cells, thus leading to emergence morphodynamic complexity. In the general context of cell extrusion in which all polarity features are generally downregulated, these are remarkable features.

Second, we inform the Reviewers that we have performed a major revision of the work on the Pard3 proteins issue the outcome of which, hopefully, substantiates significantly the idea of a tuning of cell polarity features in the HE and all along the EHT time-window, for supporting EHT pol⁻ and EHT pol⁺ types of emergence. To achieve this, we entirely revised the experimental strategy to increase specificity and sensitivity of detection of Pard3 protein isoforms expressed in the vascular system, based on endothelial FACS-sorting, qRT-PCR and single-molecule whole mount in situ hybridization using RNAscope. Importantly, we wish to stress that, by addressing Pard3 proteins, we initially aimed at substantiating our observations on the localization of our podxl2 construct (del-podxl2) used to label apical membranes. Hence, we sought to bring correlative evidence on the variation of expression of polarity proteins at early and later time points of the EHT time-window (suggesting tightly regulated expression control of polarity determinants, possibly at the mRNA level). This was clearly written and justified in the text, lines 227 or 303 of the initial manuscript. Also, this may have led to identify (a) specific isoform(s), including splicing variants as initially addressed.

As the Reviewers will see, while performing the revision of our work, we now have been able to point at a specific isoform of Pard3, namely Pard3ba, whose mRNA expression level, in aortic cells and at the single cell resolution, is uniquely and specifically enhanced in cells contacting emergence ‘hot spots’. Using our Runx1 mutant fish line (dt-Runx1), we also show that expression of Pard3ba mRNAs, in these specific aortic regions, is sensitive to interference with Runx1 activity (i.e dt-Runx1 increases Pard3ba expression). Altogether, our new results strongly support our idea, initially proposed, on the regulation of polarity features during EHT; they indicate intercellular coordination, throughout cooperative cross-talk between aortic and HE/EHT cells. This is compatible with the idea of a ‘tuning’ of apico-basal polarity during the entire EHT time-window (including maturation of the HE to become competent for emergence and the emergence process per se whose morphodynamic complexity relies on regulating apico-basal polarity associated functions (ex: for controlling the specific junctional recycling modes of EHT pol⁺ and EHT pol⁻ cells, as we suggest using JAM proteins that we have chosen owing to their function in the recruitment of Pard3 proteins for apico-basal polarity establishment)). This complements nicely our work and highlights the relevance of studying the interplay between aortic and HE/EHT cells (which we have started to dissect in the second part of our manuscript). Further work is obviously required to address local, dynamic variations of mRNAs encoding for this specific isoform of Pard3 as well as specific interference with its functions at the spatial and temporal levels (hence on live tissues), which is far beyond the scope of our currently submitted work.

Finally, this emphasizes the importance of the aortic context, at the mesoscopic level, in the regulation of the EHT.

Third, based on these major points and Reviewers suggestions, we propose to take into account the fact that the heterogeneity in emergence morphodynamics was not highlighted and propose the following title:

‘Tuning apicobasal polarity and junctional recycling in the hemogenic endothelium orchestrates the morphodynamic complexity of emerging pre-hematopoietic stem cells’

Regarding Results and Figures, the previous Figures 3 and 4 have been entirely revised, with the support of Supplement Figures (3 and 4 supplement figures, respectively as well as a supplement video to Figure 3). Supplement Figures have also been included to the revised version, for nearly all results that appeared as data not shown (Figure 1 – figure supplement 2: illustrating the maintenance of EHT pol+ and EHT pol- cells after division; Figure 1 – figure supplement 3: illustrating the expression of the hematopoietic marker CD41 by EHT pol+ and EHT pol- cells). Also, a new supplemental figure, Figure 7 – figure supplement 7, has been added to substantiate the impact of interfering with ArhGEF11/PDZ-RhoGEF alternative splicing on hematopoiesis. Finally, a Figure for the Reviewers is added at the end of this file that shows that virtually 100% of aortic floor cells that we consider as hemogenic cells are positive for the hematopoietic marker Gata2b which is upstream of Runx1 (using RNAscope which allows achieving cellular resolution unambiguously).

Reviewer #1 (Public Review):

Summary:

In this research article, the authors utilized the zebrafish embryo to explore the idea that two different cell types emerge with different morphodynamics from the floor of the dorsal aorta based on their apicobasal polarity establishment. The hypothesis that the apical-luminal polarity of the membrane could be maintained after EHT and confer different functionality to the cell is exciting, however, this could not be established. There is a general lack of data supporting several of the main statements and conclusions. In addition, the manuscript is difficult to follow and needs refinement. We present below some questions and suggestions with the goal of guiding the authors to improve the manuscript and solidify their findings.

Here, we wish to emphasize that we do not make the hypothesis that ‘...the apical-luminal polarity of the membrane could be maintained after EHT ...’ but that the apico-basal polarity establishment/maintenance controls the type of emergence and their associated cell biological features (EHT pol+ and EHT pol- cellular morphodynamics, establishment of membrane domains). Hence, our work suggests that these emergence modes, as a consequence of their intrinsic characteristics and differences, might have an impact on cellular behavior after the release (to place the work in the broader context of hematopoietic cell fate and differentiation). More specifically, the difference in the biological features of the luminal versus abluminal membrane for the two EHT types (ex: membrane signaling territories, membrane pools devoted to specific functions), might endow the cells with specific functional properties, after the release. What happens to those cells thereafter, except for illustrating the evolution of the luminal membrane for pol+ EHT cells, is beyond the scope of this paper. Here, we analyze and characterize some of the cell biological features of the EHT process per se (the emergence from the aortic floor), including the dynamic interface with adjoining endothelial cells.

Strengths:

New transgenic zebrafish lines developed. Challenging imaging.

Weaknesses:

(1) The authors conclude that the truncated version of Podxl2 fused to a fluorophore is enriched within the apical site of the cell. However, based on the images provided, an alternative interpretation is that the portion of the membrane within the apical side is less stretched than in the luminal side, and therefore the fluorophore is more concentrated and easier to identify by confocal. This alternative interpretation is also supported by data presented later in the paper where the authors demonstrate that the early HE is not polarized (membranes are not under tension and stretched yet). Could the authors confirm their interpretation with a different technique/marker like TEM?

The argument of the apparent enrichment, or exclusion, of a marker depending on membrane stretching (and hence molecular packing) would be valid for any type of molecule embedded in these membranes, including of course endogenous ones (this is one of the general biophysical principles leading to the establishment of membrane domains, structurally and functionally speaking); hence, using another marker would not solve the issue because it would depend on its behavior in regard to packing (in particular lipid packing), which is difficult to anticipate and is a topic in its own (especially in this system that has been poorly investigated in regard to its biophysical and biochemical properties in vivo (including its exposure to the hemodynamics)).

If we follow the logic of the Reviewer, it appears that it is not consistent with our results on the maturing HE. Indeed, in our dt-Runx1 mutants, mKate2-podxl2 is enriched at the luminal membrane of HE cells (HE cells are elongated, and the two membrane domains have a relative equal surface and bending); in comparison, HE cells have the same morphology in control animals than in mutants but, in controls, eGFP-podxl2 and mKate2-podxl2 are equally partitioned between the luminal and abluminal membranes (see Figure 3 – figure supplement 2 (for mKate2-podxl2) and Figure 2 – figure supplement 1 and 2 (for eGFP-podxl2)). In addition, we took care while designing the eGFP and mKate2 fusions to keep the natural podxl2 sequence containing critical cysteine residues to maintain assembly properties and distance from the transmembrane segment (hence the fluorescent protein per se is not directly exposed to membrane stretching).

Finally, electron microscopy is not the approach to use for this issue because requiring tissue fixation which is always at risk because modifying significantly membrane properties. On this line, when we fix embryos (and hence membranes, see our new Figure 4 and its Supplemental Figures), we do not appear to maintain obvious EHT pol+ and pol- cell shapes. In addition, to be conclusive, the work would require not TEM but immuno-EM to be able to visualize the marker(s), which is another challenge with this system.

(2) Could the authors confirm that the engulfed membranes are vacuoles as they claimed, using, for example, TEM? Why is it concluded that "these vacuoles appear to emanate from the abluminal membrane (facing the sub-aortic space) and not from the lumen?" This is not clear from the data presented.

The same argument regarding electron microscopy mentioned on the point before is valid here (in addition, it would require serial sectioning in the case it would be technically feasible to make sure not to miss the very tinny connection that may only suggest ultimate narrowing down of the facing adjacent bilayers, which is quite challenging). The term vacuole which we use with caution (in fact, more often, we use the term pseudo-vacuoles in the initial manuscript, lines 140, 146, 1467 (legend to Figure 1 – figure supplemental 1 or apparent vacuole-like in the same legend lines 1465 and 1476) is legitimate here because we cannot say that they are portions of the invaginated luminal membrane as we could be accused not to show that these membranes are still connected to the luminal surface; we are here at the limit of the resolution that in vivo imaging is allowing for the moment with this

system, and we drive the attention of the Reviewer on the fact that we are reaching here a sub-cellular level which is already a challenge by itself.

In addition, if there would not be at some point vacuoles (or pseudo-vacuoles) formed in this system (membrane-bounded organelles), it would be difficult to conceive how, after release of the cell, the fluid inherited from the articular lumen would efficiently be chased from these membranes/organelles (see also our model Figure 1 – figure Supplement 1B).

Why is it concluded that "these vacuoles appear to emanate from the abluminal membrane (facing the sub-aortic space) and not from the lumen?" This is not clear from the data presented.

This is not referring to our data but to the Sato et al 2023 work. For EHT undergoing cells leading to aortic clusters in mammals and avians, vacuolar structures indeed appear to emanate from the ab-luminal side facing the sub-aortic space (we cannot call it basal because we do not know the polarity status of these cells). In the Revised version of the manuscript, we have moved this paragraph referring to the Sato et al work to the Discussion, which gives the possibility to expand a bit on this issue, for more clarity (see the second paragraph of our new Discussion).

(3) It is unclear why the authors conclude that "their dynamics appears to depend on the activity of aquaporins and it is very possible that aquaporins are active in zebrafish too, although rather in EHT cells late in their emergence and/or in post-EHT cells, for water chase and vacuolar regression as proposed in our model (Figure 1 - figure supplement 1B)." In our opinion, these figures do not confirm this statement.

This part of the text has been upgraded and moved to the Discussion (see our answer to point 2), to take Reviewers concern about clarity of the Results text section and allowing elaborating a bit more on this issue. We only wished to drive the attention on the described presence of intracellular vacuolar structures recently addressed in the Sato et al 2023 paper showing EHT cell vacuoles that are proposed to contribute to cellular deformation during the emergence. We take this example to rationalize the regression of the vacuolar structures described Figure 1 - figure supplement 1B, which is why we have written ‘... it is very possible that aquaporins are active in zebrafish too’; the first part of the sentence refers to the Sato et al 2023 paper.

(4) Could the authors prove and show data for their conclusions "We observed that both EHT pol+ and EHT pol- cells divide during the emergence"; "both EHT pol+ and EHT pol- cells express reporters driven by the hematopoietic marker CD41 (data not shown), which indicates that they are both endowed with hematopoietic potential"; and "the full recovery of their respective morphodynamic characteristics (not shown)?"

To the new version of our manuscript, we have added new Supplemental information to Figure 1 (two new Supplemental Figures):

- Figure 1 - figure Supplement 2 that illustrates that both EHT pol+ and EHT pol- cells divide during the emergence as well as the maintenance of morphology for both EHT cell types. We wish also to add here that the maintenance of the EHT pol+ morphology is the most critical point, showing that dividing cells in this system do not necessarily lead to EHT pol- cells.
- Figure 1 - figure Supplement 3 that shows that both EHT cell types express CD41.

(5) *The authors do not demonstrate the conclusion traced from Fig. 2B. Is there a fusion of the vacuoles to the apical side in the EHT pol+ cells? Do the cells inheriting less vacuoles result in pol- EHT? It looks like the legend for Fig. 2-fig supp is missing.*

As said previously, showing fusion here is not technically possible, but indeed, this is the idea, which fits with the images corresponding to timing points 0-90 minutes (Figure 2A), showing (in particular for the right cell) a large pseudo-vacuole whose membrane is heavily enriched with the polarity marker podxl2 (based on fluorescence signal in a membrane-bounded organelle that, based on its curvature radius, should be more under tension than the more convoluted EHT pol+ cell luminal membrane). Also, EHT pol – cells may be born from HE cells that either inherit from less intracellular vesicles after division (or that are derived from HE cells that are less – or not - exposed to polarity-dependent signaling (see our data presented in the new Figure 4 and the new version of the Discussion (see paragraphs ‘Characteristics of the HE and complexity of pre-hematopoietic stem cell emergence’ and ‘Spatially restricted control of Pard3ba mRNAs by Runx1’).

Finally, the cartoon Figure 2B is a hypothetical model, consistent with our data, and that is meant to help the reader to understand the idea extrapolated from images that may not be so easy to interpret for people not working on this system. In legend of Figure 2 that describes this issue in the first version of our manuscript (lines 1241-1243), we were cautious and wrote, in parentheses: ‘note that exocytosis of the large vacuolar structure may have contributed to increase the surface of the apical/luminal membrane (the green asterisk labels the lumen of the EHT pol + cell’.

The legend to Figure 2 – figure supplement 1 is not missing (see lines 1492 – 1499 of the first manuscript). The images of this supplement are not extracted from a time-lapse sequence and show that as early as 30hpf (shortly after the beginning of the EHT time-window – around 28hpf), cells on the aortic floor already exhibit podxl2-containing pseudo-vacuolar structures (which we propose is a prerequisite for HE cell maturation into EHT competent cells; see also Figure 2 – figure supplement 2).

(6) *The title of the paper "Tuning apico-basal polarity and junctional recycling in the hemogenic endothelium orchestrates pre-hematopoietic stem cell emergence complexity" could be interpreted as functional heterogeneity within the HSCs, which is not demonstrated in this work. A more conservative title denoting that there are two types of EHT from the DA could avoid misinterpretations and be more appropriate.*

There was no ambiguity, throughout our initial manuscript, on what we meant when using the word ‘emergence’; it refers only to the extrusion process from the aortic floor.

Reducing our title only to the 2 types of EHT cells would be very reductionist in regard to our work that also addresses essential aspects of the interplay between hemogenic cells, cells undergoing extrusion (EHT pol+ and pol- cells), and their endothelial neighbors (not to mention what we show in terms of the cell biology for the maturing HE and the regulation of its interface with endothelial cells (evidence for vesicular trafficking, specific regulation of HE-endothelial cell intercalation required for EHT progression etc ...). However, and to take this specific comment into account, we propose a slightly changed title saying that there are emergences differentially characterized by their morphodynamic characteristics:

‘Tuning apicobasal polarity and junctional recycling in the hemogenic endothelium orchestrates the morphodynamic complexity of emerging pre-hematopoietic stem cells’

(7) *There are several conclusions not supported by data: "Finally, we have estimated that the ratio between EHT pol+ and EHT pol- cells is of approximately 2/1". "We observed that*

both EHT pol⁺ and EHT pol⁻ cells divide during the emergence and remain with their respective morphological characteristics". "We also observed that both EHT pol⁺ and EHT pol⁻ cells express reporters driven by the hematopoietic marker CD41 (data not shown), which indicates that they are both endowed with hematopoietic potential." These conclusions are key in the paper, and therefore they should be supported by data.

Most of the requests of the Reviewer in this point have already been asked in point 4 and were added to the revised version.

Regarding the EHT pol⁺/pol⁻ ratio, we will keep the ratio to approximately 2/1. The Reviewer should be aware that quantification of EHT cells is a tricky issue and a source of important variability, as can be assessed by the quantifications that we have been performing (see for example figures in which we compare the dt-Runx1 phenotype with Ctrl). This is inherent to this system, more specifically because the EHT process is asynchronous, ranging from approx. 28 hpf to 3 days post fertilization (we have even observed EHT at 5 dpf). We systematically observed heterogeneity in EHT numbers and EHT types between animals and also between experiments (some days we observe EHTs at 48 hpf, others more around 55 hpf or even later). In addition, emergence also proceeds on the lateral side of the aorta and, while it is relatively easy to identify EHT pol⁺ cells because of their highly characterized morphology, it is more difficult for EHT pol⁻ cells that can be mistaken to round HE cells preparing for division. In the current revision of our work, we provide additional facts and potential explanations on the mechanisms that control this asynchrony and the apparent stochasticity of the EHT process (see results of new Figures 3 and 4).

Reviewer #2 (Public Review):

In this study, Torcq and colleagues make careful observations of the cellular morphology of haemogenic endothelium undergoing endothelial to haematopoietic transition (EHT) to become stem cells, using the zebrafish model. To achieve this, they used an extensive array of transgenic lines driving fluorescent markers, markers of apico-basal polarity (podocalyxin-FP fusions), or tight junction markers (jamb-FP fusions). The use of the runx truncation to block native Runx1 only in endothelial cells is an elegant tool to achieve something akin to tissuespecific deletion of Runx1. Overall, the imaging data is of excellent quality. They demonstrate that differences in apico-basal polarity are strongly associated with different cellular morphologies of cells undergoing EHT from HE (EHT pol⁻ and EHT pol⁺) which raises the exciting possibility that these morphological differences reflect the heterogeneity of HE (and therefore HSCs) at a very early stage. They then overexpress a truncated form of Runx1 (just the runt domain) to block Runx1 function and show that more HE cells abort EHT and remain associated with the embryonic dorsal aorta. They identify pard3aa and pard3ab as potential regulators of cell polarity. However, despite showing that loss of runx1 function leads to (late) decreases in the expression of these genes, no evidence for their role in EHT is presented. The FRAP experiments and the 2d-cartography, albeit very elegant, are difficult to interpret and not very clearly described throughout the text, making interpretation difficult for someone less familiar with the techniques. Finally, while it is clear that ArhGEF11 is playing an important role in defining cell shapes and junctions between cells during EHT, there is very little statistical evidence to support the limited data presented in the (very beautiful) images.

As mentioned in the response to reviewer 1, we revised our whole strategy for the analysis of the role of Pard3 proteins in regulating the emergence of hematopoietic precursors. Our new data, obtained using refined gene expression analysis by qRT-PCR on FACS sorted populations and by in situ gene expression analysis at the single-cell resolution using RNAscope, show first that a unique Pard3 isoform (Pard3ba) is sensitive to runx1 activity, and that its expression is specifically localized in aortic cells contacting hemogenic(HE)/EHT cells. We

show a clear correlation between the densification of Pard3ba mRNAs and the presence of contacting HE/EHT cells, suggesting a key role for Pard3ba in a cross talk between aortic and hemogenic cells. Furthermore, we show that our *dt-runx1* mutant impacts on the maturation of HE cells; when this mutant is expressed, we observe, in comparison to control, an accumulation of HE cells that are abnormally polarized as well as unusually high numbers of EHT pol+ cells. This strongly suggests that the polarity status of HE cells controls the mode of emergence. Overall, our work shows that regulation of apico-basal polarity features is essential for the maturation of the HE and the proper proceeding of the EHT.

We made efforts to explain more clearly the FRAP experiments as well as the analysis of 2Dcartography throughout the text to facilitate readers comprehension. 2D-cartography are an invaluable tool to precisely discriminate between endothelial and hemogenic cells, and their usage was essential during the FRAP sessions, to point at specific junctional complexes accurately. Performing FRAP at cellular junctions during aortic development was extremely challenging technically and the outcome subjected to quite significant variability (which often leads to quantitative results at the limit of the statistical significance, which is why we speak of tendencies in our results section reporting on this type of experiments). Apart from constant movement and drifting of the embryos which are sources of variability, the EHT process per se is evolving over time and does so at heterogeneous pace (for example, the apical closure of EHT pol+ cells is characterized by a succession of contraction and stabilization phases, see Lancino et al. 2018) which is an additional source of variability in the measurements. Despite all this, our data collectively and consistently suggest a differential regime of junctional dynamics between EHT cell types and support the critical function of ArhGEF11/PDZ-RhoGEF in the control of junctional turnover at the interface between HE and aortic cells as well as between HE cells to regulate cell-cell intercalation.

There is a sense that this work is both overwhelming in terms of the sheer amount of imaging data, and the work behind it to generate all the lines they required, and at the same time that there is very little evidence supporting the assertion that pard3 (and even ArhGEF11) are important mediators of cell morphology and cell fate in the context of EHT. For instance, the pard3 expression data, and levels after blocking runx1 (part of Figure 3 and Figure 4) don't particularly add to the manuscript beyond indicating that the pard3 genes are regulated by Runx1.

We thank the reviewer for the comment on the Pard3 data particularly because it led us to reconsider our strategy to address with more precision and at the cellular resolution the potential function of this protein family during the time-window of the EHT. As summarized in the header of the Public Review, we identified one specific isoform of Pard3 in the zebrafish - Pard3ba – whose sensitivity to *runx1* interference and spatial restriction in expression reinforce the idea of a fine control of apico-basal polarity features and associated functions while EHT is proceeding. Our new data also reinforce the interplay between HE/EHT cells and their direct endothelial neighbors.

Weaknesses

The writing style is quite convoluted and could be simplified for clarity. For example, there is plenty of discussion and speculation throughout the presentation of the results. A clearer separation of the results from this speculation/discussion would help with understanding. Figures are frequently presented out of order in the text; modifying the figures to accommodate the flow of the text (or the other way around) - would make it much easier to follow the narrative. While the evidence for the different cellular morphologies of cells undergoing EHT is strong, the main claim (or at least the title of the manuscript) that tuning apico-basal polarity and junctional recycling orchestrate stem cell emergence complexity is not well supported by the data.

We refined our text when necessary, in particular taking care of transferring and substantiating the arguments that appeared in the Results section, to the Discussion. We also made efforts, on several occasions and for clarity, to describe more precisely the results presented in the different panels of the Figures.

As mentioned in the header of the text of the Public Review and the response to the 6th point of the Public Review of Reviewer 1, we modified slightly the title to avoid ambiguity. In addition, we added a new paragraph to the beginning of our discussion that summarizes the impact of our findings and, we believe, legitimates our title.

Recommendations for the authors:

Reviewer #1 (Recommendations For The Authors):

(1) Embryonic stages should be indicated in all images presented for clarification.

We thank the reviewer for this point, we added stages when missing on the figures (Figure 1, Figure 1 - Figure supplement 1, Figure 2, Figure 2 - Figure supplement 1, Figure 5, Figure 6, Figure 6 - Figure supplement 1, Figure 7 - Figure supplement 3, Figure 7 - Figure supplement 5, Figure 7 - Figure supplement 6)

(2) In which anatomical site/s were images from Fig 1C and D taken? The surrounding environment looks different, for example, cells in Fig1D seem to be surrounded by other cells, resembling the endothelial plexus at the CHT, while the cells in Fig. 1C seem to be in the dorsal aorta. Is there a spatial difference depending on where cells are budding off? The authors state that there are no differences, but no quantification or data demonstrating that statement is provided.

As mentioned in the figure legend (lines 1206-1209 of the original manuscript), images for Figure 1C and 1D were both taken at the boundary between the end of the AGM and the entry in the caudal hematopoietic tissue. As the images were acquired from different embryos, the labelling of the underlying vein differs between the two panels, with veinous tissues being more sparsely labelled in panel C than in panel D. These images were chosen to illustrate the clearly opposite morphology between the two EHT types that we describe. However, for the rest of the paper, all images and all analysis were exclusively acquired / performed in the dorsal aorta in the AGM, in a region spanning over approximately 10-12 inter-segmentary vessels, starting from the end of the elongated yolk up to the start of the balled yolk. In light of the work from the lab of Zilong Wen showing that only cells emerging anteriorly exhibit long-term replenishment potential (Tian et al. 2017), we specifically chose to limit our comparative analysis to the AGM region and did not quantitatively investigate emergences occurring in the caudal region of the aorta. Additionally, although we routinely observe both types of emergences occurring in the caudal region of the dorsal aorta, we did not quantify the frequency of either EHT events in this region.

Finally, the EHT pol+ cells that we show Figure 1C are of the highest quality obtained ever; one reason is that these two cells emerge at the entry of the CHT which is a region a lot easier to image at high resolution in comparison to the trunk because the sample is less thick and because we are less perturbed by heart beats.

(3) Which figure shows "EHT pol- cells were observed in all other Tg fish lines that we are routinely imaging, including the Tg(Kdrl:Gal4;UAS:RFP) parental line that was used for transgenesis, thus excluding the possibility that these cells result from an artefact due to the expression of a deleted form of Podxl2 and/or to its overexpression."? It would be informative to include this figure.

Other examples of EHT pol- cells were shown Figure 5C as well as Figure 6B using the Tg(kdrl:Jam3b-eGFP; kdrl:nls-mKate2) fish line, that was routinely used for junctional dynamic analyses by FRAP. Furthermore, we add now a new figure (New Figure 1 – figure supplement 3), to illustrate the presence of EHT pol- cells using the Tg(CD41:eGFP) transgenic background, additionally illustrating that EHT pol- cells are CD41 positive.

*(4) Are the spinning disk confocal images a single plane? Or maximum projections?
Sometimes this is not specified.*

We made sure to take into account this remark and went through all figures legends to specify the type of images presented (Figure 1 – figure supplement 1, Figure 2, Figure 2 – figure supplement 1, Figure 2 – figure supplement 2, Figure 7 – figure supplement 3) and also, when relevant, we added this information directly to the figure panels (Figure 6A – 6B).

(5) Could the expression data by RT-qPCR for the Pard3 isoforms be shown? Additionally, it would be appreciated if this expression data could be complemented using Daniocell (<https://daniocell.nichd.nih.gov/>).

As mentioned in the first paragraph of our response to Public Reviews, and based on reviewers' comments, we revised our strategy for the investigation of pard3 proteins expression in the vascular system, for their potential role in EHT and sensitivity to runx1. First, we used FACS sorting as well as tissue dissection to enrich in aortic endothelial cells and perform our qPCR analyses (see the new Figure 4 – figure supplement 1A and Figure 4 – figure supplement 3A for the strategy). As asked by the reviewers and for more transparency, we show the expression relative to the housekeeping gene ef1a in our different control samples (new Figure 4 – figure supplement 1C). Furthermore, we used single-molecule FISH to precisely characterise in situ the expression of several of the Pard3 isoforms (Pard3aa, Pard3ab and Pard3ba, which, based on qPCR, were the most relevant for our investigation in the vascular system) (see lines 386 to 412 in text relative to Figure 4 – figure supplement 2). This new addition nicely shows the different pattern of expression of 3 of the Pard3 zebrafish isoforms in the trunk of 2dpf embryos, outlining interesting specificities of each isoform expression in different tissues.

We thank the reviewer for this suggestion to complement our data with the published Daniocell dataset. However, and potentially due to the poor annotation of the different pard3 genes on public databases, gene expression information was absent for two of our isoforms of interest (pard3aa and pard3ba), that we ultimately show to be the most enriched in the vascular system in the trunk. Daniocell gene expression data for the Pard3ab isoform at 48hpf show expression in pronephric duct at 48-58hpf, as well as in intestine progenitors and neuronal progenitors, which is consistent with our in situ observations using RNAscope. However, pard3ab is poorly detected within the hematopoietic and vascular clusters. This observation is coherent with our data that do not show any enrichment of this isoform in vascular tissues compared to other structures. On the other hand, pard3bb does not seem to be particularly enriched in vascular/hematopoietic clusters at 48-58hpf in the Daniocell dataset, in accordance to what we observe with our qPCR. Finally, in the Daniocell dataset, all of the pard3 variants (pard3ab, pard3bb, PARD3 and PARD3 (1 of many)) seem to be either scarcely or not detected in the hematopoietic/vascular system. In our case, for all the isoforms we studied in control condition (pard3aa, pard3ab and pard3ba), and although the technic is only semi-quantitative due to the presence of an amplification step, RNAscope assays seem to indicate a very low expression in aortic cell (with sometime as little as one mRNA copy per cell; this explains low detection in single-cell RNAseq datasets and is coherent with the Daniocell dataset).

(6) *It would be informative to add in the introduction some information on apico-basal polarity, tight junctions, JAMs (ArhGEF11/PDZ-RhoGEF).*

We modified the introduction so as to add relevant information on Pard3 proteins, their link with our JAMs reporters in the context of polarity establishment, as well as the role of ArhGEF11/PDZ-RhoGEF and its alternative splicing variants in regulating junctional integrity in the context of epithelial-to-mesenchymal transition (lines 99 to 127). This modification of the introduction also allowed us to lighten some parts of the result section (lines 222 to 224, 345 to 349 and 454 to 456 of the original manuscript).

Reviewer #2 (Recommendations For The Authors):

(1) *There is lots of data (and lots of work) in this paper; I feel that the pard3 data doesn't substantially add to the paper, and at the same time there is data missing (see point 10, point 11 below for an example).*

To add to the clarity and substantiate our findings on Pard3, we revised entirely our investigation strategy as mentioned in previous paragraphs. We refined the characterization of Pard3 isoforms expression in the vascular tissue, using both cell enrichment by FACS for gene expression analysis as well as single-molecule FISH (RNAscope) to access to spatial information on the expression of pard3 isoforms, reaching sub-cellular resolution.

This new strategy allowed us to show the unexpected localization of Pard3ba mRNAs in mRNAs enriched regions in the vicinity of HE/EHT cells (new Figure 4, and paragraph Interfering with Runx1 activity unravels its function in the control of Pard3ba expression and highlights heterogeneous spatial distribution of Pard3ba mRNAs along the aortic axis, see the new manuscript). Overall, the new spatial analysis we performed allowed us to substantiate our findings on Pard3ba and suggests a direct interplay between hemogenic cells and their endothelial aortic neighbors; this interplay supposedly relies on apico-basal polarity features that is at least in part regulated by runx1 in the context of HE maturation and EHT.

(2) *Labelling of the figures could be substantially improved. In many instances, the text refers to a figure (e.g. Fig 6A), but it has several panels that are not well annotated (in the case of Fig 6A, four panels) or labelled sparsely in a way that makes it easy to follow the text and identify the correct panel in the figure. Even supplementary figures are sparsely labelled. Labelling to include embryonic stages, which transgenic is being used, etc should be added to the panels to improve clarity for the reader.*

We revised the figures to added relevant information, including stages, types of images and annotations to facilitate the comprehension, including Figure 6A – 6B, Figure 5B – 5C (see response to Reviewer 1, first comment, for a more complete list of all revised figures, transgenic fish lines and embryonic stages annotations). Furthermore, we revised the integrality of the manuscript to fit as much as possible to the figures and added some annotations to more easily link the text to the figures and panels.

(3) *The current numbering of supplementary figures is quite confusing to follow.*

We revised the manuscript so as to make sure all principal and supplementary figures were called in the right order and that supplementary figures appearance was coherent with the unfolding of the text. For Figure 7 only, the majority of the supplemental figures are called before the principal figure, as they relate to our experimental strategy that we comment on before describing the results.

(4) Graphs in Fig 4, Fig 7 supplement 1 and some of the supplementary figures miss statistical info for some comparison (I assume when non-significant), and sometimes present a p-value of a statistical test being done between samples across stages - but these are not dealt with in the text. Throughout all graphs, the font size used in graphs for annotation (labelling of samples, x-axis, and in some cases the p values) is very small and difficult to read.

For Figure 7 - figure supplement 1, non-significant p-values of statistical tests were not displayed (as mentioned in the Figure legend, line 1614 of the original manuscript). For the new Figure 4, all p-values are displayed. For new Figure 4 - figure Supplement 1, statistical tests were only performed to compare RFP+ and RFP- cells in the trunk condition (3 biological replicates) and not in the whole embryo condition, for which we did not perform enough replicates for statistical analysis (biological duplicates).

(5) The results are generally very difficult to follow, with a fair amount of discussion included but then very little detail of the experiments per se.

We thank the reviewers for these comments that helped us improve the clarity of the manuscript.

The Results section was revised to move some of the paragraphs to the introduction (see response to Reviewer 1, 6th comment), and some of them to the Discussion (such as lines 149 to 156 or 410 to 416 in the first version of the manuscript referring to vacuolar structures or to the recycling modes of JAMs in EHT pol+ and EHT pol- cells).

(6) The truncated version of runx1 is introduced but its expected effect is not explained until the discussion. Related to this, is it expected that blocking runx1 with this construct (leading to accumulation of cells in the aorta before they undergo EHT) then leads to increased numbers of T-cell progenitors in the thymus? Abe et al (2005, J Immunol) have used the same strategy to overexpress the runt domain in thymocytes and found a decrease in these cells, rather than an increase. Can you explain this apparent discrepancy?

We thank the reviewer for this interesting point on the effect of runx1 interference. This phenotype (increased number of thymic cells) seems to be in agreement with the phenotype that was described in zebrafish using homozygous runx1 mutants (Sood et al. 2010 PMID: 20154212), in which the authors show an increase of lymphoid progenitors in the kidney marrow of adult runx1W84X/W84X mutants compared to controls as well as a similar number of intra-thymic lck:eGFP cells in mutants and controls. Notably, the T-lymphoid lineage seems to be the only lineage spared by the mutation of runx1. This could suggest that in this case either the T-lymphoid lineage can develop independently of runx1 or that a compensation phenomenon (for example by another protein of the runx family) occurs to rescue the generation of T-lymphocytes.

Although our data shows an impact on T-lymphopoiesis, we do not elucidate the exact mechanism leading to an increased number of thymic cells. In our case, we do not know the half-life of our dt-runx1 protein in newly generated hematopoietic cells when our transgene, expressed under the control of the kdrl vascular promoter, ceases to be produced after emergence. The effect we observe could be direct, due to the presence of our mutant protein after 3 days in thymic cells, or indirect, due to the impact of our mutant on the HE, that could lead to the preferential generation of lymphoid-biased progenitors. Similarly, we do not know whether the cells we observe at this stage in the thymus are generated from long-term HSC or short-term progenitors. Indeed, cell tracing analysis from the lab of Zilong Wen (Tian et al. 2017, see our Ref list) show the simultaneous presence of short-term PBI derived and

longterm AGM derived thymic cells at 5dpf. Based on this, we can imagine for example that the sur-numerous cells we observe in the thymus are transient populations that could multiply faster in the absence of definitive populations. Conversely, based on our observation of an accumulation of EHT pol⁺ events, we can imagine that the EHT pol⁺ and EHT pol⁻ cells are indeed differentially fated and that EHT pol⁺ may be biased toward a lymphoid lineage. We also know that at the stage we observe (5dpf), RNAscope assay of runx1 show that a vast majority of thymic cells do not express runx1 (our preliminary data), suggesting that the effect we observe would be an indirect one caused by upstream events rather than by direct interference with the endogenous expression of runx1 in thymic cells.

The article referred to by the reviewer (Sato et al. 2005, PMID: 16177090) investigates on the role of runx1 during TCR selection for thymic cell maturation and shows that runx1 signaling lowers the apoptotic sensitivity of double-positive thymocytes when artificially activated, leading to a reduced number of single-positive thymic cells. Furthermore, this paper references another study from the same lab (Hayashi et al. 2000, PMID: 11120804) that used the same strategy to study the role of runx1 on the positive and negative selection steps of T lymphocytes maturation. This paper, although showing that runx1 is important for later stages of T lymphocytes differentiation — the double-positive to single-positive stage maturation —, also shows a relative increase in the amount of double-negative and double-positive thymocytes, that could be coherent with our observations. Indeed, in our case, although we show an increased number of thymic cells, we do not know the relative proportion of the different thymocyte subsets. We could explain the increased number of thymic cells by increased number of DN/DP thymocytes that would not preclude a decrease in single-positive thymocytes. Finally, the cells we observe in the thymus of our dt-runx1 mutants may also be different lymphoid populations, namely ILCs, that would react differently to runx1 interference.

(7) Lines 154-155 refer to aquaporins but are missing a reference. This is a bit of speculation right in the results section and I struggled to understand what the point of it was.

To clarify the argument and ease the flow of the text, as suggested by the reviewers, we transferred this paragraph (lines 149 to 156 of the initial manuscript) to the Discussion section lines 763-789). We additionally made sure to add the missing reference (Sato et al. 2023, see our Ref list).

(8) Lines 173-175, indicating that both EHTpol⁺ and pol⁻ express the CD41 transgenic marker - would be useful to show this data.

We provide a new supplement Figure (Figure 1 – figure supplement 3), where, using an outcross of the CD41:eGFP and kdrl:mKate2-podxl2 transgenic lines, we show unambiguously and for multiple cells that both polarized EHT pol⁺ cells and non-polarized EHT pol⁻ cells are CD41 positive. In addition, but not commented on in the main text, we can also see that an HE cell, characterized by its elongated morphology (in the middle of the field), its thickened nucleus and its position on the aortic floor, is also CD41 positive.

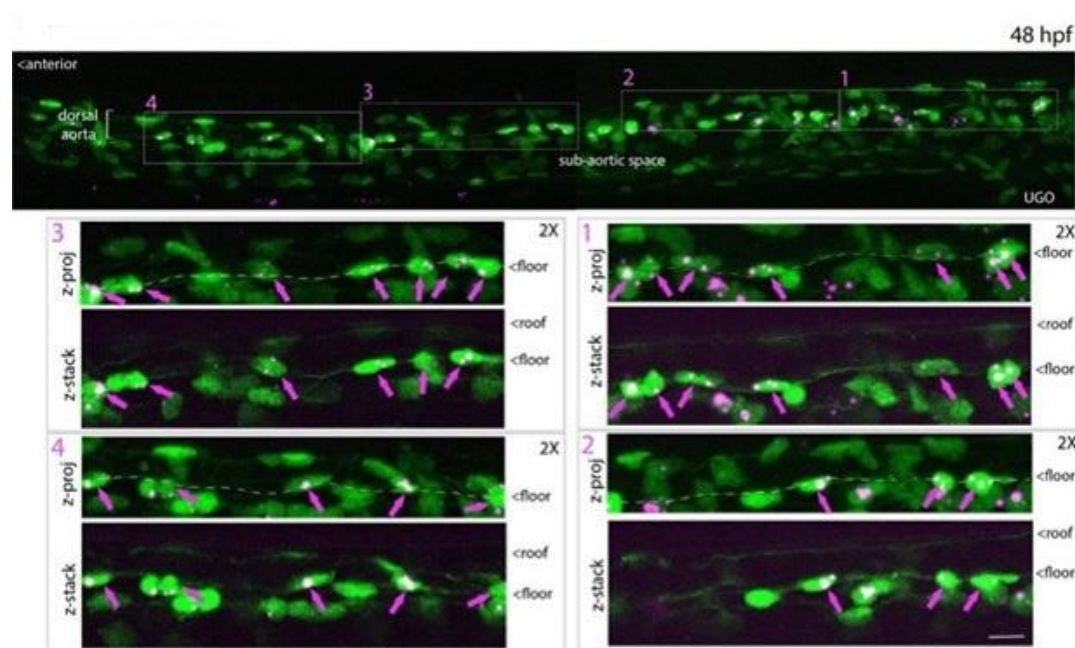
(9) Lines 181-201 - it's not clear how HE cells were identified in the first place - was it just morphology? Or were they identified retrospectively?

HE cells were identified solely on morphology and spatial criteria (as mentioned in the Methods section, lines 1073-1082 and 1108-1111 of the first manuscript). Furthermore, a recent investigation by the lab of Zilong Wen (Zhao et al. 2022, see our Ref list) questioning the common origin of HE cells and of endothelial cells as well as their respective capacity to extrude from the aorta to generate hematopoietic cells showed, by single-cell tracing, that

96% of floor cells are indeed hemogenic endothelial cells. Furthermore, as mentioned in the response to the 8th point, we show in Figure 1 – figure supplement 3 that all floor cells express CD41. Finally, we also used an alternative method to validate the true hemogenic identity of aortic floor cells and show, using RNAscope, that virtually 100% of floor cells that we consider as typical HE cells are indeed expressing an hematopoietic transcription factor upstream of Runx1, namely Gata2b (see Author response image 1).

Author response image 1.

All cells from the aortic floor, at 48hpf, express the hematopoietic marker Gata2b. 48 hpf Tg(Kdrl:eGFP) fixed embryos were used for RNAscope using a probe designed to detect Gata2b mRNAs. Subsequently, images were taken using spinning disk confocal microscopy. The image in the top panel is a z-projection of the entire aortic volume of one embryo and shows the full portion of the dorsal aorta from the anterior part (left side, at the limit of the balled yolk) down to the urogenital orifice (UGO, right side). The 4 boxes (1 - 4) delineate regions that have been magnified beneath (2X). The 2X images corresponding to each box are z-projections (top views) or z-sections (bottom views). The bottom views allow to visualize the aortic floor and to mark its position on top views). Pink arrows point at HE cells (elongated in the anteroposterior direction) and at EHT cells (ovoid/round cells; EHT pol+ cell morphology is not preserved after fixation and RNAscope; thus, it cannot be distinguished from ovoid/round EHT pol- cells). Pink dots = RNAscope spots of various sizes. The green cells in the subaortic space that are marked by RNAscope spots are newly born hematopoietic stem and progenitor cells (see for example box 1). This embryo is representative of n = 5 embryos treated and imaged.



(1) Line 276 - the difference between the *egfp-podxl2* and *mKate-podxl2* - could that be due to the fluorophore used? Also, it would be good to label Fig 3 supplement 2 better and to see a control alongside the runt overexpression.

Line 276 does not point at a difference in control conditions between eGFP-podxl2 and mKatepodxl2 (see in new Figure 1 – figure supplement 3, Figure 2 or in new Figure 3 - figure supplement 2 several examples of non-polarized HE cells in control conditions using both fluorophores) but between control and dt-runx1 conditions, both expressing the

mKate2podxl2 transgene. Similarly, the new example that we provide now in the CD41 figure (Figure 1 – figure supplement 3) clearly shows that mKate-podxl2 is enriched at the apical/luminal membrane of EHT pol+ cells while no such enrichment is observed for EHT pol- cells. The Reviewer should be informed that EHT cells are not always the most typical in shape, in particular because cells can be squeezed by underlying tissues and for example the vein; or from the luminal side by flow and tensions on the aortic wall because of heart beat (the more we image up in the trunk, the more difficult the imaging and the stability of cell shape during long time-lapse sequences). To also take into account the reviewer's comments, we added for the new Figure 3 – figure supplement 2A a control condition next to the dt-runx1 condition.

(2) There is no quantitation data on the number of excess EHT pol+ cells in the DA, or in the thymus data (Figs 3 Supp1 and Fig 3 Supp 3). Can you quantify this data? This would better support the claim that tunin apico-basal polarity alters the morphology of the emerging HE cells.

We added quantifications relative to both the emergence process itself, showing the accumulation of HE and EHT pol+ cells (new Figure 3B), and on hematopoiesis per se (new Figure 3 – figure supplement 1). Indeed, we show a diminution in the number of newly generated cmyb+ cells in the sub-aortic space. Furthermore, we improved our quantification of the later phenotype on the thymus (new Figure 3 – figure supplement 3), using improved segmentation methods, that indeed validate the increase number of thymic cells that we described.

(3) The observed changes in pard3 isoforms are just reading out changes in their expression in the runt1 transgenics, rather than demonstrating a role in apico-basal polarity.

We entirely revised our strategy regarding Pard3 expression analyses (see also the text at the beginning of this file, for the Public Review). But we wish to stress on the point that we did not intend initially to show directly a role of Pard3 proteins in controlling apico-basal polarity in the system, we just intended to provide correlative evidence supporting our observations with the polarity marker podxl2 (by interfering with their function, as written in the text, apico-basal polarity - which is essential for aortic lumenization and maintenance -, would have been impaired, blurring interpretations).

During the revision, we obtained the unexpected finding, using RNAscope, that one Pard3 isoform, namely Pard3ba, is the one Pard3 that is expressed non-homogenously along the aortic axis and, in vast majority, by aortic cells and in the direct vicinity of emergence domains of the aortic floor (see the new Figure 4 and Figure 4 – figure supplements 2, 3).

This correlative relation between expression of Pard3ba in aortic endothelial cells neighbouring HE/EHT cells suggests, as we propose, that a cross talk occurs between hemogenic and aortic cells, and that this cross talk relies, at least in part, on the expression of key components of apico-basal polarity and their associated functional features. In addition, we show that junctional recycling differs between both EHT types, based on our observations on the different dynamics in the turnover of JAM molecules, in the two EHT types. As JAM molecules are also required for the recruitment of Pard3, which initiates the establishment of apico-basal polarity, these different dynamics suggest that the control of apico-basal polarity is involved in supporting the morphodynamic complexity of EHT cell types.

(4) There is a Fig 5, Supp 2 that is neither mentioned nor described anywhere in the manuscript.

Figure 5 - figure Supplement 2 is mentioned lines 366-370 of the original manuscript, to describe the initial validation that was performed for our eGFP-JAM constructs in multiple cell types using an ubiquitous heat-shock promoter. We developed our description of this supplemental figure in the new manuscript (lines 504 to 514).

| *(5) Lines 445-456 - these read like a bit of discussion, not results. There are other similar parts of the results section that also read like a discussion (e.g. 526-533)*

Although we decided to keep this paragraph in the Results section, as it justifies the rationale behind the choice of ArhGEF11/PDZ-RhoGEF, we took the reviewers comment into account and, as mentioned in the response to reviewer 1 6th comment, lightened the Results section by transferring some of the paragraphs to the Introduction or Discussion sections.

| *(6) The description of Fig 7A (from line 505) is missing the stages at which the experiments were performed (also not labelled on the figure).*

The stages at which the experiments were performed is stated in the figure legend (line 1366) as well as in the Methods section of the original manuscript (line 1033). We added the information on top of the panels A and B for more clarity.

| *(7) Some figures have multiple panels (e.g. Fig 7Aa'), so when referred to in the text, it remains unclear which panel is being referred to.*

We modified the text so as to refer more clearly to the different panels when mentioned in the text, particularly with regards to Figure 7 and 8 but also for all the other figures.